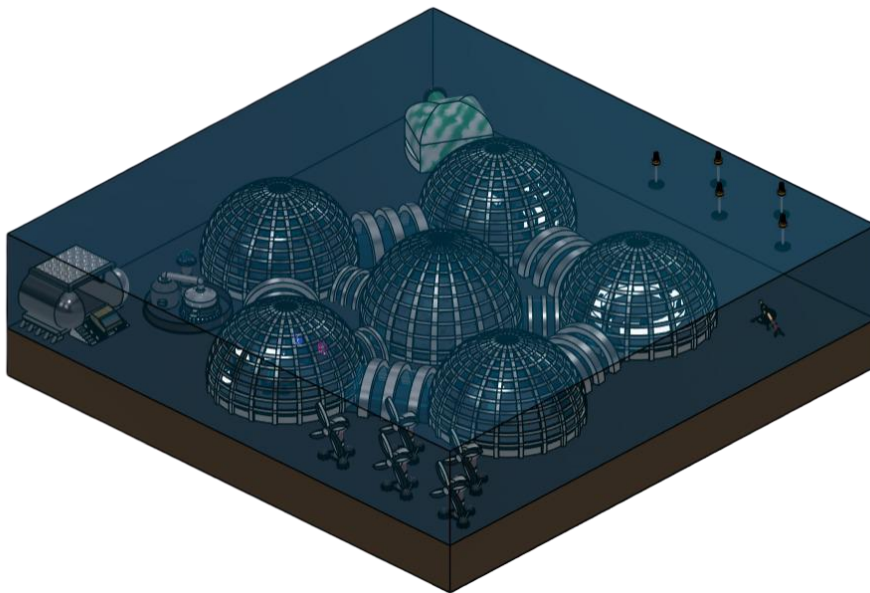


School of Engineering Integrated Design Project 2

Mechanical Engineering Detail Design 2025/26

Group: DM3 Future Ocean Habitat



04.05.2026

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Statement of Contribution

Name	Student ID	Role / Contribution in your own words. Include your assigned WP completed and all the tasks you completed in the other WPs.	Contribution level agreed with group with respect to the group average.	Signed signature with date
Ahmad Alhashmi	2716969	• WP 3 - DOC – Electrochemical Reactor / pH-Swing Stack Housing Coordinator • APPENDIX C: Research Journal Club Reports x6 • Conceptual Design Development WP3 • Design Calculations WP3 • CAD Development & Subsystem Integration WP3 • Conceptual Design Development WP8 • Overall Integration of Systems/Subsystems (Section 4): Integration narrative (how it all works as one facility) • Introduce the team & project (Presentation)	Normal	 06.03.2026
Matteo Karam	2760382	• WP5 - UDC - Thermal Management & Heat Rejection Coordinator • APPENDIX C: Research Journal Club Reports x6 • Conceptual Design Development WP5 • Design Calculations WP5 • CAD Development & Subsystem Integration WP5 • Conceptual Design Development WP7 • Overall Integration of Systems/Subsystems (Section 4): UDC thermal management loop → seawater heat rejection interface • Present the adopted (& enhanced) final Ocean Habitat concept (Presentation)	Normal	 06.03.2026

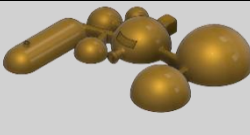
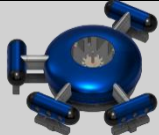
Muhammad Hammad	2840920	• WP2 - DOC - Process Flow Transmission Coordinator • APPENDIX C: Research Journal Club Reports x6 • Conceptual Design Development WP2 • Design Calculations WP2 • CAD Development & Subsystem Integration WP2 • Overall Integration of Systems/Subsystems (Section 4): Fluid Transmission System (FTS) • Design Calculations WP9 • CAD Development & Subsystem Integration WP9 • Describe system integration across the entire facility (Presentation)	Normal	 06.03.2026
Saba Falah	2847582	• Team and WP1 - DOC – Seawater Intake & Pretreatment Skid Coordinator • APPENDIX C: Research Journal Club Reports x6 • Candidate Conceptual Model Selection • Conceptual Design Development WP1 • Design Calculations WP1 • CAD Development & Subsystem Integration WP1 • Conceptual Design Development WP9 • Overall Integration of Systems/Subsystems (Section 4): DOC seawater intake → pre-treatment → DOC process modules → discharge/reject handling • Present the finalised design of the main units (Presentation)	Normal	 06.03.2026
Sarah Arafa	2715709	• WP4 - DOC – Neutralization /Mixing Coordinator • APPENDIX C: Research Journal Club Reports x6 • Conceptual Design Development WP4 • Design Calculations WP4 • CAD Development & Subsystem Integration WP4 • Design Calculations WP8	Normal	

		• CAD Development & Subsystem Integration WP8 • Overall Integration of Systems/Subsystems (Section 4): Energy/power distribution • Summarise the main systems (Presentation)		06.03.2026
Talal Abu Ghazaleh	2813867	• WP6 - UDC - Rack Chassis, Power Interface Coordinator • APPENDIX C: Research Journal Club Reports x6 • Product Design Specifications • Conceptual Design Development WP6 • Design Calculations WP6 • CAD Development & Subsystem Integration WP6 • Design Calculations WP7 • CAD Development & Subsystem Integration WP7 • Overall Integration of Systems/Subsystems (Section 4): Design rationales for integration	Normal	 06.03.2026

Section 1 – Candidate Conceptual Model Selection

The final habitat configuration is a pentagonal radial arrangement of geodesic pressure domes (HY-100 steel and Ti-6Al-4V) connected by cylindrical tunnels to a central hub, selected using a weighted decision matrix (Table 1). This layout integrates the Direct Ocean Capture (DOC) system, designed for a 1000 kg/day CO₂ capture rate requiring ~660 m³/h seawater intake, and the Underwater Data Centre (UDC) with a 1225 kW IT load. The interior is compartmentalized with a central control hub, a DOC intake and pre-treatment bay, a DOC process module, a UDC server zone with service corridor, energy storage (Li-ion BESS + hydrogen), systems communication module, and a mining workshop, all connected by transfer corridors with bulkhead doors for safety and maintenance access. The 3D CAD exterior (see Figure 1) shows the six domes on seabed footings, detailing the DOC bell-mouth intake with external pre-treatment skid, the UDC module, energy storage pods, and ROV docking interfaces. The pentagonal radial dome was selected for its superior weighted score (4.80) against the criteria. Subsequent modifications from the initial concept included adding the defined DOC intake and pre-treatment skid, a central control hub to improve circulation, and energy storage systems to enhance system integration and operational safety.

Table 1. Weighted decision matrix evaluated four Assignment 1 candidates against subsea-specific criteria.

Criteria	Weight	 Group D2	 Group D4	 Group D1	 Group D3
Manufacturability / Modularity	0.15	2	3	3	5
Maintainability	0.20	3	2	3	5
Corrosion / Biofouling Resilience	0.15	3	3	4	5
Energy Efficiency	0.10	3	4	4	4
Reliability / Redundancy	0.15	2	3	3	5
Sustainability	0.05	3	3	4	4
Safety	0.10	2	3	3	5
Ergonomics / Layout Efficiency	0.10	3	3	3	5
Weighted Score	1.00	2.60	2.95	3.35	4.80

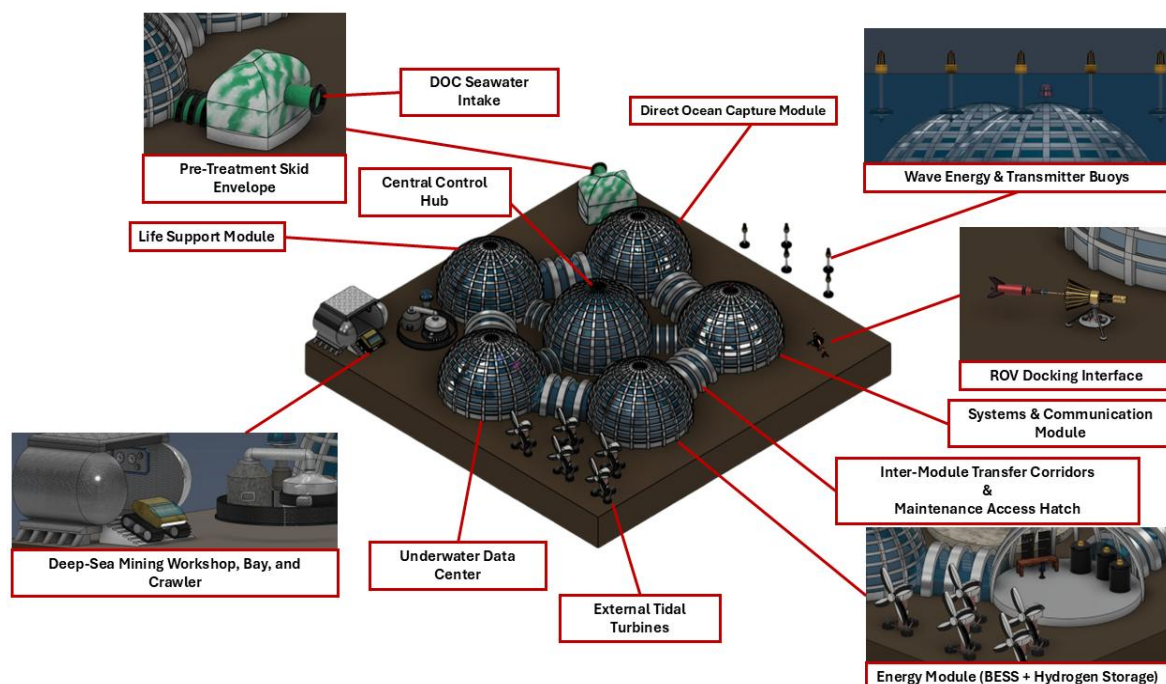


Figure 1. Annotated exterior CAD rendering of the modular underwater habitat.

Section 2 - Product Design Specifications

Table 2. Group DM3 Product Design Specification.

Design requirement	Design criteria	Means of assessment	Relevant standards
Operating environment	The Ocean Habitat shall operate in the Ionian Sea/south of Crete at a habitat depth of 100-150 m and remain functional under subsea pressure, salinity, corrosion, and biofouling conditions.	Site selection review; environmental design basis; corrosion/biofouling assessment; materials/coatings review.	ISO 19901-1; DNV-RP-C205
Structural integrity	The habitat structure shall maintain the pressure boundary at operating depth with a factor of safety ≥ 2 , leak detection active, and compartment isolation available.	Pressure-vessel calculations; structural stress analysis; FEA; leak-detection/isolation review.	ASME BPVC Section VIII Div. 2; DNV-RP-C202
Internal habitat environment	The habitat shall maintain a safe internal atmosphere for occupants: O ₂ 19.5-23.5%, CO ₂ $\leq 0.5\%$, temperature 20-25°C, humidity 30-60% RH.	Atmosphere mass-balance calculations; sensor and alarm review; life-support performance analysis.	ISO 7730
DOC seawater intake and pre-treatment	The DOC subsystem shall include seawater intake and pre-treatment suitable for continuous operation and protection of pumps and the electrochemical system.	PFD/P&ID review; hydraulic sizing calculations; component selection review; maintenance/access check.	ASME B31.3; ISO 21457; ISO 23936-1
DOC process/capture basis	The DOC subsystem shall use the selected electrochemical capture concept with associated pumping, gas separation, and CO ₂ handling.	Process-flow analysis; mass-balance calculations; equipment sizing calculations; concept design review.	ASME B31.3; IEC 61511-1
DOC performance	The DOC subsystem shall achieve CO ₂ capture uptime $\geq 95\%$.	Reliability analysis; operating-mode review; process/control assessment.	ISO 14064-2; ISO 14064-3
DOC environmental compliance	The DOC subsystem shall avoid harmful discharge, with pH change $\leq \pm 0.2$ and salinity change $\leq \pm 1$ PSU at discharge.	Discharge/mixing assessment; environmental impact review; monitoring and control review.	ISO 14001; ISO 5667-9
UDC availability and reliability	The UDC subsystem shall provide continuous data storage and processing with availability $\geq 99\%$, controlled degradation, and recovery to normal operation within 10-15 min after shutdown.	Reliability analysis; uptime/availability assessment; recovery strategy review.	ISO/IEC 22237-1; ISO/IEC TS 22237-31

Design requirement	Design criteria	Means of assessment	Relevant standards
UDC thermal management/heat rejection	The UDC shall maintain thermal levels within design limits and reject heat safely to seawater.	Heat-load calculations; thermal analysis; cooling-system review; heat-rejection assessment.	ISO/IEC 22237-1; ISO/IEC 22237-4
Energy/power continuity	The facility shall provide reliable power to critical loads with a continuous Tier 1 supply, storage for ≥ 72 h, and microgrid voltage regulation within $\pm 5\%$.	Load audit; energy-balance calculations; storage sizing calculations; power continuity assessment.	ISO/IEC 22237-3; IEC 60092-350; IEC 60092-376
Fluid Transmission System (FTS)	The system shall include routing for seawater, process streams, utilities, and heat-rejection flows between DOC, UDC, and supporting systems. Seawater circulation pumps shall deliver the required flow rate at sufficient pressure, with N+1 redundancy, NPSH margin ≥ 0.5 m above NPSHR, and all wetted materials compatible with seawater at 150 m depth.	PFD/P&ID review; routing/integration assessment; pump duty sizing calculations; system curve analysis; NPSH check; Joukowsky water hammer screening; material compatibility review.	ASME B31.3; ISO 21457; ISO 23936-1
Maintainability	The facility shall be designed for subsea maintainability with defined access routes, a modular replacement strategy where practical, and maintenance clearances.	CAD access/interference check; maintainability review; replacement strategy assessment.	IEC 60300-3-10
Deployment and retrieval	The facility shall support a defined deployment and retrieval strategy suitable for subsea installation and recovery.	Lift-case analysis; deployment/retrieval concept review; handling assessment.	DNV-ST-N001
Safety and fault response	The facility shall include safety provisions for leaks, electrical faults, overheating, and hazardous DOC deviations, with alarms, isolation, and monitoring coverage.	HAZOP review; fault-response assessment; instrumentation and interlock review.	IEC 61508; IEC 61882
Interface requirements	Interfaces between DOC, UDC, energy/power, controls, and FTS shall be clearly defined for flows, pressures, temperatures, power, and signals.	Interface matrix review; PFD/P&ID review; CAD integration check.	ISO/IEC 22237-1

Section 3 – Design & Engineering Development

Section 3.1 – Conceptual Design Development

WP1 – DOC – Seawater Intake & Pre-treatment Skid

WP Owner	Saba Falah
Contributors	Saba Falah
Key Interfaces	DOC processing unit, seawater pumps, control system, power system, discharge/flush system, subsea structural frame

Table 3. Scope and Boundaries for Seawater Intake & Pre-treatment Skid Work Package.

Included Items	Excluded Items
- Seawater intake architecture - Debris screening system - Pre-treatment filtration system - Biofouling mitigation strategy - Clean-in-Place (CIP) and flushing routing - Drain and vent points - Isolation and bypass arrangements - Concept-level instrumentation and control points - Skid layout concept and mounting interfaces	- Main seawater pumps - DOC chemical processing equipment - Electrical power supply systems - Control software and automation logic - Structural design of the main subsea facility - Detailed pipe sizing and hydraulic calculations - Detailed CAD modelling (covered in later design stages)

Table 4. Concept Options Comparison Table for Seawater Intake & Pre-treatment Skid.

Options	Concept Description	Main Components
PROJECT: SEAWATER INTAKE PILOT PLANT TITLE: P&ID, OPTION 1, OPEN INTAKE & CARTRIDGE FILTER SYSTEM DWG NO: SWI-PF-001 DATE: 2024-05-15 REVISION: 1.0 Legend: (DI) DIFFERENTIAL PRESSURE INDICATOR, (PT) PRESSURE TRANSMITTER, (CV) CHECK VALVE (CV), (RO) RESTRICTION ORIFICE (RO), (FC) BUTTERFLY VALVE (MANUAL), (P) CENTRIFUGAL PUMP, (F) CARTRIDGE FILTER.	Open seawater intake with coarse screen followed by cartridge filters mounted on a skid	Intake bell mouth, coarse screen, isolation valve, cartridge filter housing, differential pressure indicator, drain valve, flush valve, outlet pipe to DOC
Drawing No. DWG-002-P REV 1 Date 2024-05-21 Scale NTS Sheet 1 of 1 Title: Option 2: Ducted Intake - Automatic Backwash Filter Project: Seawater Filtration System Skid (PILOT PLANT) Drawn By: [Signature] App'd By: [Signature] Legend: (PT) PRESSURE TRANSMITTER, (DPI) DIFFERENTIAL PRESSURE INDICATOR, (CV) CHECK VALVE (CV), (RO) RESTRICTION ORIFICE (RO), (FC) BUTTERFLY VALVE (MANUAL), (P) CENTRIFUGAL PUMP, (F) AUTOMATIC SELF-CLEANING BACKWASH FILTER, (V) VENT VALVE, (D) DRAIN VALVE, (CIP) CIP CONNECTION, (S) SOLIDS SCREEN, (N) NOZZLES, (S) SCANNERS, (R) ROTATING SUCTION.	Ducted intake with coarse screen and automatic self-cleaning backwash filter	Ducted intake pipe, coarse debris screen, isolation valve, automatic backwash filter, backwash discharge line, flow meter, pressure sensors, drain valve, vent valve, CIP connection, skid frame

Table 5. Selection of the Final Concept Based on Product Design Specification (PDS) Requirements and Subsea Constraints.

Criteria	Option 1	Option 2
Maintainability	Medium	High
Biofouling resistance	Low	High
Reliability	Medium	High
Pressure loss	Low	Medium
Power consumption	Low	Medium
Modularity	Medium	High
Safety	Medium	High
Subsea suitability	Low	High

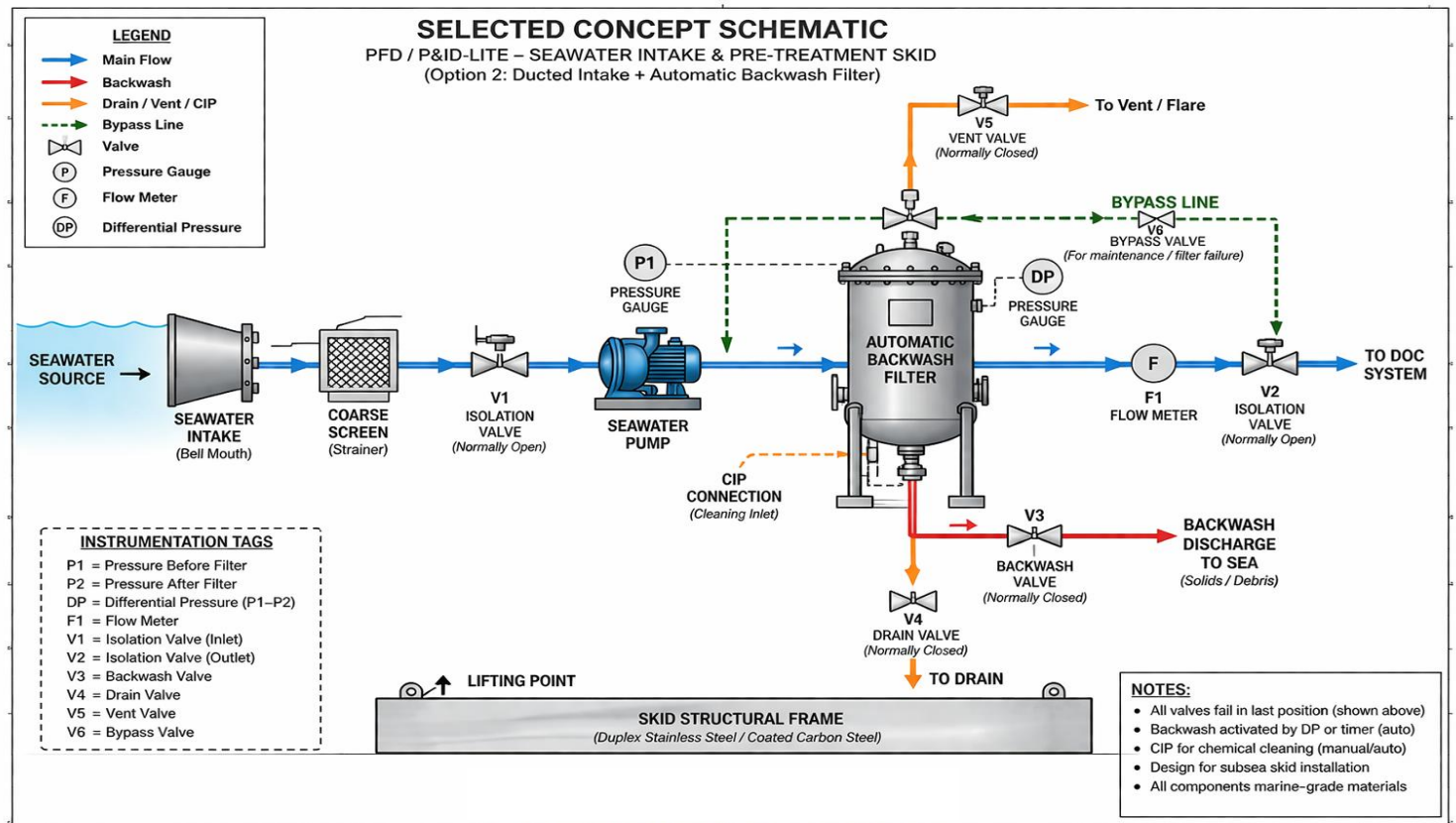


Figure 2. Conceptual PFD/P&ID-Lite DOC - Seawater Intake & Pre-treatment Skid.

Table 6. Maintainability and Safety Considerations for Seawater Intake & Pre-treatment Skid Work Package.

Maintainability	Safety
- Isolation valves upstream and downstream of the filter to allow maintenance without shutting down the full system - Automatic backwash filter to reduce manual cleaning frequency - Drain points at the lowest locations for system emptying - Vent points at high locations for air removal - CIP connections for chemical cleaning and biofouling removal - Adequate access clearance around the filter housing for removal and replacement - Modular skid design allowing retrieval and replacement as a single unit - Lifting points integrated into the skid frame - Standard flange connections for easy disconnection	- Isolation valves for emergency shutdown - Non-return valve to prevent backflow - Pressure relief protection to prevent over-pressurisation - Corrosion-resistant materials such as duplex stainless steel or HDPE - Anti-biofouling coating at intake screen - Smooth intake geometry to reduce flow turbulence - Emergency bypass line to maintain flow if the filter is blocked - Proper skid mounting interfaces to withstand subsea loads - Clear access zones for ROV or maintenance operations

WP2 – DOC – Process Flow Transmission (Pumps, Manifolds & Piping)

WP Owner	Muhammad Hammad
Contributors	Muhammad Hammad
Key Interfaces	WP1 pre-treatment skid outlet, WP3 DOC reactor inlet, WP4 discharge return line, electrical power supply, VFD control signals

Table 7. Scope and Boundaries for Process Flow Transmission Work Package.

Included Items	Excluded Items
- Main seawater pump selection, sizing, and N+1 redundancy - Pump baseplate, flexible couplings, and vibration isolation - Distribution manifold, valve station, and bypass line arrangement - System curve, TDH, and NPSH analysis - Water hammer screening and mitigation strategy - Pipe/flowline sizing and wall thickness calculations - Pipe supports, clamps, and expansion allowances - Material selection for seawater and galvanic compatibility - Concept-level instrumentation within WP2 boundary - Maintainability and safety provisions	- Seawater intake and pre-treatment skid (WP1) - DOC electrochemical reactor/stack housing (WP3) - Neutralisation and discharge diffuser (WP4) - Electrical power supply and motor drive hardware - Control software and automation logic - Structural design of subsea habitat frame

Table 8. Concept Options Comparison for Process Flow Transmission – Pumps, Manifolds & Piping.

Concept Schematic	Concept Description	Main Components
Option 1: Vertical Inline Centrifugal Pump	Positive displacement pump modules (such as twin screw pumps) are distributed on the surface in a modular manner alongside the pipeline. The pump takes suction from the flow of the DOC and delivers a constant flow rate to the downstream system. Power and controls can be provided through a wet mate connection.	Positive displacement pumps with twin screws, dry mounted electric motor in pressurized housing, wet mate electrical connection, gearbox coupling assembly, discharge manifold, isolation valve, non-return valve, bypass line, structural skid, subsea control module (SCM), pressure/temperature sensor, flowmeter.
Option 2: Submersible Centrifugal Pump (Selected)	Hermetically sealed submersible centrifugal pump/motor unit rated for 150 m external hydrostatic pressure, deployed directly in seawater adjacent to habitat. ROV-retrievable frame. Shared suction header and discharge manifold with isolation, bypass, and non-return valves. VFD-controlled.	Submersible pump/motor units (e.g. Demart Pumps 200WQ300-40-55), subsea wet-mate electrical connectors, suction/discharge manifold, isolation valves, non-return valves, bypass valve and line, flexible jumper hoses, VFD panel, pressure transmitters, flow meter.

Table 9. Selection of Final Concept Based on PDS Requirements and Subsea Constraints.

Criteria	Option 1	Option 2
Maintainability	Medium	High
Biofouling/Corrosion resistance	Medium	High
Reliability ($\geq 95\%$ uptime)	High	High
Pressure loss	Low	Low
Power consumption	Low	Low
Modularity	High	High
Safety	Medium	High
Subsea suitability at 150 m	Medium	High

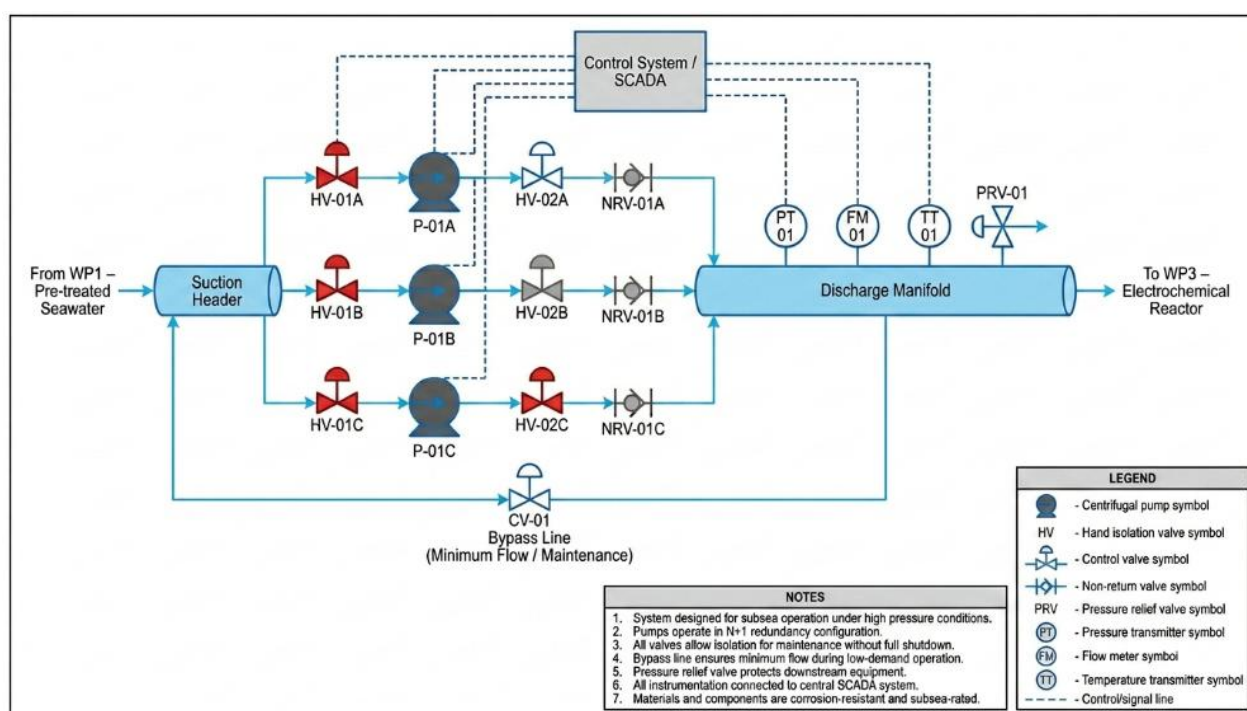


Figure 3. Conceptual PFD/P&ID-Lite — DOC Process Flow Transmission System (Option 2: Submersible Centrifugal Pump, N+1 Arrangement).

Table 10. Maintainability and Safety Considerations for Process Flow Transmission Work Package.

Maintainability	Safety
- Isolation valves upstream and downstream of each pump for online removal without full shutdown - N+1 arrangement enables duty pump maintenance while the standby maintains DOC flow - Bypass line allows pumps to run at minimum flow during DOC module isolation - ROV-retrievable pump modules with standardised wet-mate connectors - Flexible jumper hoses with quick-disconnect fittings for pump decoupling - Drain/vent points at lowest/highest locations for system decommissioning - All components pressure-rated for 150 m external pressure per ASME B31.3 (American Society of Mechanical Engineers, 2014)	- Non-return valves on each pump discharge to prevent backflow on pump trip - Isolation valves for emergency shutdown of individual pump trains - Water hammer mitigation: VFD ramp-down, slow valve actuators - Pressure relief on discharge manifold to protect WP3 reactor from overpressure - Super duplex SS (UNS S32750) and titanium fasteners per ISO 23936-1 (ISO 23936-1, 2009). - Cathodic protection anodes on ROV frame and pump casing - Anti-biofouling coating on external pump surfaces and jumper hoses

WP3 – DOC – Electrochemical Reactor / pH-Swing Stack Housing (and membrane/contactor if used)

WP Owner	Ahmad Alhashmi
Contributors	Ahmad Alhashmi
Key Interfaces	DOC pre-treatment outlet; DOC process pumping unit; DC power supply; CO2 vent and gas line; CIP and flush supply; drain and leak collection; control and instrumentation system; structural skid frame.

Table 11. Scope and Boundaries for WP3 - Electrochemical Reactor / pH-Swing Stack Housing Work Package.

Included Items	Excluded Items
- Mechanical housing for pH-swing electrochemical stack - Modular cassette philosophy for maintenance and swap-out - Sealing strategy, compression method and leak containment - Optional HFMC membrane/contactor side-loop with bypass and isolation - Vent, drain, CIP/flush, integrity-test and instrument points - Seawater-compatible materials, coatings and galvanic isolation concept - Service access, lifting points, inspection and retrieval considerations	- Brine neutralization, mixing and discharge diffuser hardware - Seawater intake, pre-treatment and process transfer pump design - Detailed electrochemical kinetics, catalyst design or membrane chemistry - DC rectifier design, automation logic and detailed electrical controls - Detailed bolt preload calculations, stress calculations and CAD tolerances - Full facility pipe routing, whole-habitat layout and procurement costing

Table 12. Concept Options Comparison for WP3 DOC Electrochemical Reactor / pH-Swing Stack Housing.

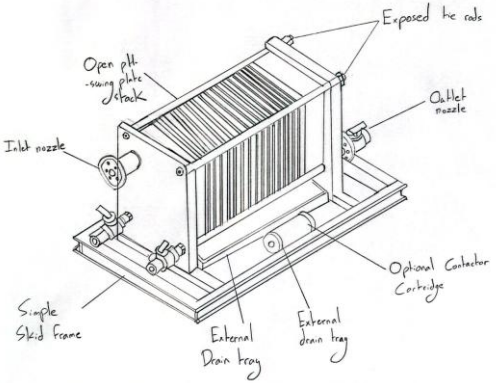
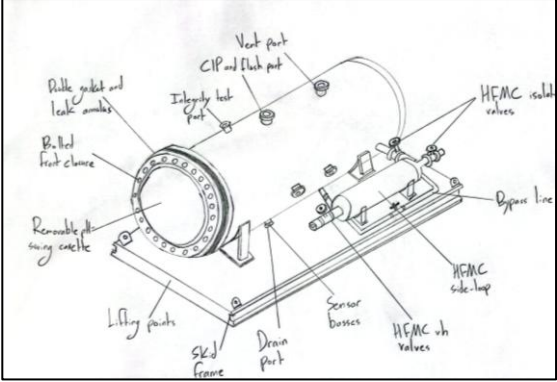
Options	Concept Description	Main Components
Option 1. Open filter-press stack 	Open plate-and-frame pH-swing stack with exposed tie rods and gasket edges. The concept gives direct access to the stack and is simple to assemble, but exposed compression hardware increases leakage risk and is less suitable for long-duration subsea operation. The pH-swing basis is technically feasible, but the housing is less robust for seawater deployment (Sharifian et al., 2022; Moreno Soriano et al., 2021).	Open plate pack, end plates, tie rods, visible gasket edges, inlet and outlet nozzles, external drain tray, simple skid frame, optional external contactor cartridge and manual isolation points.
Option 2. Sealed housing + HFMC side-loop 	Pressure-rated cylindrical housing with a removable cassette inside a sealed shell. The optional HFMC side-loop is isolated, bypassable and cleanable. This concept provides better sealing, controlled compression, leak capture and subsea material protection while retaining cassette serviceability (Aliaskari et al., 2024; Rivero et al., 2023; Sharifian et al., 2023).	Pressure shell, removable cassette, bolted front closure, double gasket/leak annulus, vent, drain, CIP and integrity ports, HFMC side-loop, isolation valves, bypass line, sensor bosses, skid frame and lifting points.

Table 13. Selection of the Final Concept Based on Product Design Specification Requirements and Subsea Constraints.

Criteria	Option 1	Option 2
Sealing reliability	Medium	High
Compression control	Medium	High
Leak containment	Low	High
Serviceability	Medium	High
HFCM bypass and cleanability	Low	High
Fouling management	Medium	High
Subsea material suitability	Medium	High
Integration simplicity	High	Medium

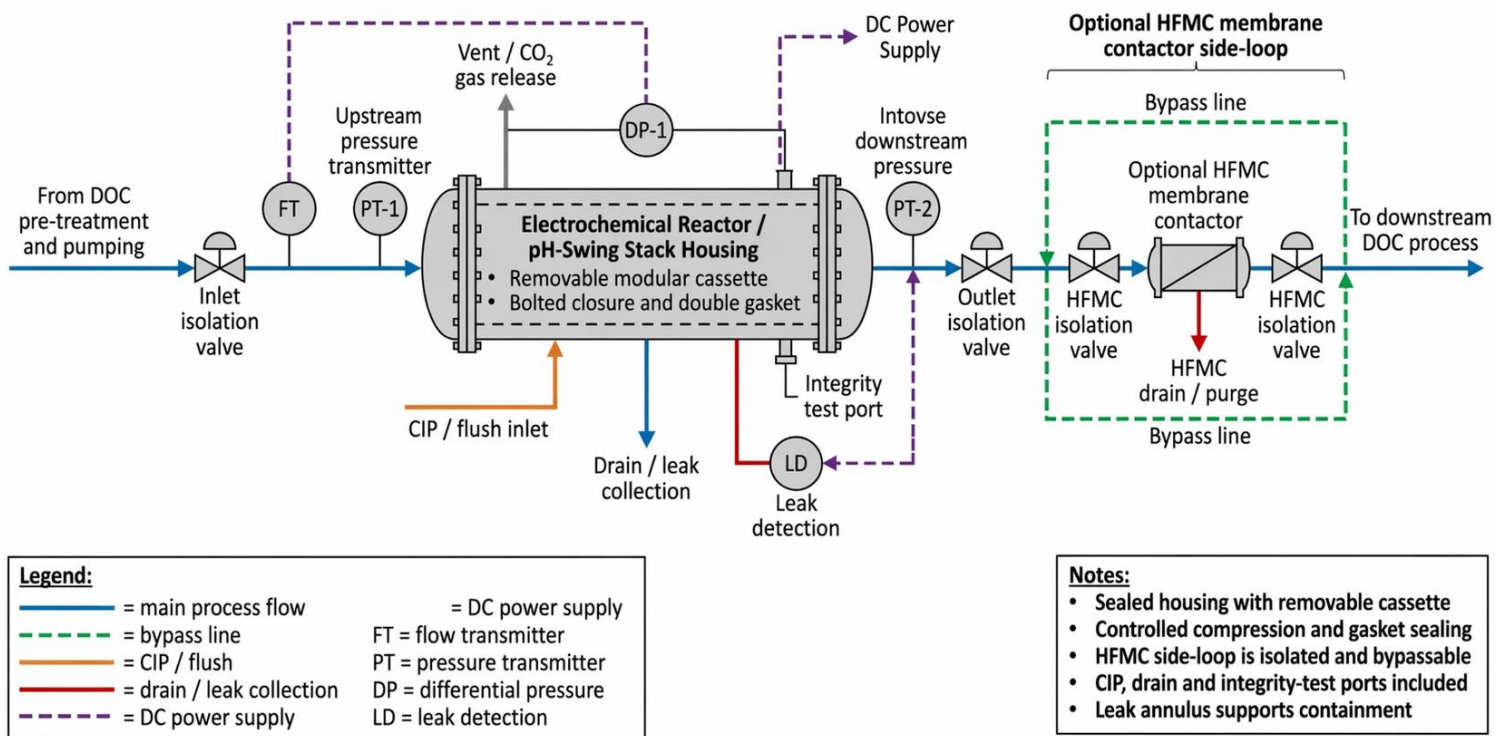


Figure 4. Selected Concept Schematic for WP3 DOC Electrochemical Reactor / pH-Swing Stack Housing and Optional HFCM Contactors (Sharifian et al., 2022; Rivero et al., 2023; Sharifian et al., 2023).

Table 14. Maintainability and Safety Considerations for WP3 Electrochemical Reactor / pH-Swing Stack Housing Work Package.

Maintainability	Safety
- Front closure allows the pH-swing cassette to be removed as one replaceable module - Cassette guide rails and compression stops reduce reassembly error - Isolation and bypass valves allow the HFCM module to be cleaned or replaced without dismantling the stack (Rivero et al., 2023) - CIP/flush and drain points support fouling removal because seawater BPMED can suffer carbonate scaling (Sharifian et al., 2023) - Standard flanged nozzles and lifting points support retrieval and shop maintenance	- Sealed shell, double gasket land and captured drain reduce uncontrolled acid or alkaline leakage - Integrity-test port and leak annulus support pressure checks before operation - Controlled clamp compression improves repeatable gasket seating and reduces local over-compression (Moreno Soriano et al., 2021) - Titanium or super-duplex wet parts with dielectric breaks reduce galvanic corrosion risk in seawater (Hao et al., 2024) - Vent line provides safe CO2 gas release from the acidified stream

WP4 – DOC – Neutralisation/Mixing + Discharge Diffuser

WP Owner	Sarah Arafa
Contributors	Sarah Arafa
Key Interfaces	Inputs: Acid + base from WP3. Outputs: Neutralised seawater to ocean via multiport diffuser. Interfaces: WP2 (pump head), WP1 (upstream), WP9 (SCADA).

Table 15. WP4 Scope and Boundaries.

	Included	Excluded
Systems	Neutralisation mixing chamber (V-401); pH control & recirculation loop; discharge pipeline (Line 401-C, DN300, 50 m); ESD valve; 12-port multiport diffuser manifold; seabed base frame; anti-scour mattresses; environmental monitoring cluster; all flanges, gaskets, bolting, supports, corrosion protection.	WP3 reactor internals; WP1 intake/pre-treatment; WP2 pumps; CO ₂ handling; WP7 pressure hull; WP8 energy generation/storage; WP9 SCADA hardware.

Table 16. WP4 Concept Options Comparison.

	Option 1: Baffled Tank + Linear Multiport Diffuser (Selected)	Option 2: Static Mixer + Single-Port Rosette
Neutralisation	Horizontal cylindrical vessel (SD 2507, ID 1600 mm, L 4800 mm), 4 alternating baffles, 52.7 s residence time, pH feedback + recirculation.	Kenics-type helical static mixer in pipeline; ~5–10 s residence time; no recirculation.
Diffuser	18 m DN350 header, 12×DN65 ports at 60° from horizontal, 1500 mm spacing, ROV-replaceable nozzle inserts.	Rosette head, 6×DN80 ports radially arranged at pipeline terminus.
pH Control	PLC-modulated base flow valve (FCV-401) + duplex recirculation pumps for polishing.	pH analyser downstream; limited trim authority.

Table 17. Selection Rationale Linked to PDS and Subsea Constraints.

Criteria	Option 1	Option 2
Maintainability	High – isolable vessel, CIP flush, ROV-replaceable nozzles	Medium – mixer fixed in pipeline
Biofouling / corrosion	High – SD 2507 (PREN > 40), CIP connection, CP anodes (Norsok M-001)	High – same materials, no CIP
Reliability / Redundancy	High – 52.7 s residence, N+1 recirc pumps, deterministic mixing	Medium – short residence, no back-up
Pressure loss	Low – 23.2 kPa (11.6% of pump budget, Darcy–Weisbach)	Low – comparable (~20 kPa)
Power Demand	Low – recirc pump ~1.5 kW (duty cycled)	Lower – no recirc pump
Modularity	High – vessel on skid + diffuser on independent frame	Medium – mixer integral to pipeline
Safety / ESD	High – hardwired ESD (SIL-1, IEC 61511), fail-closed XV-401	Medium – same ESD, no recirc polishing

Selected Concept Description: The selected design comprises five main components that are modelled separately in CAD and then assembled: (1) Neutralisation Mixing Chamber (V-401) – horizontal cylinder with 4 alternating internal baffles creating a serpentine flow path, 8 nozzles, and 2 saddle supports (ASME BPVC VIII Div. 1); (2) Discharge Pipeline (Line 401-C) – DN300 pipe with manual gate valve, motorised ESD valve, check valve, and flanged connections with regular pipe supports (ASME B31.3); (3) Multiport Diffuser Manifold – DN350 header with 12 angled outlet ports and removable nozzle inserts with sealing and retaining mechanism (Fischer et al., 1979); (4) Seabed Base Frame – structural frame using square hollow sections with cross-members, pipe support cradles, levelling feet, pad eyes, ballast pockets,

and anti-scour mattresses (DNV-OS-H101); (5) Monitoring Bracket – vertical mast with sensor mounting rail, cable routing provision, and subsea junction box.

Environmental constraints addressed: $\Delta pH \leq \pm 0.2$ at boundary via 49:1 near-field dilution (Fischer et al., 1979); anti-scour mattresses prevent seabed disturbance; corrosion managed through SD 2507 (PREN > 40, Alleima, 2024) + CP anodes (Farwest); biofouling controlled by CIP flush capability.

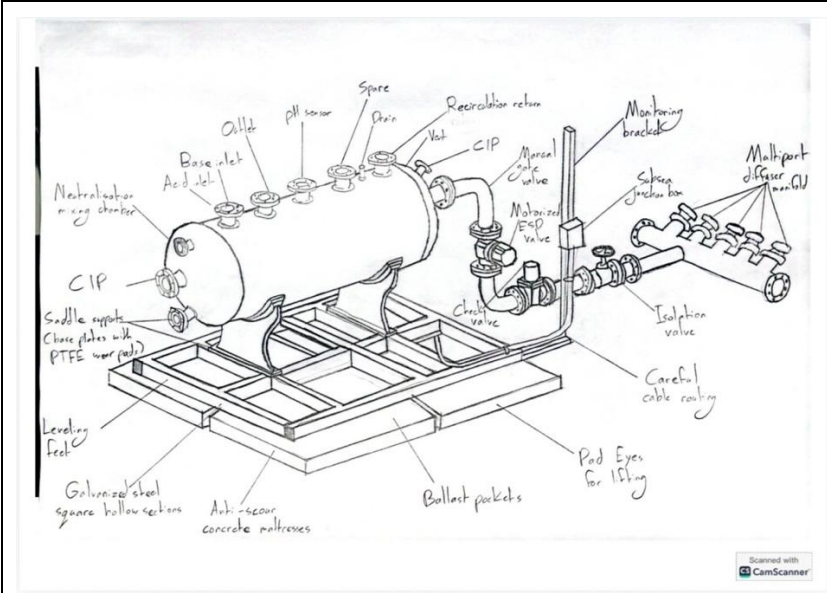


Figure 5. WP 4 Conceptual Design Image (Option 1: Baffled Tank + Linear Multiport Diffuser)

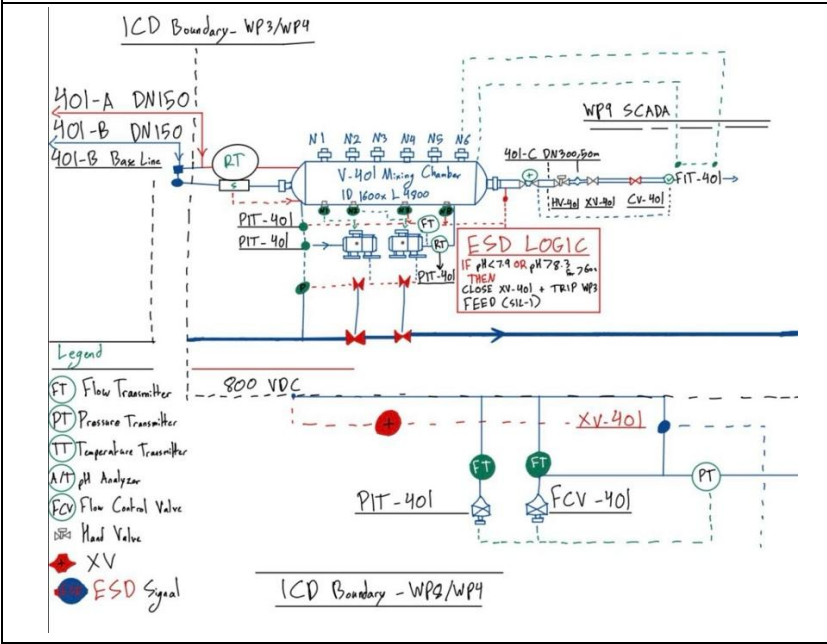


Figure 6. WP4 Conceptual PFD/P&ID-Lite (Option 1: Baffled Tank + Linear Multiport Diffuser)

Table 18. WP4 Maintainability and Safety Considerations.

Maintainability	Safety
Isolation via HV-401/XV-401; drain via HV-402; CIP flush via N8 (ASME B31.3).	Hardwired ESD: pH outside 7.9–8.3 > 60 s → XV-401 close + WP3 trip (SIL-1, IEC 61511).
ROV-replaceable diffuser nozzle inserts (4×M12, no hot work).	All wetted materials SD 2507 (PREN > 40, Alleima, 2024); Inconel 625 bolting; PTFE gaskets.
≥ 600 mm clearance around V-401; instruments individually swappable.	Fail-closed spring-return ESD valve; anti-scour mattresses beneath frame (DNV-OS-H101).

WP5 – UDC – Thermal Management & Heat Rejection

WP Owner	Matteo Karam
Contributors	Matteo Karam
Key Interfaces	DOC/process unit, internal heat-generating equipment, seawater pump/cooling supply, control and instrumentation system, power system, subsea frame/enclosure

Table 19. Scope and Boundaries for Thermal Management & Heat Rejection Work Package.

Included Items	Excluded items
- Thermal management architecture - Internal cooling system - Heat rejection system - Cooling fluid selection and management - Biofouling and corrosion mitigation for heat rejection surfaces - Isolation, bypass, and redundancy arrangements - Concept-level instrumentation - Cooling system layout and integration within UDC - Interface definition with seawater and internal equipment - Emergency/degraded thermal management philosophy	- Main seawater pumps - DOC processing/chemical system - Electrical power system design - Control software and automation logic - Structural design of subsea frame - Detailed thermal/hydraulic calculations - CFD/FEA simulations - Detailed CAD modelling - Installation procedures - Vendor-specific equipment selection

Table 20. Concept Options Comparison Table for Thermal Management & Heat Rejection (Images Generated by Gemini AI).

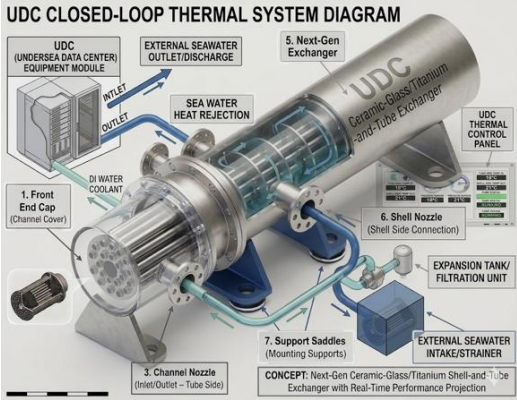
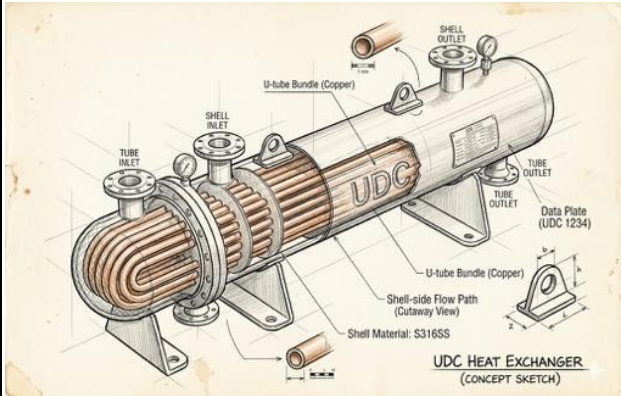
Options	Concept Description	Main Components
	Seawater used directly to remove heat from UDC components via heat exchanger/cooling channel	Seawater interface, strainer, heat exchanger, isolation valves, discharge line, sensors
	Internal coolant loop removes heat and transfers it to seawater through a heat exchanger	Cold plates, coolant loop, pumps, expansion volume, filters, heat exchanger, valves, sensors

Table 21. Selection of Final Concept Based on PDS Requirements and Subsea Constraints.

Criteria	Option 1	Option 2
Maintability	Medium	High
Biofueling resistance	Low	High
Reliability	Medium	High
Pressure loss	Low	Medium
Power consumption	Low	Medium
Modularity	Medium	High
Safety	Medium	High
Subsea suitability	Low	High

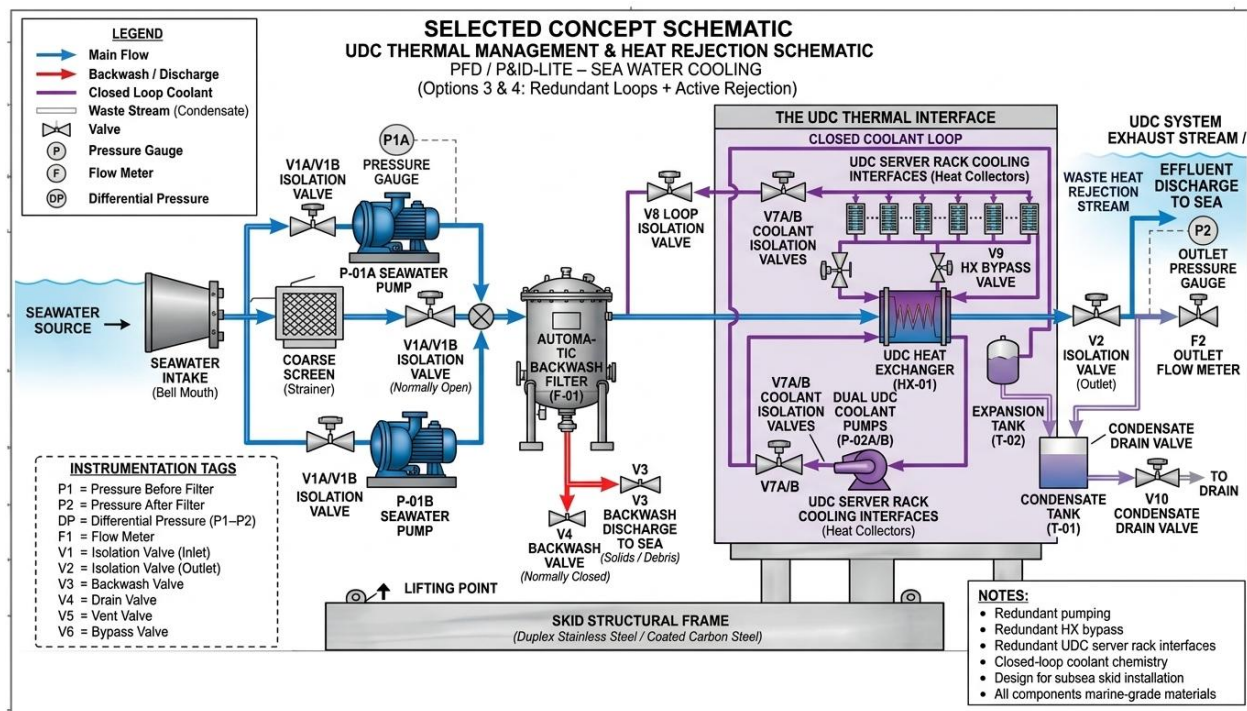


Figure 7. P&ID - UDC Cooling and Heat Rejection Process.

Table 22. Maintainability and Safety Considerations for Thermal Management & Heat Rejection Work Package.

Maintainability	Safety
- Isolation valves for maintenance without full shutdown - Duty/standby pumps for redundancy - Filters/strainers for debris control - Drain points for system emptying - Vent points for air removal - Modular components for replacement - Bypass lines for continued operation during faults - Accessible layout for intervention - Monitoring (temperature, pressure, flow)	- Isolation valves for emergency shutdown - Non-return valves to prevent backflow - Pressure relief protection - Over-temperature protection - Leak detection system - Corrosion-resistant materials - Anti-biofouling measures - Emergency/degraded cooling operation - Structural integrity for subsea conditions

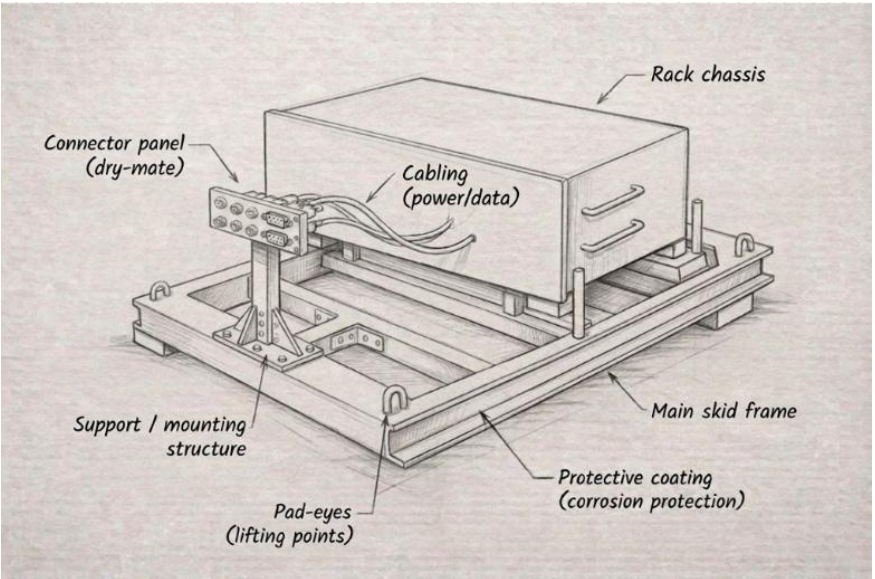
WP6 – UDC – Rack/Chassis, Power Interface, Corrosion Protection, Deployment/Retrieval Base frame

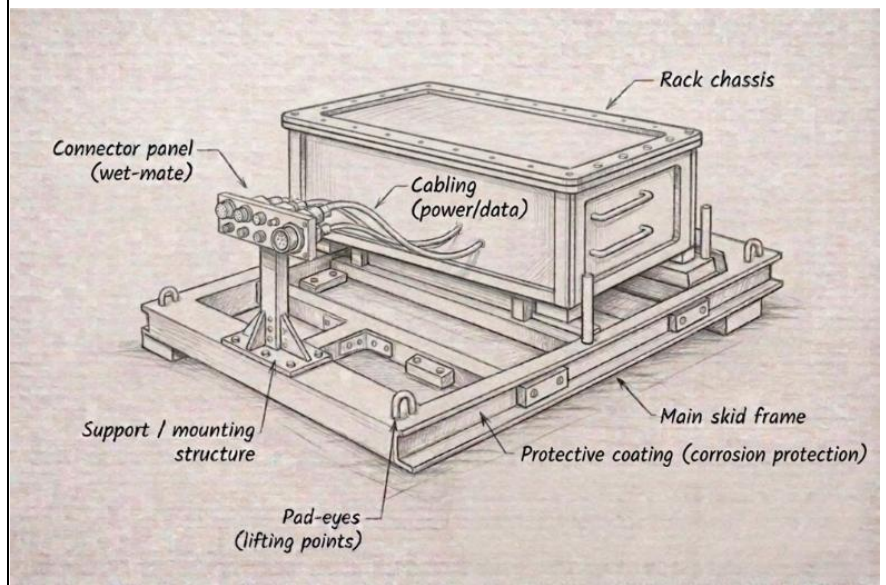
WP Owner	Talal Abu Ghazaleh
Contributors	Talal Abu Ghazaleh
Key Interfaces	electrical power supply, data/communication signals, mechanical mounting to pressure hull, structural load transfer, thermal dissipation, control/monitoring signals

Table 23. Scope and Boundaries for UDC Support System Work Package.

Included Items	Excluded Items
UDC internal rack/chassis layout Mounting and support structure for electronics Connector panel (power/data interface) – dry-mate concept Cable routing arrangement between panel and rack Corrosion protection strategy (protective coating) Deployment/retrieval skid frame design Lifting points (pad-eyes) and handling features Concept-level structural layout and integration	Internal electronic module design within rack Detailed electrical circuit design Wet-mate connector design (alternative only discussed) Detailed structural calculations / FEA Material selection optimisation Detailed manufacturing processes Full system-level integration (other WPs) Detailed CAD modelling (later stage)

Table 24. Concept Options Comparison Table for UDC Support System.

Options	Concept Description	Main Components
Option 1: 	Modular UDC rack mounted on a skid frame with a dry-mate connector panel. Electrical and data connections are made prior to deployment, with cabling routed between the panel and rack for simple integration and retrieval-based maintenance.	Modular rack chassis, dry-mate connector panel, power/data cabling, support/mounting structure, main skid frame, pad-eyes, protective coating (corrosion protection)

Option 2: 	Integrated sealed UDC rack mounted on a skid frame with a wet-mate connector panel. Electrical and data connections can be made underwater, reducing the need for retrieval but increasing	Sealed rack chassis, wet-mate connector panel, power/data cabling, support/mounting structure, main skid frame, pad-eyes, protective coating (corrosion protection)
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The proposed concepts above use a rack mounted on a skid frame with an external connector panel. More racks can be added if needed, so the system can be expanded easily.

Table 25. Selection of the Final Concept Based on Product Design Specification (PDS) Requirements and Subsea Constraints.

Criteria	Option 1	Option 2
Maintainability	High	Medium
Biofouling / corrosion	Medium	High
Reliability	High	Medium
Power consumption	Low	Medium
Modularity / scalability	High	Low
Safety	High	Medium
Connector suitability	Medium	High
Deployment & retrieval	High	Medium
Overall suitability	High	Medium

Table 26. Maintainability and Safety Considerations for UDC Support System Work Package.

Maintainability	Safety
- Modular rack design allows easy removal and replacement of components - Dry-mate connector panel enables simple disconnection before retrieval - Accessible connector panel for inspection and maintenance - External cabling routing for easy inspection and repair - Retrieval-based maintenance approach simplifies repairs - Standard mounting interfaces for easy installation and replacement - Adequate spacing around rack for access during maintenance	- Pad-eyes designed for safe lifting and load handling - Skid frame designed to withstand subsea loads during deployment - Corrosion protection (coating + anodes) to prevent structural failure - Stable base frame to prevent tipping on seabed - Secure mounting of rack to prevent movement during operation - ROV-friendly features for safe handling and positioning - Electrical connections protected to avoid failure underwater

WP7 – UDC – Pressure Hull / Enclosure (Pressure vessel under external pressure)

WP Owner	Talal Abu Ghazaleh – Matteo Karam
Contributors	All Group Members
Key Interfaces	Internal UDC equipment, electrical penetrations/connectors, piping and fluid penetrations, subsea structural frame/mounting interface, sealing and connector systems, corrosion protection system.

Table 27. Scope and Boundaries for UDC Pressure Hull Work Package.

Included Items	Excluded Items
- Pressure hull / enclosure concept - Structural integrity for external pressure - Material selection for pressure containment - Sealing philosophy - Penetration interfaces - Corrosion protection strategy - Internal layout envelope definition - Access provisions - Mounting and interface with subsea frame - Concept-level structural design	- Internal process systems - Electrical system design - Control software and automation logic - Detailed CFD/thermal analysis - Detailed manufacturing procedures - Installation and deployment procedures - Vendor-specific equipment selection - Detailed CAD modelling

Table 28. Scope and Boundaries for UDC Pressure Hull Work Package.

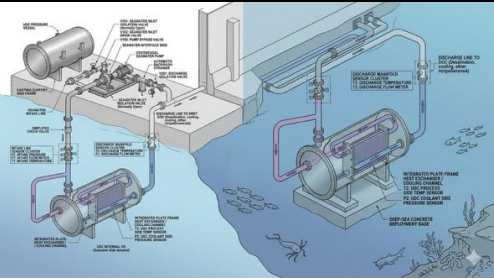
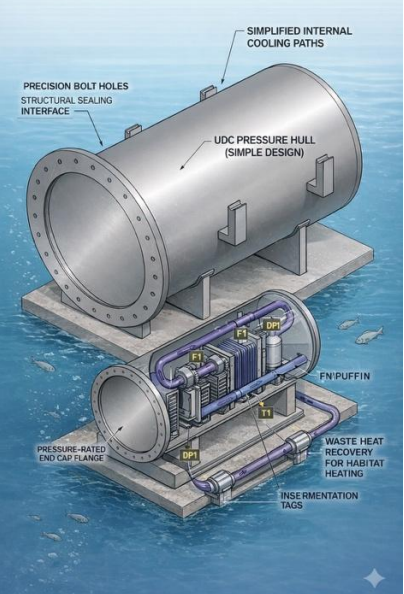
Options	Concept Description	Main Components
	Cylindrical pressure hull with hemispherical or flat end caps designed to withstand external hydrostatic pressure.	Cylindrical shell, end caps, seals/O-rings, bolted flanges, penetrators, support mounts
	Box-shaped enclosure using pressure compensation or reinforced structure to handle external pressure.	Rectangular housing, seals, pressure compensation system, penetrators, support frame

Table 29. Scope and Boundaries for UDC Pressure Hull Work Package.

Criteria	Options 1	Option 2
Maintainability	High	Medium
Biofouling resistance	High	High
Reliability	High	Medium
Pressure resistance	High	Medium
Complexity	Medium	High
Modularity	Medium	Medium
Safety	High	Medium
Subsea suitability	High	Medium

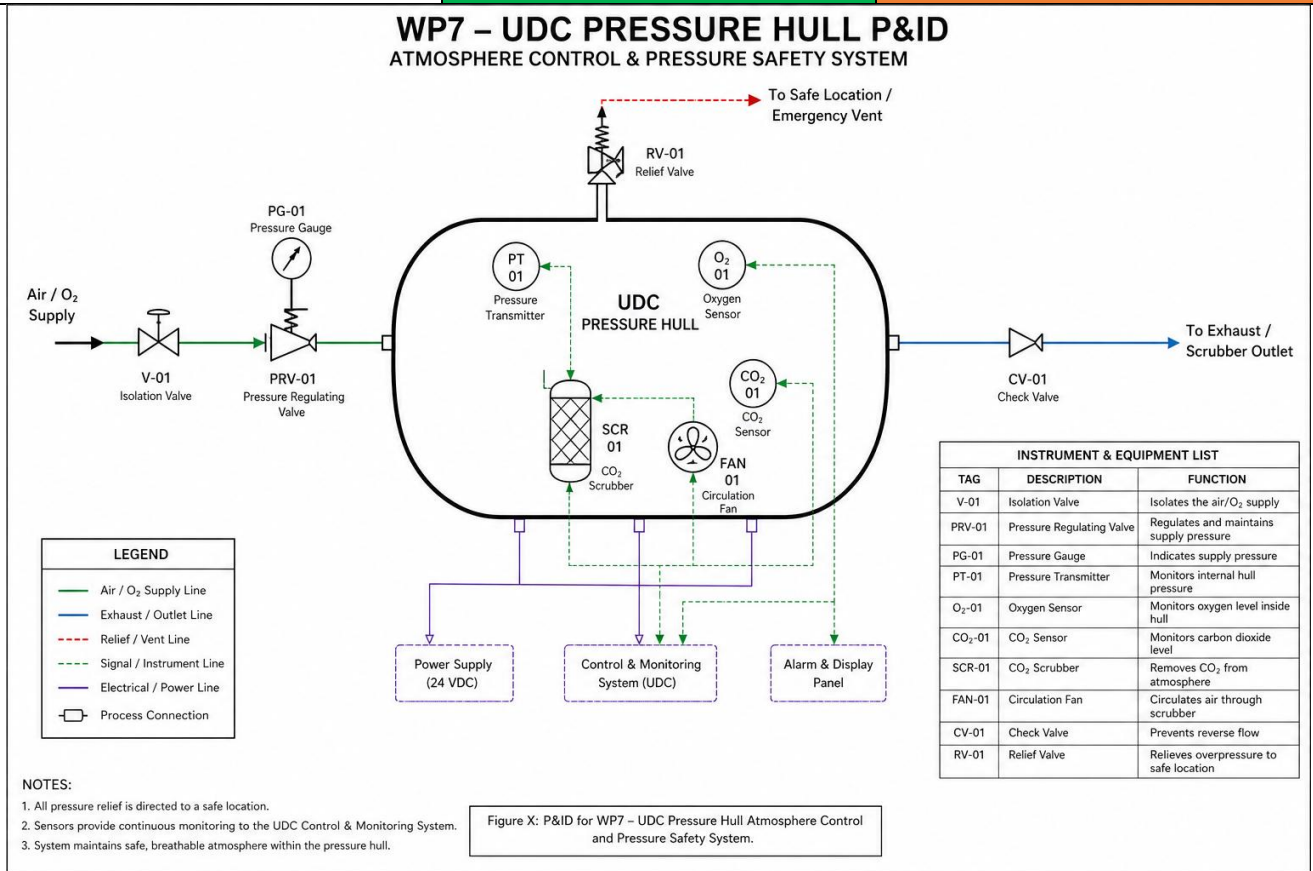


Figure 8. P&ID – UDC Pressure Hull.

Table 30. Scope and Boundaries for UDC Pressure Hull Work Package.

Maintainability	Safety
- Removable flat end cap for access to internal components - Standardised penetrators for easy replacement - Modular internal mounting frame - Accessible sealing interfaces for maintenance - Replaceable seals - Clear internal layout for equipment access - Lifting points integrated into enclosure - Standard flange/bolt connections	- High-integrity sealing system to prevent water ingress - Pressure-rated design to prevent collapse under external pressure - Leak detection within the enclosure - Pressure testing capability before deployment - Corrosion-resistant materials - Redundant sealing barriers - Structural design to withstand subsea loads - Safe handling and lifting provisions

WP8 – Energy System – Load Audit + Sustainable Supply + Storage selection/sizing

WP Owner	Sarah Arafa – Ahmad Alhashmi
Contributors	Sarah Arafa – Ahmad Alhashmi
Key Interfaces	Inputs: Power demand from DOC (~300 kW), UDC (~1225 kW), habitat (~35 kW). Outputs: 800 V DC bus, battery energy storage (BESS), distribution. Interfaces: All WPs

Table 31. WP8 Scope and Boundaries.

Included	Excluded
Load audit (DOC+UDC+habitat); renewable generation sizing (tidal+WECs); BESS in pressure-rated Ti-Gr2 enclosures on gravity frames; 800 V DC bus architecture; distribution hub w/ wet-mate connectors; cable routing to DOC, UDC, controls; protection & load shedding; thermal management via seawater cooling; energy enclosure concept + mounting.	Individual internal electrical design; BMS firmware; SCADA software; habitat structural design.

Energy Architecture

All components permanently underwater at 200 m. Architecture: (1) Generation — 7×100 kW tidal turbines on gravity bases + 5×100 kW submerged CETO-type WECs (Carnegie Clean Energy, 2024); (2) Storage — 4.0 MWh LFP BESS in 6 cylindrical Ti-Gr2 pressure-rated, N₂-purged enclosures on 2 gravity frames with passive seawater cooling (Forum Subsea Technologies, 2024); (3) Distribution — pressure-compensated oil-filled DC switchgear at central junction hub with wet-mate connectors (Teledyne ODI Nautilus); (4) Redundancy — sectionalised 800 V DC bus (Port/Starboard) with automated load shedding: Tier 3 (comfort) → Tier 2 (DOC) → Tier 1 (life support), per Xu et al. (2022); (5) Autonomy — 8 h full-load via BESS.

Table 32. WP8 Concept Options Comparison.

	Option 1: Tidal-Dominant Hybrid (Selected)	Option 2: Wave-Dominant Hybrid
Generation	7×100 kW tidal turbines (CF 0.35, Walker & Thies, 2021) on gravity bases + 5×100 kW submerged CETO-type WECs (CF 0.25). Avg. 370 kW.	3×100 kW submerged WECs (primary) + 1×50 kW tidal. Higher installed capacity, lower avg.
Storage	BESS 4.0 MWh (LFP) in 6 cylindrical Ti-Gr2 pressure enclosures on 2 gravity frames; 8 h full-load autonomy.	BESS 6.0 MWh to compensate wave intermittency in 9 enclosures.
Distribution	800 V DC (Xu et al., 2022), sectionalised Port/Starboard, N+1 converters in oil-filled pressure-compensated pods, wet-mate connectors at central hub.	Same 800 V DC architecture; more cable runs due to WEC spread.

Table 33. WP8 Selection Rationale Linked to PDS and Subsea Constraints.

Criteria	Option 1	Option 2
Maintainability	High: tidal on gravity bases, ROV-assisted retrieval; fewer enclosures	Medium : more WEC units increase intervention frequency
Biofouling/corrosion	Medium: anti-fouling coatings; Ti-Gr2 enclosures; CP anodes on all frames	Medium : same challenges, more units
Reliability	High : tidal predictable (CF 0.35); fewer BESS enclosures = fewer seal points	Medium : wave CF 0.25 requires more cycling and larger BESS
Deployment feasibility	High : proven tidal gravity-base deployment (Simec Atlantis); compact BESS layout	Medium : submerged WEC mooring complexity at depth
Safety	High : fewer units = simpler protection; N ₂ -purged enclosures	Medium : more fault points and seal interfaces

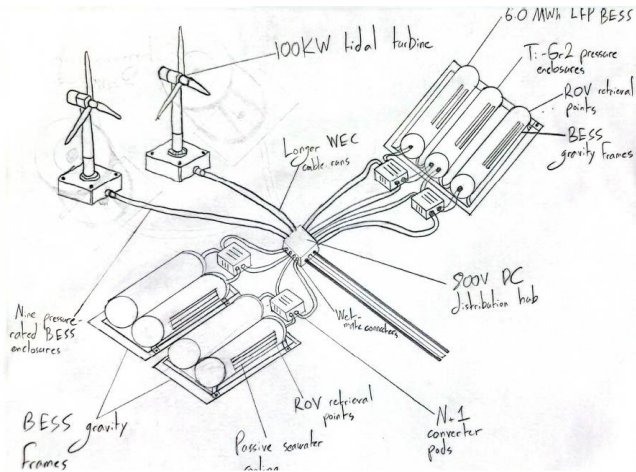


Figure 9. WP 8 Conceptual Design Image (Option 1: Tidal-Dominant Hybrid)

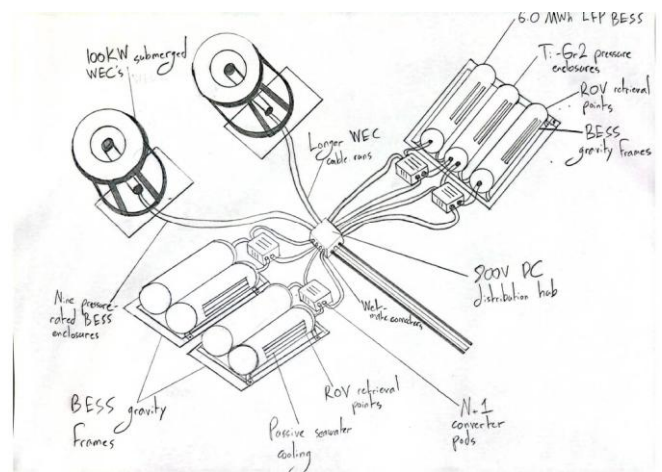


Figure 10. WP 8 Conceptual Design Image (Option 1: Wave-Dominant Hybrid)

Table 34. WP8 Maintainability and Safety.

Maintainability	Safety
Tidal turbines on gravity bases with ROV-guided lift points; 6-month retrieval cycle via surface vessel.	Sectionalised 800 V DC bus (Port/Starboard) with tie breaker and fault isolation (Xu et al., 2022).
WECs retrievable via buoyancy-assisted ROV disconnect at wet-mate interface; PTO service on surface.	Automated load shedding: Tier 3 → 2 → maintain Tier 1 (life support 20 kW).
BESS enclosures: individual string contactors allow isolation; enclosure retrieval via crane + ROV for cell swap.	N ₂ -purged 1-atm BESS enclosures eliminate fire/explosion risk; pressure relief valves vent to ocean (DNV-ST-0359).
Wet-mate connectors (Teledyne ODI Nautilus) at junction hub enable tool-free ROV cable swap.	Thermal runaway detection: per-cell thermocouples; auto-disconnect + N ₂ flood on $\Delta T > 10^{\circ}\text{C}$.
Seawater cooling on BESS enclosures: passive convection, no moving parts, barnacle-tolerant design.	DC breakers, differential/earth-leak detection per section; ground fault < 50 ms trip.

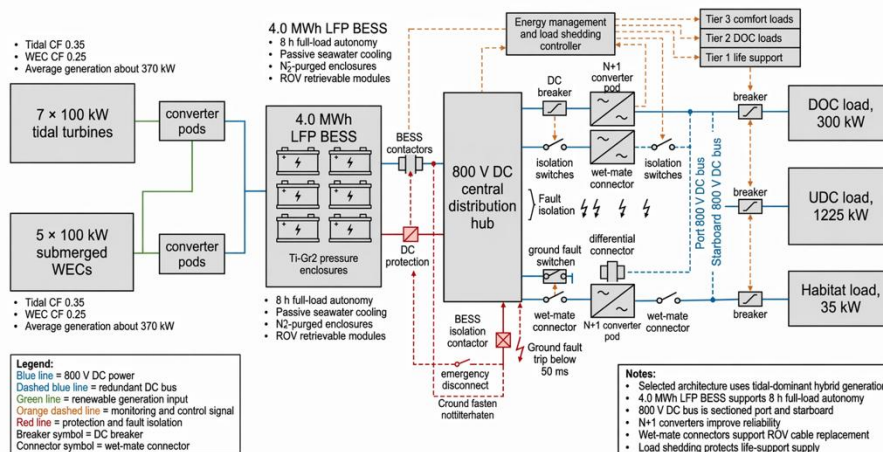


Figure 11. WP8 Conceptual PFD/P&ID-Lite (Option 1: Tidal-Dominant Hybrid).

WP9 – Integration – ICD + Integrated CAD + V&V + Risk

WP Owner	Muhammad Hammad – Saba Falah
Contributors	All Group Members
Key Interfaces	Seawater intake and pre-treatment skid, pumping system, electrochemical DOC reactor, discharge system, underwater data centre (UDC) thermal management system, energy system, control and instrumentation system, subsea structural frame.

Table 35. Scope and Boundaries for Integration Work Package.

Included Items	Excluded Items
- Interface Control Document (ICD) defining mechanical, hydraulic, electrical, and control interfaces between subsystems - Integrated process flow diagram and thermal management schematic - Integrated mechanical layout and equipment arrangement - Mounting interfaces and connection philosophy - Verification and Validation (V&V) plan - System-level risk register and mitigation strategy	- Detailed design of individual subsystems - Electrical system detailed design - Control software and automation programming - Structural analysis of the main subsea frame - Manufacturing and installation planning - Detailed fabrication drawings

Table 36. Concept Options Comparison Table for Integration (Images Generated by Gemini AI).

Options	Concept Description	Main Components
Option 1: 'Centralised Skid Integration' 1. Intake 2. Industrial Pumps 3. Electrochemical Reactor 4. Main Manifolds 5. Valve Station Massive scooped intake, tile, coarse-mesh screens, anti-biofouling mechanism. Large skid frame anti-corrosion Super Duplex Stainless Steel, painted, anti-fouling. Technical seabed interface stabilizing feet. Main Manifolds to UDC Main Manifolds to Energy Systems	All DOC and supporting equipment are mounted on a single large skid connected to UDC and energy systems via main manifolds	Large skid frame, intake, pumps, reactor, manifolds, valve station
Option 2: 'Modular Distributed Integration' 1. Intake (COARSE SCREENING & BIOFOULING MGMT) 2. PROCESS FLOW TRANSMISSION PUMPS 3. ELECTROCHEMICAL REACTOR (modular cassettes) 4. NEUTRALISATION & DISCHARGE DIFFUSER 5. COOLING LOOP (cooling loop pipeline exchangeable subseas and seawater filters) 6. ENERGY MODULE (battery stacks and hydrogen storage concepts); internal thermal management 1. INTAKE (Double-O-ring flanges, external heat exchangers, in ROV-friendly features) Scale reference ROV NEUTRALISATION & DISCHARGE DIFFUSER UNDERWATER DATA CENTRE (UDC) PRESSURE HULL (with ROV-friendly features) TO WIDER HABITAT INTERFACE TO WIDER HABITAT INTERFACE FUTURE OCEAN HABITAT FUTURE HABITAT	Each subsystem is mounted on separate modular skids connected via pipelines and cables	Intake module, pump module, reactor module, discharge module, UDC module, energy module

Table 37. Selection of the Final Concept Based on Product Design Specification (PDS) Requirements and Subsea Constraints.

Criteria	Option 1	Option 2
Maintainability	Medium	High
Reliability	Medium	High
Modularity	Low	High
Installation complexity	Medium	Medium
Expandability	Low	High
Safety	Medium	High
Integration flexibility	Low	High

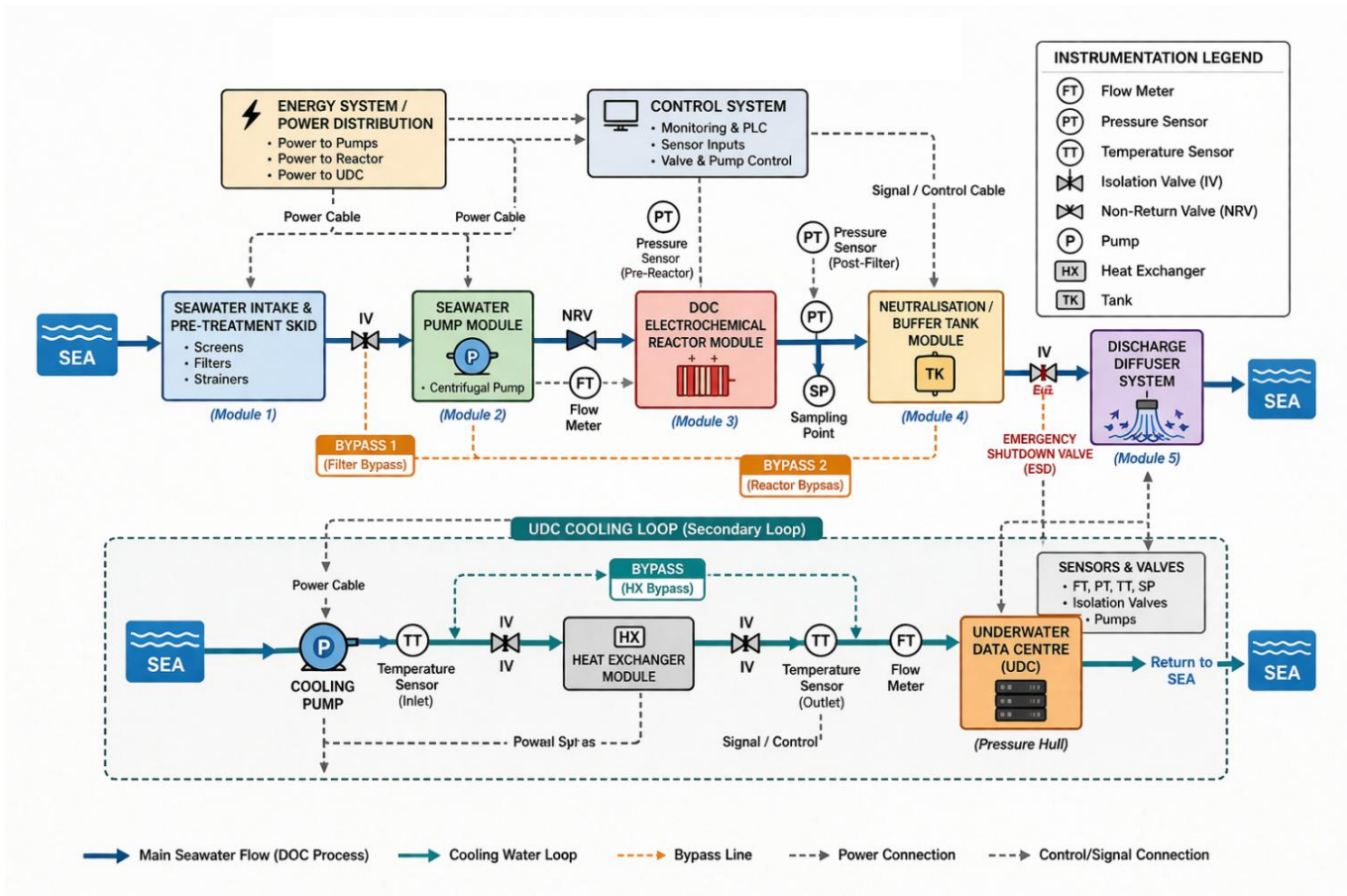


Figure 12. Conceptual Integrated PFD/P&ID-Lite (Modular Distributed Subsea DOC and UDC System Integration).

Table 38. Maintainability and Safety Considerations for Integration Work Package.

Maintainability	Safety
- Modular skid-mounted subsystems allowing retrieval and replacement - Isolation valves between all major subsystems - Bypass lines to maintain operation during maintenance - Standard flange connections for easy disconnection - Access zones for ROV inspection and maintenance - Cable routing trays for cable management and replacement - Lifting points integrated into all modules - Central manifolds to reduce piping complexity - Inspection and testing ports throughout the system	- Emergency shutdown valves on the main seawater supply line - Pressure relief valves to prevent over-pressurisation - Double sealing on pressure hull penetrations - Corrosion-resistant materials and protective coatings - Cathodic protection system - Leak detection sensors and alarms - Temperature and pressure monitoring systems - Structural mounting designed for subsea loads and stability - Separation between electrical systems and seawater systems

Section 3.2 – Design Calculations

1. WP1 – DOC – Seawater Intake & Pre-treatment Skid

WP Owner	Saba Falah
Contributors	Saba Falah
Key Interfaces	- Inlet seawater from the intake structure - Outlet pre-treated seawater to the DOC unit - Electrical power to pump - Instrumentation signals (flow, pressure, differential pressure)

Table 39. Design requirements, assumptions, and constants for DOC – Seawater Intake & Pre-treatment Skid.

System Requirements and Design Goals	Requirements	Goals	
	- Continuous seawater supply to the DOC unit - Removal of debris and suspended solids - Low pressure drop to reduce pump power - Anti-biofouling capability via backwash and CIP - Subsea compatibility and corrosion resistance - Maintainability via isolation valves, drain, and vent points	Parameter	Target
		Flow rate	10-20 m ³ /h
		Pipe velocity	1-2 m/s
		Screen approach velocity	≤ 0.15 m/s
		Pressure losses	Minimise
Assumptions and Constants	Parameter	Symbol	Value
	Density	ρ	1025 kg/m ³
	Gravity	g	9.81 m/s ²
	Vapour pressure	P_{vap}	2 kPa
	Friction factor	f	0.02
	Flow rate	Q	0.004 m ³ /s (14.4 m ³ /h)
	Pipe diameter	D	0.05 m
	Pipe length	L	10 m
	Minor loss coefficient	ΣK	8
	Screen approach velocity	v_{app}	0.15 m/s
	Pump efficiency	η	0.7

Table 40. Core Sizing Calculations for DOC – Seawater Intake & Pre-treatment Skid.

Core Sizing Calculations	Pipe Flow Velocity	Flow velocity in the intake pipe is calculated using: $v = \frac{Q}{A} \quad \text{(Eq. 1)}$ $A = \frac{\pi D^2}{4}$ $A = \frac{\pi(0.05)^2}{4} = 0.00196 \text{ m}^2$ $v = \frac{0.004}{0.00196} = 2.04 \text{ m/s}$ ∴ Pipe velocity ≈ 2.0 m/s, which is within the recommended range for seawater piping (1-2 m/s).
	Intake Screen Sizing	The intake screen open area must satisfy: $A_{open} \geq \frac{Q}{v_{app}} \quad \text{(Eq. 2)}$ $A_{open} \geq \frac{0.004}{0.15} = 0.0267 \text{ m}^2$ ∴ The intake screen must have a minimum open area of approximately 0.027 m² to prevent excessive intake velocity.

Pressure Loss Calculation	Total pressure loss in the intake and skid piping is calculated using the Darcy-Weisbach equation with minor losses:	
	$\Delta P = f \frac{L}{D} \frac{pv^2}{2} + \sum K \frac{pv^2}{2} \quad (\text{Eq. 3})$	
	$\frac{pv^2}{2} = \frac{1025(2.04)^2}{2} = 2133 \text{ Pa}$	
	Major and minor losses:	
	$\Delta P_{major} = 0.02 \times \frac{10}{0.05} \times 2133 = 8532 \text{ Pa}$ $\Delta P_{minor} = 8 \times 2133 = 17064 \text{ Pa}$	
Component		Pressure Loss
Major losses		8.5 kPa
Minor losses		17.1 kPa
Total pressure drop		25.6 kPa

Table 41. Cavitation Check, Pump Power Requirement, and Energy Implications Table for Seawater Intake & Pre-treatment Skid.

Cavitation Check	Parameter	Value
	Pump inlet pressure	100 kPa
	Pressure losses	25.6 kPa
	Local pressure	74.4 kPa
	Vapour pressure	2 kPa
	∴ To avoid cavitation, the local pressure must remain above the vapour pressure; hence, cavitation will not occur.	
Pump Power Requirement	Pump power is calculated using: $P = \frac{\Delta P \cdot Q}{\eta}$$P = \frac{25596 \times 0.004}{0.7} = 146 \text{ W} \approx 0.15 \text{ kW}$ ∴ A 0.37 kW pump is selected to provide a safety margin and account for additional losses in the filter and fouling.	
Energy Implications	Parameter	Value
	Pressure drop	25.6 kPa
	Pump power	0.15 kW
	Selected pump	0.37 kW
	Daily energy use	≈ 3.6 kWh/day

Table 42. Final Design Parameters for CAD and Integration.

Parameter	Flow Rate	Pipe Diameter	Screen Open Area	Pump Rating	Materials	Backwash connection	Drain connection	Vent connection	CIP connection	Bypass line	Skid footprint
Value	14.4 m ³ /h	50 mm (DN50)	0.027 m ²	0.37 kW	Duplex SS / HDPE	Included	Included	Included	Included	Included	~3.5 m × 2 m

Table 43. Component Selection Table (OEM-Backed).

Component	Example Supplier	Model / Series	Reference
Automatic backwash filter	Amiad	SAF / Omega Series	https://amiad.com
Seawater pump	Grundfos	CR / NB Series	https://www.grundfos.com
Flow meter	Endress+Hauser	Promag Series	https://www.endress.com
Pressure sensors	WIKA	S-10 Series	https://www.wika.com
Valves	Emerson	Fisher Valves	https://www.emerson.com

2. WP2 – DOC – Process Flow Transmission (Pumps, Manifolds & Piping)

WP Owner	Muhammad Hammad
Contributors	Muhammad Hammad
Key Interfaces	WP1 pre-treatment skid outlet, WP3 DOC reactor inlet, WP4 discharge return line, electrical power supply, VFD control signals

Table 44. Design Requirements and Goals for DOC – Process Flow Transmission.

	Requirement	Parameter	Target
System Requirements and Design Goals	Deliver 657.6 m ³ /hr seawater to WP3 DOC reactor to achieve 1000 kg CO ₂ /day; receive pre-filtered flow from WP1 V2 outlet	Total flowrate / per pump	657.6 / 328.8 m ³ /hr
	N+1 redundancy (2 duty + 1 standby); DOC uptime ≥95%; NPSH available ≥ NPSH required + 0.5 m margin (PDS UOBD,2026)	Redundancy / uptime / NPSH margin	N+1 / ≥95% / ≥0.5 m above NPSHr
	Water hammer ΔP mitigated by VFD ramp-down, slow actuators, surge volume; online maintenance via individual pump isolation (ASME B31.3, 2-14)	TDH / delivery pressure	30.02 m / 3 bar min
	All wetted components super duplex SS UNS S32750; galvanically compatible with WP1 duplex SS; all components rated for 150 m depth (ISO 23936-1,2009)	Pipe material / depth	UNS S32750 / 150 m Ionian Sea

	Parameter	Symbol / Value	Parameter	Symbol / Value
Assumptions and Constants	Seawater density	$\rho = 1025 \text{ kg/m}^3$	Allowable hoop stress — UNS S32750	$\sigma_{allow} = 550 \text{ MPa}$
	DIC / conversion efficiency / degasser recovery	2 mol/m ³ / 0.80 / 0.90	Minor loss coefficient ΣK	$K = 3.58$
	Target pipe velocity	$v = 1.0\text{--}1.5 \text{ m/s}$	Pump / motor / VFD efficiency	$\eta_p / \eta_m / \eta_v = 0.70 / 0.92 / 0.97$
	Dynamic viscosity (seawater 10°C)	$\mu = 1.2 \times 10^{-3} \text{ Pa}\cdot\text{s}$	Delivery pressure to WP3 reactor inlet	$P_{del} = 3 \text{ bar}$
	Pipe roughness — super duplex SS	$\varepsilon = 0.015 \text{ mm}$	Pipe length — pump to WP3	$L = 25 \text{ m}$
	Friction factor (Colebrook)	$f \approx 0.0038$	Wave speed/ vapour pressure	$a = 1200 \text{ m/s} / P_v = 1400 \text{ Pa}$

Table 45. Core Sizing Calculations for DOC – Process Flow Transmission.

Core Sizing Calculations	Seawater Flowrate	First, the CO ₂ capture per unit volume is determined: $\alpha = \text{DIC} \cdot MW_{\text{CO}_2} \cdot \eta_{\text{conv}} \cdot \eta_{\text{rec}} = 2 \cdot 0.044 \cdot 0.80 \cdot 0.90 = 0.0634 \text{ kg CO}_2/\text{m}^3 \quad (\text{Eq.5})$ This is then used to calculate the required seawater flowrate: $Q_{sw} = \frac{\dot{m}_{\text{CO}_2}}{\alpha} = \frac{1000}{0.0634} \Rightarrow V_{\text{day}} = 15,783 \text{ m}^3/\text{day} \cdot \frac{15,783}{24} = 657.6 \text{ m}^3/\text{hr} = 0.183 \text{ m}^3/\text{s} \quad (\text{Eq.6})$ Flow per pump (2 duty pumps): $Q_{\text{per pump}} = \frac{657.6}{2} = 328.8 \text{ m}^3/\text{hr} = 0.091 \text{ m}^3/\text{s}$
	Pipe Sizing - Velocity Method	Pipe diameter is obtained from the continuity equation: Main header: $D = \sqrt{\frac{4Q}{\pi v}} = \sqrt{\frac{4 \cdot 0.183}{\pi \cdot 1.2}} = 0.44 \text{ m} \Rightarrow \text{Select DN450 } (v \approx 1.15 \text{ m/s}) \quad (\text{Eq.7})$ Branch pipe: $D = \sqrt{\frac{4 \cdot 0.091}{\pi \cdot 1.2}} = 0.311 \text{ m} \Rightarrow \text{Select DN350 } (v \approx 0.95 \text{ m/s})$
	Pipe Wall Thickness	Internal pressure: $t = \frac{P \cdot D}{(2 \cdot (S \cdot E + P \cdot Y))} \rightarrow \text{DN350: } \frac{(3 \times 10^5 \times 0.3556)}{(2 \times 207 \times 10^6)} = 0.26 \text{ mm}; \text{ DN450} = 0.33 \text{ mm} \quad (\text{Eq. 4})$ External pressure: $t = \frac{D}{2} \times \left[\frac{P_{cr,req} \times (1 - \nu^2)}{(2E)} \right]^{1/3} \rightarrow \text{DN350: } \frac{355.6}{2} \times \left[\frac{4.829 \times 10^6 \times 0.91}{(2 \times 200 \times 10^9)} \right]^{1/3} = 3.95 \text{ mm}; \text{ DN450} = 5.08 \text{ mm. So } t = 10 \text{ mm} \text{ satisfies a minimum thickness requirement}$
	Friction and Minor Losses	Flow regime is first verified: $Re = \frac{\rho v D}{\mu} = \frac{1025 \cdot 0.95 \cdot 0.350}{1.2 \times 10^{-3}} = 2.84 \times 10^5 \quad (\text{Eq.8})$ Major losses: $h_f = f \cdot \frac{L}{D} \cdot \frac{v^2}{2g} = 0.0038 \cdot \frac{25}{0.350} \cdot \frac{0.95^2}{2 \cdot 9.81} = 0.012 \text{ m} \quad (\text{Eq.9})$

	Minor losses: $h_m = \sum K \cdot \frac{v^2}{2g} = 3.58 \cdot \frac{0.95^2}{2 \cdot 9.81} = 0.164 \text{ m}$			
System Curve	System head is defined as: $H_{sys}(Q) = H_{static} + H_{loss}(Q)$ (Eq. 10)			
	Three operating points establish the curve and confirm the duty point intersection:			
	Flow condition	Q per pump (m³/hr)	Hloss (m)	Hsys (m)
	50% — minimum	164.4	0.044	29.88
	100% — design duty	328.8	0.176	30.02 ← duty point
	120% — maximum	394.6	0.254	30.09
	Losses remain small relative to pressure head, so the curve is nearly flat and stable around the duty point.			
Total Dynamic Head	Static head is zero (same depth), and pressure head is: $h_{comp} = \frac{P_{del}}{\rho g} = \frac{3.0 \times 10^5}{1025 \cdot 9.81} = 29.84 \text{ m}$ (Eq.11) Total dynamic head: $H_{TDH} = 0 + 0.012 + 0.164 + 29.84 = 30.02 \text{ m}$			
Pump Power	Hydraulic power: $P_{hyd} = \rho g Q H = 1025 \cdot 9.81 \cdot 0.091 \cdot 30.02 = 27.56 \text{ kW}$ (Eq.12)			
	Electrical power: $P_{elec} = \frac{P_{hyd}}{\eta_p \cdot \eta_m \cdot \eta_{VFD}} = \frac{27.56}{0.70 \cdot 0.92 \cdot 0.97} = 44.1 \text{ kW}$ (Eq.13)			
	Total (2 pumps): $P_{total} = 2 \cdot 44.1 = 88.2 \text{ kW}$			
	Motor selection: $P_{rated} = 88.2 \cdot 1.15 = 101.5 \text{ kW} \Rightarrow \text{Select } 2 \times 55 \text{ kW}$			

Table 46. NPSH Check, Water Hammer Screening for DOC – Process Flow Transmission.

NPSH	Absolute suction pressure: $P_{abs} = P_{atm} + \rho g h = 101325 + (1025 \cdot 9.81 \cdot 150) = 1.6098 \times 10^6 \text{ Pa}$ (Eq.14)
	Available NPSH: $NPSH_a = \frac{P_{abs}}{\rho g} - \frac{P_v}{\rho g} - h_f - \frac{v^2}{2g} = 159.9 \text{ m}$ (Eq.15) Since $NPSH_a \gg NPSH_r$, cavitation is avoided.
Water Hammer Screening	Worst-case surge pressure: $\Delta P = \rho a \Delta v = 1025 \cdot 1200 \cdot 0.95 = 116.9 \text{ bar}$ (Eq.16)
	Minimum closure time: $t_c = \frac{2L}{a} = \frac{2 \cdot 25}{1200} = 0.042 \text{ s}$ (Eq.17) Since actual closure time $\gg t_c$, surge effects are negligible.

Final Design Parameters for CAD and Integration

Table 47. Final Design Parameters for CAD and Integration.

	Flow Rate	Header Pipe	Branch Pipe	TDH	Pump Rating	Redundancy	Valve Closure	Materials
Value	657.6 m ³ /hr total; 328.8 m ³ /hr per pump	DN450, ID=450 mm, t=10 mm	DN350, ID=350 mm, t=10 mm	30.02 m	2×55 kW (200WQ300-40-55)	N+1 (2 duty + 1 standby)	10 s design (>> 0.042 s min)	Super duplex SS UNS S32750; Ti-6Al-4V fasteners

	WP1→WP2 Connection	WP2→WP3 Connection	Operating Depth	Standards
Value	DN450 flange at V2 outlet — ASME B16.5	DN350 flange at reactor inlet — ASME B16.5	150 m — all components depth-rated	ASME B31.3, ISO 23936-1 (ISO 23936-1, 2009)

Table 48. Component Selection Table (OEM-Backed).

Component	Supplier	Model / Series	Reference
Submersible centrifugal pump	Dermart Pumps	200WQ300-40-55	https://shorturl.at/bOsmT
Variable frequency drive (VFD)	ABB	ACS880 — marine duty, IP66	https://new.abb.com/mechanical-power-transmission
Non-return valves	Emerson (Fisher)	Fisher Valves — super duplex SS	https://www.emerson.com
Flow meter (FM-01)	Endress+Hauser	Promag series	https://www.endress.com
Pressure transmitters (PT-01/02)	WIKA	S-10 / A-10 series	https://www.wika.com

3. WP3 – DOC – Electrochemical Reactor / pH-Swing Stack Housing (and membrane/contactor if used)

WP Owner	Ahmad Alhashmi
Contributors	Ahmad Alhashmi
Key Interfaces	Inlet from DOC pre-treatment and pumping system, outlet to downstream DOC process line, DC power supply to pH-swing stack, CO2 vent and gas line, CIP and flush supply, drain, leak collection and instrument signals

The detailed WP3 design basis is provided in Appendix D, Table 49. This appendix lists the full set of requirements, assumptions and constants used for the DOC electrochemical reactor and pH-swing stack housing, including the process flow rate, housing design pressure, housing radius, wall thickness, bolt preload inputs, seal land area, leak annulus dimensions and optional HFMC side-loop hydraulic assumptions. In the main calculations, the key values used are $Q = 0.004 \text{ m}^3/\text{s}$, $P = 300 \text{ kPa}$ gauge, $r = 0.25 \text{ m}$, $t = 0.012 \text{ m}$, $8 \times \text{M12}$ closure bolts at 65 N m , and $Q_c = 0.002 \text{ m}^3/\text{s}$ for the optional HFMC branch. These values are used to screen housing stress, end-cap load, thermal expansion, gasket compression, leak annulus volume, cassette guide support and HFMC pressure drop. The assumptions are concept-level screening values based on the WP3 design brief and published DOC pH-swing and HFMC studies, so final verification would be completed during detailed pressure-vessel and component selection stages (Sharifian et al., 2022; Rivero et al., 2023; Aliaskari et al., 2024).

Table 50. Core sizing calculations for WP3 - DOC electrochemical reactor and pH-swing stack housing.

Core Sizing Calculations	Housing Stress Screen	Housing wall stress is screened using thin-cylinder relations $\sigma_h = \frac{Pr}{t} = \frac{300000 \times 0.25}{0.012} = 6.25 \text{ MPa} \quad (\text{Eq.18})$ $\sigma_a = \frac{Pr}{2t} = \frac{300000 \times 0.25}{2 \times 0.012} = 3.13 \text{ MPa}$ $\therefore$ Calculated shell stresses are low for titanium or super-duplex screening and the 12 mm wall is retained for concept design. Final verification would follow a selected pressure-vessel code basis (Sharma and Pachauri, 2023).
	End Cap Load	The removable front closure load is estimated from pressure acting on the vessel area $A_{\text{end}} = \pi r^2 = \pi (0.25^2) = 0.196 \text{ m}^2 \quad (\text{Eq.19})$ $F_{\text{end}} = P \cdot A_{\text{end}} = 300000 \times 0.196 = 58.9 \text{ kN}$ $\therefore$ The closure flange and bolt group must resist approximately 59 kN before proof and safety factors. A bolted circular end cap is retained for cassette access and integrity testing.
	Thermal Expansion	Thermal strain and restrained thermal stress are checked using $\epsilon_{\text{th}} = \alpha \Delta T = 8.6 \times 10^{-6} \times 15 = 1.29 \times 10^{-4} \quad (\text{Eq.20})$ $\delta = \alpha \Delta T \cdot L_h = 8.6 \times 10^{-6} \times 15 \times 1.10 = 0.000142 \text{ m} \approx 0.14 \text{ mm}$ $\sigma_{\text{th}} = E \cdot \alpha \Delta T = 110 \times 10^9 \times 8.6 \times 10^{-6} \times 15 = 14.2 \text{ MPa}$ $\therefore$ A 1 mm cassette and guide-rail clearance is adequate at this screening stage and slotted support details are retained.
	Bolt Preload and Gasket	Bolt preload is estimated from the torque-tension relationship $F_b = \frac{T}{K \cdot d} = \frac{65}{0.20 \times 0.012} = 27083 \text{ N} \approx 27.1 \text{ kN per bolt} \quad (\text{Eq.21})$ $F_{\text{total}} = n \cdot F_b = 8 \times 27.1 = 216.8 \text{ kN}$ $p_{\text{gasket}} = \frac{F_{\text{total}}}{A_{\text{seal}}} = \frac{216800}{0.016} = 13.6 \text{ MPa}$ $\therefore$ The preload exceeds the $\geq 2 \text{ MPa}$ gasket seating target. Torque control and a cross-tightening sequence are retained because gasket material and clamping pressure strongly affect electrochemical stack sealing (Moreno Soriano et al., 2021).

Table 51. Core sizing calculations continued for WP3 - DOC electrochemical reactor and pH-swing stack housing.

Core Sizing Calculations Continued	Leak Annulus	The double seal leak annulus volume is screened as a collection path $A_{\text{ann}} = w \cdot h = 0.006 \times 0.004 = 2.4 \times 10^{-5} \text{ m}^2 \quad (\text{Eq.22})$ $D_{\text{ann}} \approx 0.45 \text{ m}; V_{\text{ann}} = \pi \cdot D_{\text{ann}} \cdot A_{\text{ann}} = \pi \times 0.45 \times 2.4 \times 10^{-5} = 3.4 \times 10^{-5} \text{ m}^3$ $V_{\text{ann}} \approx 34 \text{ mL}$ $\therefore$ The annulus is not a storage volume. It is a detection and routing feature that directs leakage to a captured drain and sensor before external release.
	Cassette Guide Support	The cassette support bearing pressure is screened using an assumed wet cassette mass $\text{Weight (W)} = m_c \cdot g = 45 \times 9.81 = 441 \text{ N} \quad (\text{Eq.23})$ $A_{\text{guide}} = n_{\text{pad}} \cdot b_{\text{pad}} \cdot l_{\text{pad}} = 4 \times (0.020 \times 0.100) = 0.008 \text{ m}^2$ $p_{\text{bearing}} = \frac{W}{A_{\text{guide}}} = \frac{441}{0.008} = 55 \text{ kPa}$ $\therefore$ Bearing load is low for metallic or polymer guide pads, so two rails with four bearing pads are sufficient for concept-level cassette withdrawal.

	HFMC Side-Loop ΔP	HFMC side-loop pressure drop is screened using Darcy-Weisbach with minor losses $A \text{ (HFMC pipe flow area)} = \frac{\pi D_h^2}{4} = \frac{\pi (0.041^2)}{4}$ $v = \frac{Q_c}{A} = \frac{4 \times 0.002}{\pi (0.041^2)} = 1.51 \text{ m/s}$ $\Delta P_{pipe} = \left(\frac{f \cdot L_{HFM}}{D_h} + \Sigma K \right) \cdot \left[\frac{\rho v^2}{2} \right]$ $\Delta P_{pipe} = \left(\frac{0.003 \times 5}{0.041} + 8 \right) \times \left[\frac{1025 (1.51^2)}{2} \right] = 13.6 \text{ kPa}$ $\Delta P_{side-loop} = \Delta P_{pipe} + \Delta P_{module} = 13.6 + 20 \approx 34 \text{ kPa with module allowance}$ $\therefore$ The HFMC branch remains bypassable and isolatable because HFMC DOC performance depends on local pH, flow and fouling control (Rivero et al., 2023; Gabelman and Hwang, 1999).	(Eq.24)
	HFMC Hydraulic Power	The hydraulic power assigned to the optional HFMC branch is $P_{hyd} = \frac{\Delta P_{side-loop} \cdot Q_c}{\eta} = \frac{34000 \cdot 0.002}{0.70} = 97 \text{ W} \approx 0.10 \text{ kW}$ $E_{day} = P_{hyd} \cdot t_{day} = 0.10 \times 24 = 2.4 \text{ kWh/day}$ $\therefore$ The optional contactor branch adds a small hydraulic power allowance but requires bypass and cleaning access to manage fouling and pressure drop.	(Eq.25)

Table 52. Design verification and integration implications for WP3 reactor housing.

Check	Result	Design action
Housing stress	$\sigma_h = 6.25 \text{ MPa}$; $\sigma_{th} = 14.2 \text{ MPa}$	Retain 12 mm shell and complete code-based verification in detailed design.
End cap load	$F_{end} \approx 58.9 \text{ kN}$ at 300 kPa	Use bolted flange with proof test and integrity-test port.
Gasket seating	$p_{gasket} \approx 13.6 \text{ MPa}$	Use controlled torque sequence, compression stops and double gasket land.
Leak management	$V_{ann} \approx 34 \text{ mL}$ detection annulus	Route leak annulus to captured drain and leak sensor.
Cassette service	$p_{bearing} \approx 55 \text{ kPa}$	Use guide rails and front pull-out access for module replacement.
HFMC branch	$\Delta P_{side-loop} \approx 34 \text{ kPa}$	Keep isolation valves, bypass line, drain and CIP route.
Corrosion control	Ti or super-duplex wet parts	Use dielectric breaks at dissimilar joints and avoid galvanic contact.

Table 53. Final calculated design parameters for WP3 CAD and integration.

Parameter	Flow Rate	Housing ID	Wall t	Design Pressure	Closure Bolts	Gasket Pressure	Leak Annulus	HFMC Branch	HFMC ΔP	HFMC Power	Service Ports
Value	14.4 m ³ /h	500 mm	12 mm	300 kPa	8 × M12 at 65 N m	13.6 MPa	34 mL path	DN40 side-loop	34 kPa	0.10 kW	Vent, drain, CIP, test

Table 54. Component selection table for WP3 reactor housing.

Component	Selected basis	Calculated design value	Reference
pH-swing stack housing	Titanium Grade 2 or super-duplex wet body	300 kPa screen; 12 mm wall	(Sharifian et al., 2022; Aliaskari et al., 2024).
Compression closure	Bolted removable flange with cassette access	8 × M12, 65 N m; $p_{gasket} = 13.6 \text{ MPa}$	(Moreno Soriano et al., 2021).
Gasket and seal land	EPDM or FKM double seal with leak annulus	$\geq 2 \text{ MPa}$ target; 13.6 MPa screen	(Moreno Soriano et al., 2021).
Optional HFMC module	Bypassable side-loop module with isolation valves	$\Delta P \approx 34 \text{ kPa}$ with module allowance	(Rivero et al., 2023).
CIP and integrity ports	Flush, drain and pressure test access	Vent, drain, CIP and integrity ports included	(Sharifian et al., 2023).
Corrosion control	Dielectric isolation between Ti, SS and frame	Isolation washers and coatings at dissimilar joints	(Hao et al., 2024).

4. WP4 – DOC – Neutralisation/Mixing + Discharge Diffuser

WP Owner	Sarah Arafa
Contributors	Sarah Arafa
Key Interfaces	Inputs: Acid + base from WP3. Outputs: Neutralised seawater to the ocean via multiport diffuser. Interfaces: WP2 (pump head), WP1 (upstream), WP9 (SCADA).

System Requirements and Design Goals

Neutralise combined acid/base rejects from WP3 BPMED reactor (Eisaman, 2024) and discharge within environmental limits ($\Delta\text{pH} \leq \pm 0.2$, $\Delta T \leq 2^\circ\text{C}$, $\Delta\text{salinity} \leq \pm 1$ PSU). All components rated for 200 m depth ($P_{\text{ext}} = 2.112$ MPa, ASME BPVC VIII Div. 1), 25-year design life. $Q = 0.183$ m³/s from Assignment 1. WP4 ΔP must fit within WP2 pump head budget of 200kPa.

Table 55. WP4 Assumptions and Constants.

Parameter	Sym.	Value	Source	Parameter	Sym.	Value
SW density	ρ	1025 kg/m ³	Sharqawy (2010)	Erosion limit	v_{max}	7 m/s
Kin. viscosity	ν	1.19×10^{-6} m ² /s	Sharqawy (2010)	Ambient current	u_a	0.05–0.15 m/s
Discharge flow	Q	0.183 m ³ /s	Assign. 1 DOC	Allowable ΔpH	ΔpH	± 0.2
Pipe roughness	ϵ	0.045 mm	Moody – SS	Dilution target	S	49:1 min
SD 2507 yield	σ_y	550 MPa	Alleima (2024)	FoS (ASME VIII)	FoS	2.4
Corrosion allow.	CA	3.0 mm	Norsok M-001	Design life	–	25 years

Table 56. WP4 Core Sizing Calculations – Results Summary.

#	Calculation	Equation / Substitution	Result	Check
1	Chamber volume ($t \geq 45$ s)	$V = Q \times t = 0.183 \times 45 = 8.24$ m ³ min. Selected $D=1.6$ m, $L=4.8$ m	$V = 9.65$ m ³ , $t = 52.7$ s	✓ 17% margin above 45 s
2	Wall thickness ($P_{\text{ext}} = 200$ m)	$P_{\text{ext}} = 2.112$ MPa; $t = PD/2\sigma + CA = 7.35 + 3$	$t = 12$ mm selected	✓ > 10.35 mm (ASME VIII)
3	Pipeline velocity (DN300)	$v = Q/A = 0.183/0.07069$	$v = 2.59$ m/s	✓ Within 2–3 m/s range
4	Reynolds number	$Re = vD/\nu = 2.59 \times 0.300/1.19 \times 10^{-6}$	$Re = 652,941$	✓ Fully turbulent
5	Friction factor	Colebrook–White, $\epsilon/D = 1.5 \times 10^{-4}$	$f = 0.0155$	Moody verified
6	Major pressure loss (Darcy–Weisbach)	$\Delta P = f(L/D)(\rho v^2/2) = 0.0155 \times 166.7 \times 3439$	8.9 kPa	–
7	Minor losses	$\Sigma K = 4.15$ (gate 0.15, check 2.0, 2×elbow 0.6, tee 0.9, entry 0.5)	14.3 kPa	–
8	Total ΔP	$8.9 + 14.3$	23.2 kPa	✓ 11.6% of 200 kPa budget
9	Port velocity (12×DN65)	$v_{\text{port}} = Q/(N \times A_{\text{port}}) = (0.183/12) / 3.318 \times 10^{-3}$	4.60 m/s	✓ 66% of $v_{\text{max}}=7$ m/s
10	Jet momentum flux	$J = \rho Q v$ per port = $1025 \times 0.01525 \times 4.60$	$J_{\text{total}} = 863$ N	–

11	Near-field dilution (z=10 m)	$S = 0.32(z/d_p) = 0.32 \times (10/0.065)$ (Fischer et al., 1979)	$S = 49:1$	✓ $\Delta pH \rightarrow 0.002$ at 10 m
12	Header/port velocity ratio	$v_{hdr}/v_{port} = 1.90/4.60$ (Roberts et al., 1989)	$0.41 < 0.5$	✓ $\pm 5\%$ port uniformity
13	Seabed stability FoS	$F_{grav}/F_{drag} = 20601/72.6$ (DNV-OS-H101)	FoS = 284	✓ $\gg 1.5$ required

Energy/Power Implications

WP4 continuous demand: recirculation pump ~1.5 kW (duty-cycled), instrumentation ~50 W, ESD valve ~0.3 kW (stroking only). Total ≈ 2.0 kW (< 1% of DOC system load). Pressure drop of 23.2 kPa consumes 11.6% of the WP2 pump head budget (200 kPa available).

Table 57. WP4 Final Design Parameters for CAD and Integration.

Parameter	Value
Neutralisation Mixing Chamber (V-401)	ID 1600 mm, L 4800 mm, t 12 mm, SD 2507 (Alleima, 2024), V = 9.65 m³, t_res = 52.7 s, 8 nozzles, 4 internal baffles, 2 saddle supports
Discharge Pipeline (Line 401-C)	DN300, 50 m, Sch 10S, SD 2507, v = 2.59 m/s; manual gate + ESD + check valves (ASME B31.3); flanged connections; pipe supports at 3 m
Multiport Diffuser Manifold	DN350 header, 18 m, 12 angled outlet ports (DN65 at 60°), removable nozzle inserts, reinforcing pads, v_port = 4.60 m/s
Seabed Base Frame	20×2×0.8 m, SHS 200×200×10, cross-members, pipe support cradles, levelling feet, pad eyes (5t SWL), ballast pockets, anti-scour mattresses (DNV-OS-H101)
Monitoring Bracket	Vertical mast (DN80, 1500 mm), sensor mounting rail (5 sensors), cable routing provision, junction box
Total ΔP / pump budget	23.2 kPa / 11.6% of 200 kPa
pH control	Setpoint 8.1; alarm 8.0/8.2; ESD trip 7.9/8.3 (IEC 61511)

Table 58. WP4 Component Selection (OEM-Backed).

Component Group	Selection	Manufacturer
Piping (DN300/350)	SAF 2507 Sch 10S	Sandvik Alleima
Flanges	A182 F53, ASME B16.5 Cl.150 RF	Ulma Forged
Valves (HV/XV/CV/FCV)	Parker BV-310; Rotork IQ3; Fisher GX; Flowserve Duo-Chek II	parker.com; rotork.com; emerson.com; flowserve.com
Recirc. pump (P-401A/B)	CRN 15-2 SD, duplex N+1	Grundfos
Gaskets / Bolting	Flexitallic CGI spiral-wound (316SS/PTFE); Inconel 625	flexitallic.com
Instruments	E+H CPS171D; Aanderaa 4319; WIKA TR10-F; Hach Solitax; Nortek Aquadopp	endress.com; aanderaa.com; wika.com; hach.com; nortekgroup.com
Seabed protection	Maccaferri ACB 150 mm mattress; Farwest Al-Zn-In CP anodes	maccaferri.com; farwestcorrosion.com

5. WP5 – UDC – Thermal Management & Heat Rejection

WP Owner	Matteo Karam
Contributors	Matteo Karam
Key Interfaces	Fluid inlet connection, Fluid outlet connection, Skid/base frame mounting, Support saddle load transfer, Flange maintenance connection, Nozzle piping interface, Pressure sealing interface, Lifting/handling interface

Table 59. WP5 Assumptions and Constants.

Parameter	Symbol	Value
Total mass	m	1200 kg
Overall length	L	3.46 m
Overall height	H	1.10 m
Base width	b	1.0 m
Projected area	A	3.0 m^2
Current velocity	v	1.0 m/s
Drag coefficient	C_D	1.1
Seawater density	ρ	1025 kg/m^3
Gravity	g	9.81 m/s^2
Safety factor	SF	2

Table 60. Calculated Saddle Loads and Vibration Screening.

Saddle Load	$W = mg \quad (\text{Eq.26})$ $W = (1200) \cdot (9.81) = 11772N$ $R = \frac{W}{3} = 3924N$ $R = 3.9kN \text{ Per saddle}$
Vibration Screening	$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (\text{Eq.27})$ $f_n = \frac{1}{2\pi} \sqrt{\frac{360000}{1200}}$ $f_n = 2.8Hz$
Hydrodynamic Drag Force	$F_D = \frac{1}{2} \rho C_D A v^2 \quad (\text{Eq.28})$ $F_D = \frac{1}{2} (1025)(1.1)(3)(1)^2$

	$F_D = 1.69kN$
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Table 61. Comparison of Overturning and Restoring Moments with Derived Stability Margin.

Overturning Moment	$M_o = F_D h$ $M_o = 1691(0.55)$ $M_o = 930Nm$	(Eq.29)
Restoring Moment	$M_r = W \cdot \frac{b}{2}$ $M_r = 11772 \cdot \frac{1}{2}$ $M_r = 5886Nm$	(Eq.30)
Stability Margin	$SM = \frac{M_r}{M_o}$ $SM = \frac{5886}{930}$ $SM = 6.3$	(Eq.31)

Table 62. Buoyancy Analysis and Net Submerged Weight Verification.

Buoyancy Calculations	$F_b = \rho g V$ $F_b = (1025)(9.81)(1.32)$ $F_b = 13.3kN$	(Eq.32)
Net Submerged Weight	$W_{wet} = 22.6kN$ $F_{net} = W_{wet} - F_b$ $F_{net} = 22.6 - 13.3$ $F_{net} = 9.3kN$ Heat exchanger remains negatively buoyant	

Table 63. Summary of Calculated Lifting Loads per Lifting Point.

Lifting Load Calculations	$F_{lift} = W(SF)$ $F_{lift} = 11.8(2)$ $F_{lift} = 23.5kN$	(Eq.33)
Load Per Pad-eye	$F_{pad} = \frac{23.5}{4}$ $F_{pad} = 5.9kN$	

6. WP6 – UDC – Rack/Chassis, Power Interface, Corrosion Protection, Deployment/Retrieval Base frame

WP Owner	Talal Abu Ghazaleh
Contributors	Talal Abu Ghazaleh
Key Interfaces	Electrical power; data/comms; skid/base-frame interface; structural load transfer; deployment/retrieval; power/data connector.

The following calculations are based on a scaled system consisting of four rack units, extending the single-rack conceptual design.

Table 64. WP6 Design Requirements and Goals

	Requirement	Parameter	Target
System Requirements and Design Goals	Provide stable support for four rack modules and maintain retrieval-based maintenance through an external connector panel.	Rack configuration / connector philosophy	4-rack modular support frame with dry-mate connector interface
	Protect sensitive electronics from handling/deployment vibration and avoid resonance in the support structure.	Vibration isolation target	$f_n \approx 3.2$ Hz (screening level)
	Maintain cable integrity between connector panel and rack through bend-radius control and physical restraint.	Cable routing / strain relief	$R_{min} = 150$ mm; clamp spacing = 600 mm straight / 300 mm near bends
	Provide subsea stability, corrosion protection, and safe lifting during deployment/retrieval.	Stability / CP / lifting	SM = 3.99; sacrificial anodes + coating; 3.9 kN per pad-eye

Table 65. Core WP6 Design Calculations, Assumptions, and Constants.

Assumptions and Constants	Parameter	Symbol	Value
	Total rack mass (4 units)	m	800 kg
	Number of mounts	—	16
	Stiffness per mount	k_i	20,000 N/m
	Total stiffness	k	320,000 N/m
	Cable diameter	d	15 mm
	Water density	ρ	1025 kg/m ³
	Gravity	g	9.81 m/s ²
	Drag coefficient	C_D	1.0
	Frontal area	A	4.8 m ²
	Current velocity	v	1.0 m/s
	Height of force application	h	1.0 m
	Base width	b	2.5 m
	Volume (approx.)	V	4.8 m ³
	Safety factor	SF	2
Core Design Calculations	Vibration Isolation Screening	The natural frequency of the rack support system is calculated using: $f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{320000}{800}} = \frac{1}{2\pi} \sqrt{400} = \frac{20}{6.283} = 3.18 \text{ Hz (Eq. 34)}$ $\therefore$ The natural frequency is approximately 3.2 Hz, which is acceptable for preliminary vibration isolation.	
	Cable strain relief	The minimum cable bend radius is estimated using: $R_{min} = 10d = 10(15) = 150 \text{ mm (Eq. 35)}$ $\therefore$ The minimum bend radius is 150 mm. Cable Clamp Spacing:	

		Cable clamp spacing was taken as 600 mm along straight runs and reduced to 300 mm near bends, based on typical cable-ladder / cable-cleat support practice and the need to maintain cable restraint and minimum bend-radius compliance. Straight-run cleat spacing is commonly treated as a multiple of ladder pitch, with 900 mm preferred as a general upper spacing in trefoil installations and 300 mm maximum at bends and risers ; your chosen 600 mm / 300 mm values are therefore a conservative layout assumption.
	Hydrodynamic Drag Force	The drag force acting on the exposed rack assembly is calculated using: $F_D = \frac{1}{2} \rho C_D A v^2 = \frac{1}{2} (1025)(1.0)(4.8)(1.0)^2 = 2460 \text{ N (Eq. 36)}$ $\therefore$ The drag force is approximately 2460 N .

Table 66. WP6 Stability, Buoyancy, and Lifting Calculations.

Stability, Buoyancy and Lifting Calculations	Stability Check	The overturning and restoring moments are compared to check skid stability. $M_{\text{overturn}} = F_D h = 2460(1.0) = 2460 \text{ Nm (Eq. 37)}$ $W = mg = 800(9.81) = 7848 \text{ N}$ $M_{\text{restore}} = W \frac{b}{2} = 7848 \left(\frac{2.5}{2} \right) = 9810 \text{ Nm}$ $\therefore$ Since 9810 Nm > 2460 Nm , the skid frame is stable under the assumed current. The stability margin is: $SM = \frac{M_{\text{restore}}}{M_{\text{overturn}}} = \frac{9810}{2460} = 3.99$ $\therefore$ The skid frame provides a stability margin of approximately 4.0 , indicating that the restoring moment is about four times the overturning moment under the assumed current loading.
	Buoyancy Check	The buoyancy force acting on the submerged structure is calculated using: $F_b = \rho g V = (1025)(9.81)(4.8) = 48252 \text{ N (Eq. 38)}$ $\therefore$ The buoyancy force is approximately 48.3 kN .
	Lifting Load per Pad-eye	The lifting load is calculated using a safety factor of 2. $F_{\text{lift}} = W(SF) = 7848(2) = 15696 \text{ N (Eq. 39)}$ $F_{\text{per}} = \frac{F_{\text{lift}}}{4} = \frac{15696}{4} = 3924 \text{ N (Eq. 40)}$ $\therefore$ Each pad-eye must safely withstand approximately 3.9 kN .

Table 67. Energy Implications and Final Design Parameters for CAD and Integration.

	Rack Config.	Skid Footprint	Rack Height	Bay Width	Connector	Cable Basis	Pad-Eyes	CP Basis	Materials
Value	4 bays	2800 × 1500 mm	1819 mm	≈594 mm	Dry-mate	Rmin 150 mm; 600/300 mm	3.9 kN /eye	Coating + anodes	Coated steel skid; 316L cleat hardware

Energy / power implications: WP6 is a passive structural subsystem; its main impacts are connector access, cable routing, and handling loads.

Table 68. Component Selection Table (OEM-Backed)

Component	Example Supplier	Model / Series	Reference
Dry-mate subsea connector / penetrator	Teledyne Marine	Dry Mate Connectors	https://www.teledynemarine.com/products/product-line/connectors-and-penetrators
Shock / vibration isolator mount	Hutchinson	Super VHC	https://hutchinsonai.com/high-deflection-mounts-super-vhc/
Cable cleat / clamp	CMP Products	Cyclone III SS cleat	https://www.cmp-products.com/cleats/products/metallic-cable-cleats/cyclone-iii-stainless-steel-cable-cleats/
Sacrificial anode	Cathwell	Al anode, pre- cabled	https://cathwell.com/product/aluminium-anode-with-pre-mounted-cable/

7. WP7 – UDC – Pressure Hull / Enclosure (Pressure vessel under external pressure)

WP Owner	Talal Abu Ghazaleh
Contributors	Talal Abu Ghazaleh Matteo Karam
Key Interfaces	Hydrostatic pressure; internal enclosure pressure; power/data signals; rack/skid interface; sealing and penetrators.

Table 69. WP7 Design Requirements and Goals

	Requirement	Parameter	Target
System Requirements and Design Goals	Provide a pressure-resistant enclosure with retrieval access.	Depth rating / access	150 m; removable flat end + fixed domed end
	Prevent yielding and buckling at design depth.	Material / thickness basis	$\sigma_{allow} = 185 \text{ MPa}$; $t \approx 62 \text{ mm}$
	Maintain reliable serviceable sealing at the access interface.	Sealing philosophy	Double O-ring static face seal; 5 mm CS; 20% squeeze
	Accommodate penetrators without compromising endcap integrity.	Penetrators / NDT basis	8 penetrators; area fraction $\approx 0.077\%$; VT + weld NDT

Table 70. Core Pressure Hull Calculations

	Parameter	Symbol	Value
Assumptions and Constants	Design depth	h	150 m
	Water density	ρ	1025 kg/m^3
	Gravity	g	9.81 m/s^2
	Atmospheric pressure	P_{atm}	101325 Pa
	Hull radius	R	1.63 m
	Hull length	L	5.0 m
	Young's modulus (steel)	E	205 GPa
	Poisson's ratio	ν	0.29
	Yield strength	σ_y	370 MPa
	Factor of Safety	FOS	2
	Allowable stress	σ_{allow}	185 MPa
	O-ring cross section	CS	5 mm
	O-ring squeeze	S	20%
	Number of penetrators	—	8
Core Design Calculations	Hydrostatic Pressure	The external hydrostatic pressure is calculated using: $P = P_{atm} + \rho gh = 101325 + (1025)(9.81)(150) = 1.61 \text{ MPa (Eq. 41)}$ $\Delta P = P_{abs} - P_{atm} = 1.61 - 0.101 \approx 1.51 \text{ MPa}$ $\therefore$ The pressure hull must withstand approximately 1.51 MPa external pressure .	
	Thickness Screening (Membrane Stress)	The required thickness is estimated from the hoop stress relation: $\sigma_h = \frac{Pr}{t} \text{ (Eq. 42)}$ For safe design, the hoop stress is limited to the allowable stress σ_{allow} . Rearranging: $t = \frac{Pr}{\sigma_{allow}} = \frac{(1.51 \times 10^6)(1.63)}{185 \times 10^6} = 0.0133 \text{ m}$ $\therefore$ The estimated thickness is approximately 13 mm (screening only) .	
	External Pressure Buckling Check	The chosen method is classical shell buckling screening for a cylindrical hull under uniform external pressure: $p_{cr} = \frac{E}{4(1-\nu^2)} \left(\frac{t}{R}\right)^3 \text{ (Eq.43)}$ Applying the required factor of safety : $FOS = 2$ $p_{target} = FOS \times \Delta P = 2 \times 1.51 = 3.02 \text{ MPa (Eq.44)}$ Set $p_{cr} = p_{target}$ and solve for thickness:	

		$t = R \left(\frac{4(1-\nu^2)p_{target}}{E} \right)^{1/3}$ $t = 1.63 \left(\frac{4(1-0.29^2)(3.02 \times 10^6)}{205 \times 10^9} \right)^{1/3} = 0.0616 \text{ m} = 61.6 \text{ mm}$ Thus, the minimum required thickness is: $t_{min} \approx 62 \text{ mm}$ ∴ A hull thickness of 62mm satisfies the external pressure buckling requirements with FOS = 2
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Table 71. WP7 Sealing and Interface Design Calculations.

Sealing and Interface Calculations	O-Ring Sealing	The groove depth is calculated using: $G = CS(1 - S) = 5(1 - 0.2) = 4 \text{ mm (Eq.45)}$ ∴ A groove depth of 4 mm provides adequate O-ring compression.
	Gland Fill Design Basis	The double O-ring interface is treated as a static axial face seal at the removable flange/endcap interface. Groove depth was determined analytically from the selected 5 mm O-ring cross-section and 20% squeeze requirement, giving a groove depth of 4 mm. Gland-fill compliance was then assessed using standard static axial face-seal gland guidance, with gland proportions selected to provide sufficient void volume for seal compression and installation while maintaining acceptable gland-fill limits for static sealing. Accordingly, the adopted sealing arrangement satisfies the required O-ring squeeze and gland-fill design basis for a double O-ring static face seal.
	Penetrator Area Check	The penetrator area is calculated using: $A_p = \pi \left(\frac{d}{2} \right)^2 = 8.04 \times 10^{-4} \text{ m}^2$ $A_{total} = 8A_p = 0.00643 \text{ m}^2 \text{ (Eq.46)}$ Endcap area: $A = \pi(1.63)^2 = 8.35 \text{ m}^2$ Fraction $\approx 0.077\%$ ∴ Penetrators occupy a negligible area and do not significantly weaken the structure.

Energy / power implications: WP7 is passive; key impacts are sealed penetrations, access, and proof/leak testing.

Table 72. Final WP7 Design Parameters for CAD and Integration.

Hull Type	Depth	Shell Length	Shell Radius	Shell Thickness	Sealing	Penetrators	Inspection / NDT
Cylindrical shell; removable flat end + fixed domed end	150 m	5.0 m	1.63 m	62 mm	Double O-ring face seal; CS 5 mm; G 4 mm	8 power/data	VT + weld NDT; proof/leak test

Table 73. Component Selection Table (OEM-Backed).

Component	Example Supplier	Model / Series	Reference
Dry-mate subsea connector / penetrator	Teledyne Marine	Connectors & Penetrators / Dry Mate	https://www.teledynemarine.com/products/product-line/connectors-and-penetrators
Static face-seal O-ring design basis	Parker	O-Ring Handbook / Face Seal Glands	https://www.parker.com/content/dam/Parker-com/Literature/O-Ring-Division-Literature/ORD-5700.pdf
Pressure vessel steel plate (example)	SSAB	Pressure vessel steel / ASTM A516	https://www.ssab.com/en-us/brands-and-products/commercial-steel/pressure-vessel

8. WP8 – Energy System – Load Audit + Sustainable Supply + Storage selection/sizing

WP Owner	Sarah Arafa – Ahmad Alhashmi
Contributors	Sarah Arafa – Ahmad Alhashmi
Key Interfaces	Inputs: Power demand from DOC (~300 kW), UDC (~1225 kW), habitat (~35 kW). Outputs: 800 V DC bus, battery energy storage (BESS), distribution. Interfaces: All WPs

Table 74. WP8 Complete System Load Audit.

Subsystem	Rated kW	Duty	Avg kW	Tier	Source
WP3 BP MED stack	250	0.83	208	2	Eisaman (2024)
WP3 auxiliary	25	0.83	20.8	2	–
WP2 DOC pumps	70	0.87	61	2	WP2 sizing
WP1 intake pump	0.37	1.0	0.37	2	WP1 sizing
WP4 recirc+instruments	2.0	0.75	1.5	2	WP4 sizing
WP5 UDC IT (50 racks)	1225	1.0	1225	2	Alkrush et al. (2024)
WP5 UDC cooling	15	1.0	15	2	WP5 sizing
Life support/SCADA/comms	20	1.0	20	1	Critical – shed last
Water treatment/galley	15	0.60	9	2	–
Comfort/HVAC	15	0.40	6	3	Shed first
TOTAL	1637	–	1567	–	–

Note: UDC (1225 kW, 78% of total) requires a shore power cable or a scaled offshore array. Calculations below size the tidal/wave hybrid for DOC + habitat loads only (342 kW average).

Table 75. WP8 Energy System Assumptions.

Parameter	Value	Source	Parameter	Value	Source
DoD	0.8	LFP mfr – 6000 cycles	V _{cell}	3.2 V	CATL 280 Ah
η _{rt}	0.9	LFP industry std	Ah _{cell}	280 Ah	CATL datasheet
Safety factor	1.2	DNV-ST-0359	Tidal CF	0.35	Walker & Thies (2021)
V _{sys}	800 V DC	Xu et al. (2022)	Wave CF	0.25	Submerged WEC – conservative
SW temp @200m	13°C	Sharqawy (2010)	Max cell T	45°C	CATL datasheet

Table 76. WP8 Key Calculations/ Generation and Storage Sizing.

#	Calculation	Equation / Working	Result	Check
1	Energy audit: E _{day}	$E_{day} = \sum(P_{i \times t_i}) = 342 \times 24$	8,208 kWh/day	–
2	Peak power: P _{peak}	$P_{peak} = \max(\sum P_i) = \text{all Tier 1+2 at rated}$	≈ 450 kW	–
3	Tidal per turbine	$P = \frac{1}{2} \rho A C_{pv}^3 = 0.5 \times 1025 \times 78.5 \times 0.40 \times 2^3$	128.5 kW peak	✓
4	Wave per WEC	$P = (\rho g^2 / 64 \pi) H_s^2 T_e$; capture 4 m	~50 kW/WEC	✓
5	Generation sizing	$7 \times 100(CF_{.35}) = 245 + 5 \times 100(CF_{.25}) = 125$	P _{avg} = 370 kW	✓ 8% margin over 342
6	Storage: E _{bat}	$E_{req} / (DoD \times \eta_{rt}) \times \text{Margin} = (342 \times 8) / (0.8 \times 0.9) \times 1.2$	4,560 → 4.0 MWh	✓
7	N _{series}	$V_{sys} / V_{cell} = 800 / 3.2$	250 cells in series	✓
8	N _{parallel}	$E_{str} = 250 \times 3.2 \times 280 = 224 \text{ kWh}; 4000 / 224$	18 parallel strings	✓

9	Total cells	250S × 18P	4,500 LFP cells	–
10	Cable sizing	$\Delta V = I \times R = 500 \times 0.070$ (500mm ² Cu, 1km)	35 V = 4.4%	✓ < 5%

Table 77. BESS Enclosure Thermal and Structural Screening.

#	Calculation	Working	Result
11	Heat loss per pod	$Q = E_{\text{pod}} \times (1 - \eta_{\text{rt}}) / t_{\text{cycle}} = 667 \times 0.1 / 8$	$Q = 8.3$ kW/pod
12	Enclosure P_{ext}	$P_{\text{ext}} = \rho g h + P_{\text{atm}} = 1025 \times 9.81 \times 200 + 101325$	$P_{\text{ext}} = 2.112$ MPa
13	Ti-Gr2 wall	$t = PD / (2\sigma_{\text{allow}}) + CA = 2.112 \times 2000 / (2 \times 275) + 2$	$t = 9.7 \rightarrow 10$ mm ✓
14	SW cooling ΔT	$Q = hA\Delta T$; $A = \pi \times 2 \times 6 = 37.7$ m ² ; $h = 500$ W/m ² K	$\Delta T = 0.44^\circ\text{C}$ ✓
15	Max cell temp	$T = T_{\text{sw}} + \Delta T_{\text{wall}} + \Delta T_{\text{int}} = 13 + 0.44 + 5$	$18.4^\circ\text{C} \ll 45^\circ\text{C}$ ✓

Table 78. Energy Conversion Efficiency Chain.

Stage	η	Cumulative (tidal / wave)
Generation (mech→elec)	85% / 75%	85% / 75%
Cable losses	96%	81.6% / 72%
DC/DC converter	97%	79.2% / 69.8%
BESS round-trip	90%	71.3% / 62.9%
End-use converter	95%	67.7% / 59.7%

Overall wire-to-load: 60-68% depending on source, consistent with marine renewable microgrid benchmarks (Xu et al., 2022).

Table 79. WP8 Final Design Parameters.

Parameter	Value
Total demand	342 kW avg / 450 kW peak (DOC+habitat)
Generation	1200 kW installed (700 tidal+500 wave); avg 370 kW; all underwater
BESS	4.0 MWh LFP, 250S×18P (CATL 3.2V/280Ah); 6 Ti-Gr2 pods on 2 gravity frames; 8h full/24h Tier 1
Bus / cable	800 V DC sectionalised; 500 mm ² Cu XLPE, 1 km, $\Delta V = 4.4\%$ (DNV-ST-0359)
Enclosure	Ti-Gr2 cyl. (Ø2000×L6000), $t = 10$ mm, 1-atm N ₂ ; passive SW cooling; $T_{\text{cell}} = 18.4^\circ\text{C}$
UDC power	1225 kW – separate shore cable

Table 80. WP8 Component Selection (OEM-Backed).

Component	Selection	OEM
Tidal turbine (100 kW)	AR500-class on gravity base	Simec Atlantis
Submerged WEC (100 kW)	CETO-type submerged buoyant actuator	Carnegie Clean Energy
LFP cell	3.2 V / 280 Ah prismatic	CATL
BESS enclosure	Ti-Gr2 cylindrical PV, 1-atm N ₂	Forum Subsea Technologies
DC bus / breakers	MVDC; SACE Emax 2	ABB
Cable + connectors	XLPE 36 kV armoured; ODI Nautilus wet-mate	Nexans; Teledyne ODI
Junction hub	Oil-filled pressure-compensated dist. unit	Siemens Subsea

9. WP9 – Integration – ICD + Integrated CAD + V&V + Risk

WP Owner	Muhammad Hammad
Contributors	Muhammad Hammad Saba Falah
Key Interfaces	Seawater intake and pre-treatment skid, pumping system, electrochemical DOC reactor, discharge system, underwater data centre (UDC) thermal management system, energy system, control and instrumentation system, subsea structural frame.

Interface Control Document (ICD) — Modular Distributed Integration

The selected integration concept is Option 2: Modular Distributed Integration, in which each subsystem is mounted on a separate skid connected to adjacent modules via flanged pipework, armoured cables, and signal conduits. The ICD defines the mechanical, hydraulic, electrical, and control interface envelope at every subsystem boundary.

Table 81. Interface Control Document — All Subsystem Interface Boundaries (WP1–WP9).

Interface	From → To	Medium / Parameter	Connection Type	Acceptance Criterion
ICD-01	WP1 → WP2	Seawater: Q, P, T, cleanliness	DN450 flanged spool — ASME B16.5	Q within ±5%; P confirmed by PT-01; T ≈ 10°C; no particulate >1mm
ICD-02	WP2 → WP3	Pressurised seawater: Q, P delivery	DN450 flanged spool — ASME B16.5	Q within ±5%; P ≥ 19.10 bar abs confirmed by PT-03
ICD-03	WP3 → WP4	Acidified seawater + CO ₂ gas	DN300 flanged spool + CO ₂ vent line	pH within DOC target range; CO ₂ capture rate ≥ 1000 kg/day
ICD-04	WP4 → Ocean	Neutralised seawater discharge	DN300 diffuser return line	pH change ≤ ±0.2 at diffuser outlet
ICD-05	WP5 → UDC hull	Cooling seawater loop: flow, T in/out	DN150 flanged cooling loop; HX inside hull	Drag force is fully resisted by supports with SF ≥ 2
ICD-06	WP6 → Habitat Interface	Mechanical loads, DC power, seawater exposure, deployment loads	Bolted mounts, wet-mate connectors, coated + CP frame, lifting lugs	No structural failure; stable deployment without overturning
ICD-07	WP7 → All internal WPs	Pressure hull: mechanical envelope, penetrations	Hull penetrations: double O-ring sealed spools	FOS ≥ 2 at P _{ext} = 15.10 bar; proof test passed at 30.2 bar
ICD-08	WP8 → WP1/2/3/5	7×100 kW tidal turbines 800 V DC distribution	Armoured cables; hull penetration glands	Power within rated load
ICD-10	WP9 → All WPs	Integration envelopes; mounting interfaces; ROV access zones	Standard flange connections; frame bolted interfaces	All subsystem interfaces verified against ICD prior to fabrication; zero undefined boundaries

Proof Pressure and Leak Test Concept

Table 82. Proof Pressure and Leak Test Summary.

Parameter	Expression / Method	Value	Notes / Acceptance Criteria
External Design Pressure	From WP7 hydrostatic calculation	1.51 MPa	Based on 150 m depth (seawater)
Proof Pressure	(1.5*Delta P)	2.27 MPa	Increased safety margin aligned with subsea standards
Test Method	Hydrostatic pressure hold test	30–60 min	System pressurised to proof pressure
Leak Acceptance	Pressure decay + visual inspection	≤ 1% pressure drop	No leakage, no seal failure, no permanent deformation

The proof pressure is derived directly from the WP7 external pressure design case ($\Delta P = 1.51$ MPa), ensuring consistency with pressure hull structural and buckling requirements (FOS = 2). Leak-critical interfaces, including O-ring seals and penetrators, are aligned with WP6 sealing and connector design.

Verification Matrix

Each PDS requirement is traced to verification evidence. Methods: C = Calculation, T = Test, R = Review, I = Inspection.

Table 83. Verification Matrix — PDS Requirements Traced to Evidence Across WP1–WP9.

PDS Requirement	WP	Method	Evidence	Pass Criterion
Operating environment & structural integrity	WP7 / WP6	C, R, T	Hydrostatic pressure calc, buckling analysis, material/corrosion review, proof test	Withstands $\Delta P = 1.51$ MPa; FOS ≥ 2 ; no leakage
Pressure integrity & sealing	WP7 / WP9 / WP6	C, T, I	Proof pressure (2.27 MPa), leak test, O-ring + penetrator design	$\leq 1\%$ pressure drop; no seal failure
Internal environment & thermal management	WP5	C, R	Heat transfer calculations, cooling system design	Temp 20–25°C; safe operating conditions maintained
DOC intake, process & performance	WP1 / WP3	C, R	Hydraulic sizing, mass balance, process design	Continuous operation; $\geq 95\%$ uptime
DOC environmental compliance	WP4	C, R	Discharge and mixing analysis	$\text{pH} \leq \pm 0.2$; salinity $\leq \pm 1$ PSU
UDC reliability & availability	WP5 / WP6	R	Reliability and recovery analysis	$\geq 99\%$ availability; recovery ≤ 15 min
Fluid Transmission System (FTS)	WP2	C, R	Pump sizing, NPSH, system curve	N+1 redundancy; NPSH margin ≥ 0.5 m
Energy & power continuity	WP8	C, R	Load audit, storage sizing	≥ 72 h backup; voltage $\pm 5\%$
Deployment, maintainability & safety	WP6 / WP9	C, R, I	Lift analysis, CAD access checks, HAZOP	Safe deployment; accessible maintenance; fault response active
System integration & interfaces	WP9	R, I	ICD, P&ID, integrated CAD	All interfaces consistent and functional

System-Level Risk Register [Likelihood × Consequence — State Method]

Risk scored using L × C matrix (1=Low, 5=High). RAG: Green ≤ 7 | Amber 8–14 | Red ≥ 15 . Residual risk is shown after mitigations are applied.

Table 84. System Risk Register — WP1 to WP9 with Likelihood × Consequence Scoring and Mitigations.

Risk / WP	Description	L	C	Score	Mitigation	Residual Risk
Biofouling blockage (WP1)	Intake screen blocked; loss of flow to WP2	4	4	16	Auto backwash filter; CIP connection; anti-biofouling coating on screen (WP1)	Green (4)
Pump failure (WP2)	Duty pump trips; DOC halted if standby fails to start	3	5	15	N+1 redundancy; standby auto-start < 30 s; continuous VFD health monitoring (WP2)	Green (6)
Reactor seal failure (WP3)	Reactor housing seal fails; seawater ingress into electrical components	2	5	10	Modular cassette design; double O-ring sealing; CIP ports; isolation valves (WP3)	Green (4)
pH exceedance (WP4)	Discharge pH change exceeds ± 0.2 ; environmental non-compliance	3	4	12	Continuous pH monitoring; ESD auto-shutdown on exceedance (WP4)	Green (6)
UDC overheating (WP5)	Cooling loop fails; IT equipment exceeds thermal limits; data loss	2	4	8	Redundant cooling pumps; seawater HX with fouling allowance (WP5)	Green (4)
Maintainability / access limitations (WP6)	Limited physical access to subsea components may restrict inspection or repair, leading to increased downtime	3	3	9	Modular design, CAD access checks, maintenance planning	Green (5)
Pressure hull failure (WP7)	Buckling or seal failure at 150 m; catastrophic flooding of habitat	2	5	10	FOS ≥ 2 buckling check; double O-ring endcaps; Corrosion resistant materials (WP7)	Green (4)
Energy depletion (WP8)	Battery depleted during energy gap; pumps and DOC lose power	3	4	12	Load shedding protocol (life support > DOC > UDC); tidal turbine redundancy (WP8)	Amber (8)
Interface mismatch (WP9)	Flange, cable, or signal incompatibility discovered at integration	3	3	9	ICD issued before fabrication; interface review gate; pre-assembly dimensional checks (WP9)	Green (3)

Section 3.3 - CAD Development & Subsystem Integration

1. WP1 – DOC – Seawater Intake & Pre-treatment Skid

WP Owner	Saba Falah
Contributors	Saba Falah
Key Interfaces	Inputs: Raw seawater via Ducted Intake. Outputs: Filtered seawater to the DOC Process; Waste water via Backwash Outlet. Interfaces: All WPs.

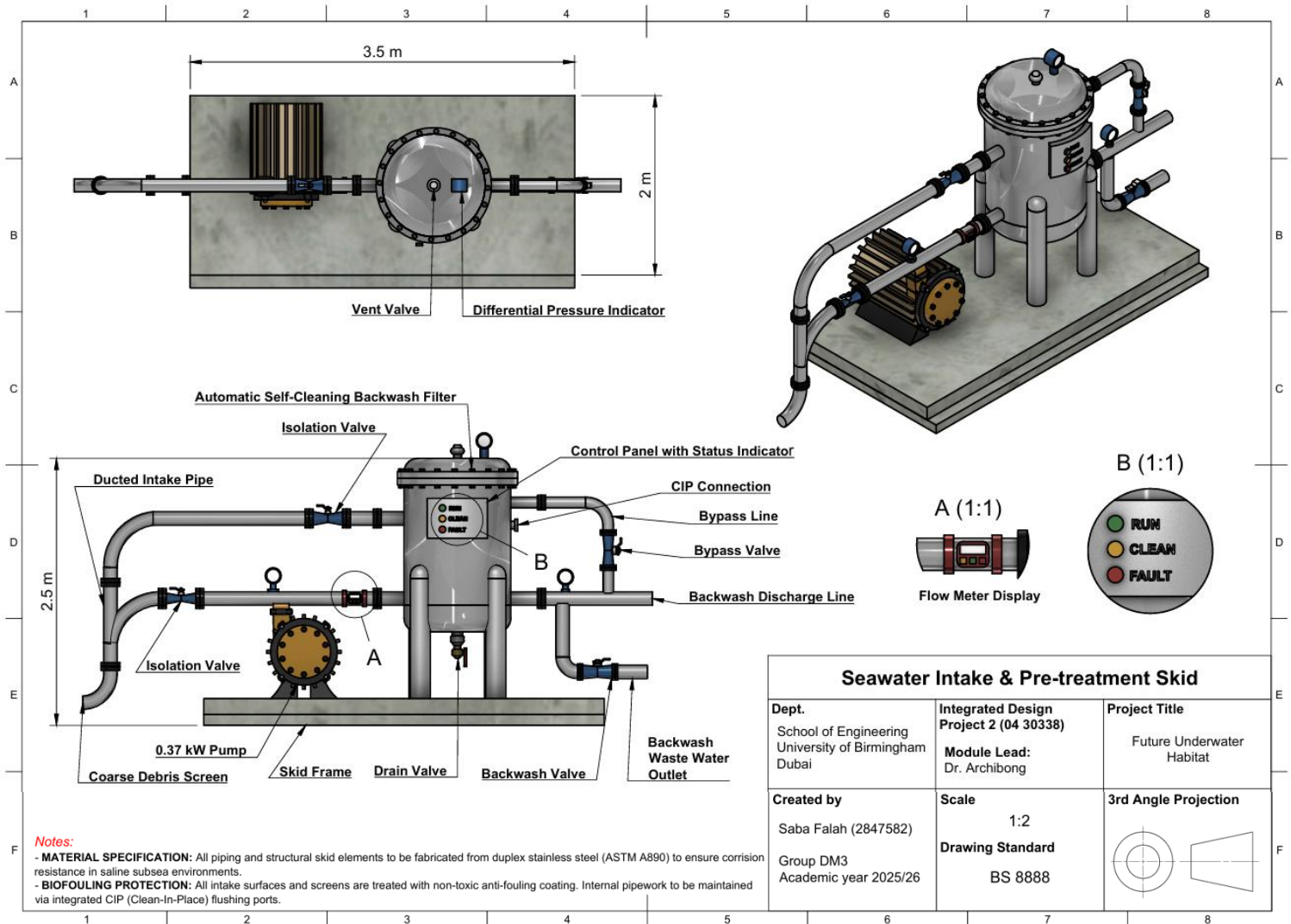


Figure 13. Comprehensive Engineering Drawing of the Pre-treatment Skid, presented in 3rd Angle Projection to BS 8888 standards. The drawing integrates Isometric, Plan, and Front views to define the 3.5 m x 2.0 m footprint and total envelope height of 2.5 m.

Table 85. Main Components and Sub-components derived from the Engineering Drawing, illustrated in Figure 13.

Component	Key Subcomponent
Automatic Self-Cleaning Filter	Pressure vessel body; Filter lid with Vent Valve; Gaskets; Internal Wedge Wire Screen; Rotating Backwash Mechanism with Suction Nozzles; Differential Pressure Indicator
Intake & Pump Assembly	Ducted Intake Pipe (DN50); 0.37 kW Seawater Pump; Coarse Debris Screen with anti-fouling coating
Manifold & Valve Network	Isolation valves (Inlet/Outlet); Bypass line; Bypass Valve; CIP (Clean-in-Place) flushing ports; Flow Meter Display
Structural Skid Frame	Duplex Stainless Steel (ASTM A890) structural members; Skid Frame Drain Valve; Backwash Discharge Line

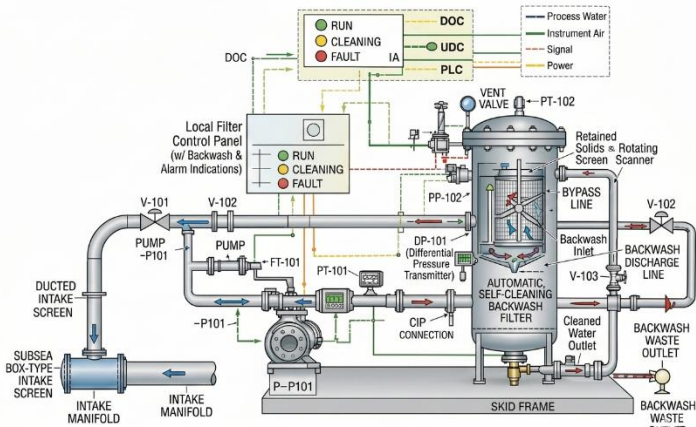
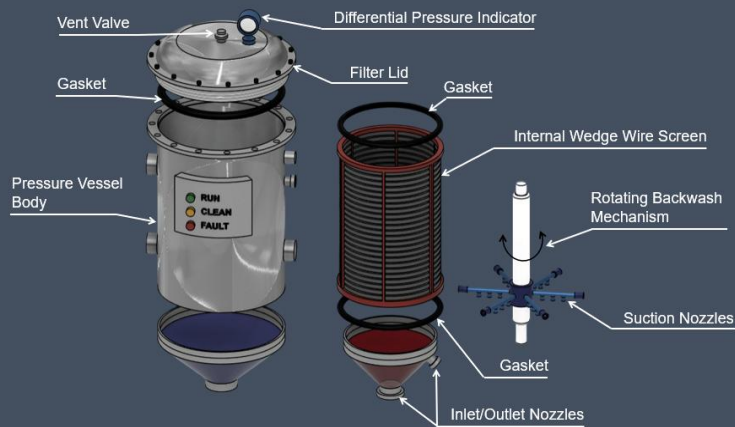
Table 86. Key Dimensional Assumptions for Seawater Intake & Pre-treatment Skid CAD.

Component	Envelope (L × W × H mm)	Notes
Main Skid Footprint	3500 × 2000 × 2500	Boundary for subsea habitat integration
DN50 Piping Network	50 mm OD	Sized for 2.0 m/s velocity to meet 14.4 m³/h flow requirement
Filter Vessel	800 × 800 × 1800	Clearance accounted for Lid removal and internal screen cleaning
Maintenance Clearance	≥ 500 mm all sides	Assumed envelope for valve actuation and ROV tool access
ROV Access Corridor	≥ 1500 mm	Clear space above the Control Panel for visual monitoring and manual override

Table 87. Integration and Clash Check Results for Seawater Intake & Pre-treatment Skid.

Check	Outcome	Action
Ports vs Manifold	DN50 flanges on vessel aligned with manifold; coordinates confirmed via CAD model	No change required
Pump vs Structural Braces	Potential clash identified between the 0.37 kW pump motor and a diagonal frame member	Shifted pump 150 mm laterally; verified clearance for motor replacement
CIP Connections vs ROV	CIP ports located at a 45° angle to the frame edge for ease of ROV hose attachment	Verified accessibility in a 3D environment
Control Panel Visibility	Status Indicators (RUN, CLEAN, FAULT) were obstructed by the Filter vessel	Relocated the Control Panel to the skid's forward face for direct line-of-sight
Intake vs Discharge	Verified ≥ 2 m separation between Intake Screen and Backwash Waste Outlet	Ensures zero re-ingestion of filtered debris

Table 88. Integration Logic and Visualization Summary.

Technical P&ID of the Seawater Pre-treatment System	Exploded Assembly of the Automatic Self-Cleaning Filter
	
Subsystem Connectivity and Logic: This diagram maps the fluidic path from the Ducted Intake Screen through the 0.37 kW pump (P-101) to the Automatic Self-Cleaning Filter. It illustrates the integration of the Bypass Line (V-102) and the CIP Connection to ensure uninterrupted flow during maintenance. The control logic is implemented by the Differential Pressure Transmitter (DP-101), which triggers the Backwash Discharge Line (V-103) when it detects screen fouling.	Component Stack and Maintenance Envelope: Provides a detailed breakdown of the internal filtration assembly, illustrating the concentric arrangement of the Internal Wedge Wire Screen and the Rotating Backwash Mechanism with Suction Nozzles. The visualization of the Filter Lid and Gasket separation justifies the 1800 mm vertical clearance requirement, ensuring sufficient overhead space for the modular extraction of the internal screen and scanner assembly during subsea servicing.

2. WP2 – DOC – Process Flow Transmission (Pumps, Manifolds & Piping)

WP Owner	Muhammad Hammad
Contributors	Muhammad Hammad
Key Interfaces	Input: Filtered Seawater from WP1 Output: WP3 DOC reactor inlet Interfaces: Electrical power supply, VFD control signals

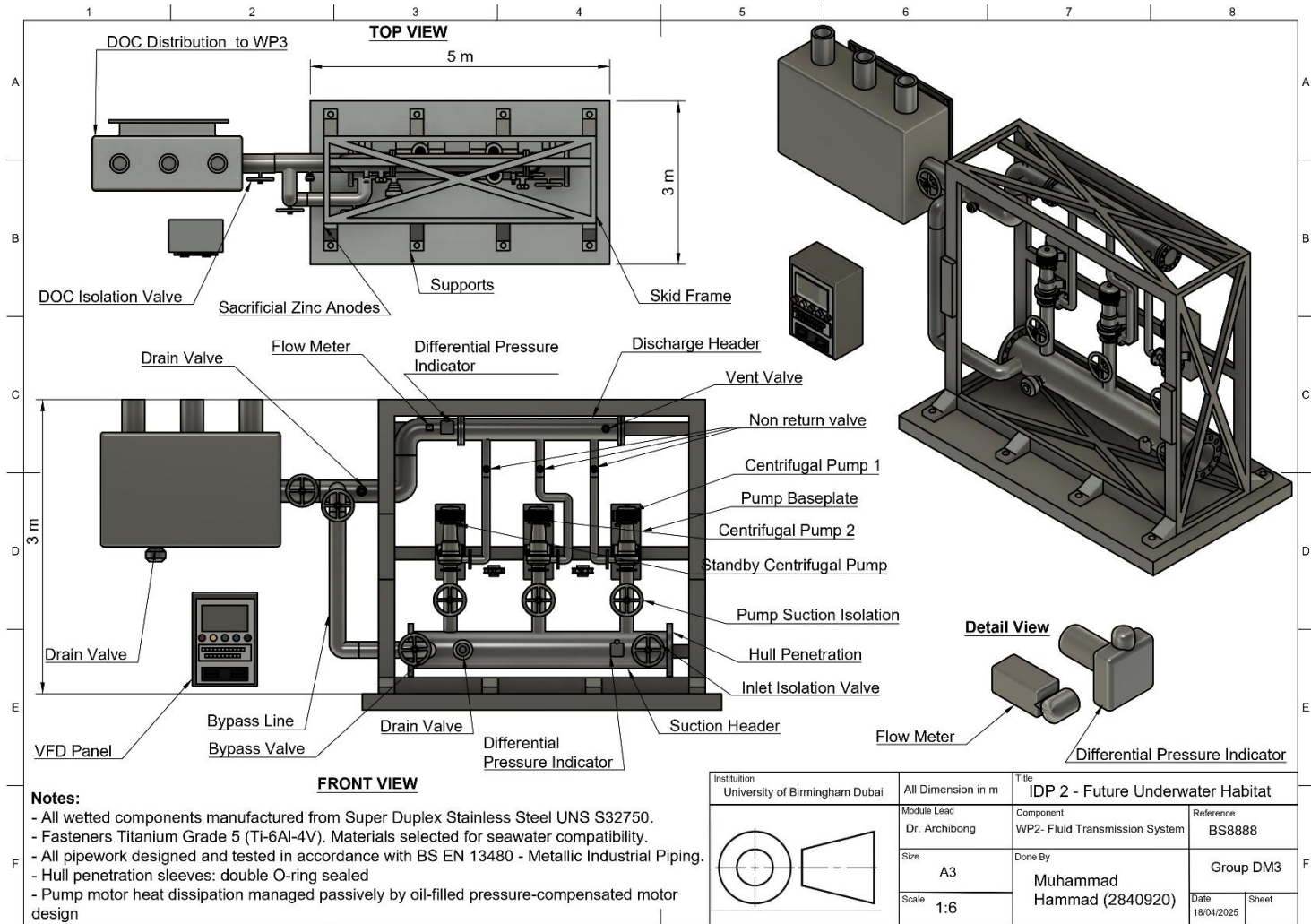


Figure 14. Comprehensive Engineering Drawing of the Fluid Transmission System in 3rd Angle Projection to BS 8888 standards.

Table 89. Main Components and Sub-components for WP2 – DOC Process Flow Transmission.

Component	Key Subcomponents
Submersible Centrifugal Pump Assembly	Pump casing; oil-filled pressure-compensated motor; wet-mate electrical connector; suction inlet nozzle; discharge outlet nozzle; Pump Baseplate; Pump Impeller; Pump Coupling; Drive Shaft; Drive Coupling
Skid Frame Structure	Super Duplex SS UNS S32750 box section frame (4 × 2 × 3 m); structural members; diagonal cross-bracing; 6× sacrificial anodes (cathodic protection); 8× skid base support
Suction Header & Valve Station	DN900 suction header (Ø100mm OD×500mm); V1 inlet isolation valve; Differential pressure indicator; V2A/V2B/V2C pump suction isolation valves; V5 drain; bypass line branch; 3× suction spools DN700×800mm
Discharge Header & Internal Components	DN900 discharge header (Ø100mm OD×500mm); NRV-1/2/3 non-return valves; Discharge isolation valve; Differential pressure indicator; flow meter; vent valve (Normally Closed); Drain Valve; DOC Distribution Module
Bypass & Control	Bypass valve on bypass pipe; VFD Panel (inside habitat); armoured cable bundle from skid to hull penetration; 3× dashed cable routes to pump motors; Hull penetration spool piece

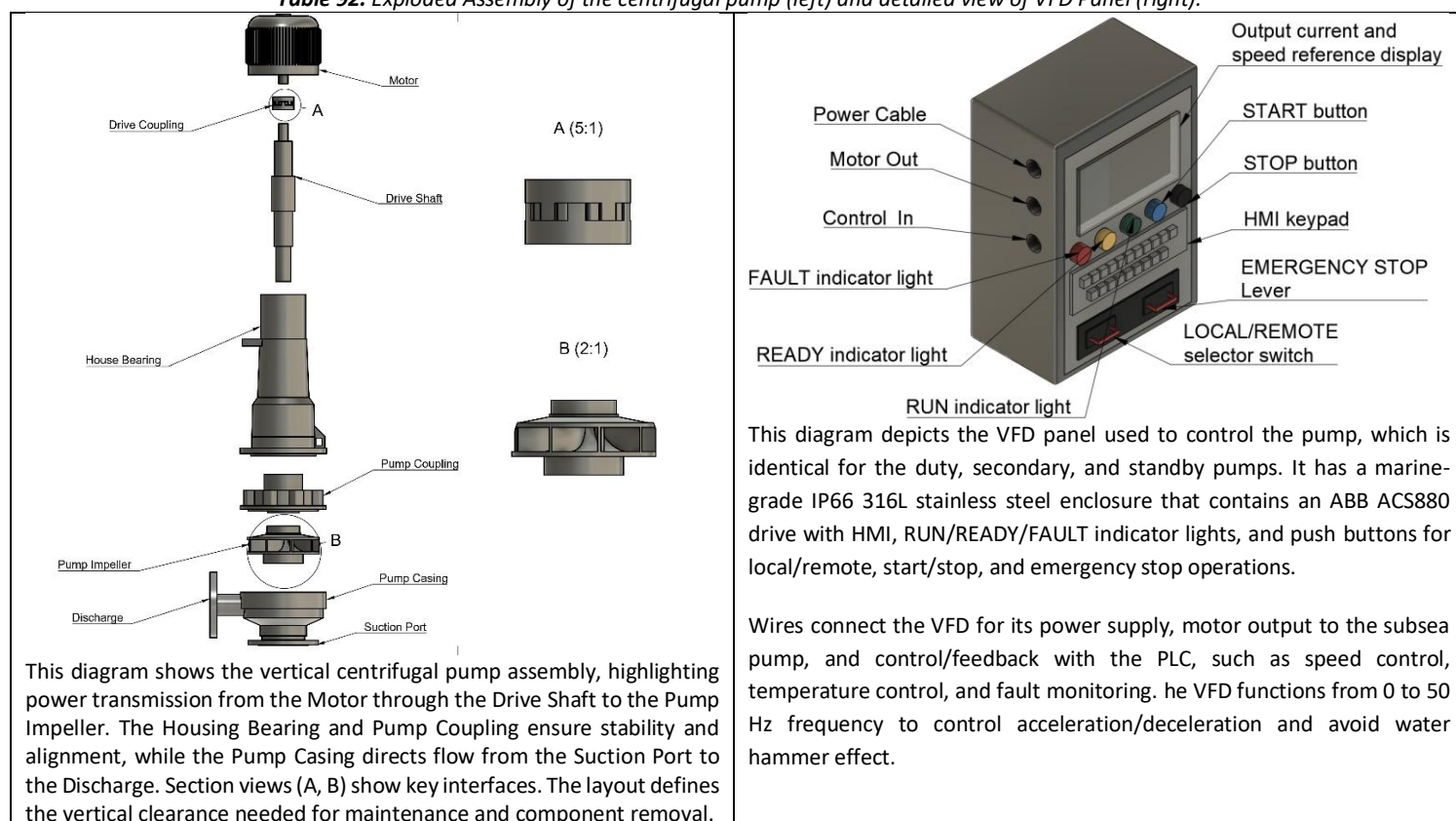
Table 90. Key Dimensional Assumptions for WP2 Process Flow Transmission CAD.

Component	Envelope / Value	Notes
Skid Footprint	4000 × 2000 × 3000 mm	Super Duplex SS UNS S32750 frame; fits the whole system and has clearances.
Header Pipes (suction & discharge)	Ø100 mm OD × 500 mm; t = 10 mm	DN900 sized for Q = 657.6 m³/hr at v = 1.15 m/s; 10 mm wall includes corrosion allowance (ASME B31.3)
Branch Pipes & Pump Envelope	DN350 branches; pump Ø600mm × 1400mm (approx.)	Branches sized for Q = 328.8 m³/hr per pump at v = 0.95 m/s; pump envelope based on Demart Pumps 200WQ300-40-55
Pump Spacing & Manifold Height	1000 mm CL spacing	No clash with frame; manifold height clears Ø90 mm floor penetration sleeves
ROV Access & Retrieval	≥ 500 mm all sides	600 mm clearance for ROV tool operation and valve actuation
Hull Penetration	DN450 spool	Double O-ring sealed; proof tested at 28.65 bar per ASME B31.3

Table 91. Integration and Clash Check Results for WP2 Process Flow Transmission.

Check	Outcome	Action Taken
WP1 to WP2 Interface	DN450 flanged connection at WP1 V2 outlet confirmed; pipe centreline heights matched between WP1 outlet and WP2 suction header	No change required — suction header positioned at matching elevation.
Pump Motor vs Discharge Header	Initial header height clashed with pump motor tops at h=1400mm, header at 1200mm insufficient	Header raised to h=1600mm and clearance confirmed.
VFD Cable Routing	Initial VFD panel at dome centre caused cable clash with suction header	VFD panel relocated to left dome wall
Pipe Supports vs Valve Flanges	Pipe support spacing caused one support leg to clash with discharge isolation valve flange body	Support spacing revised and optimum clearances provided

Table 92. Exploded Assembly of the centrifugal pump (left) and detailed view of VFD Panel (right).



3. WP3 – DOC – Electrochemical Reactor / pH-Swing Stack Housing (and membrane/contactor if used)

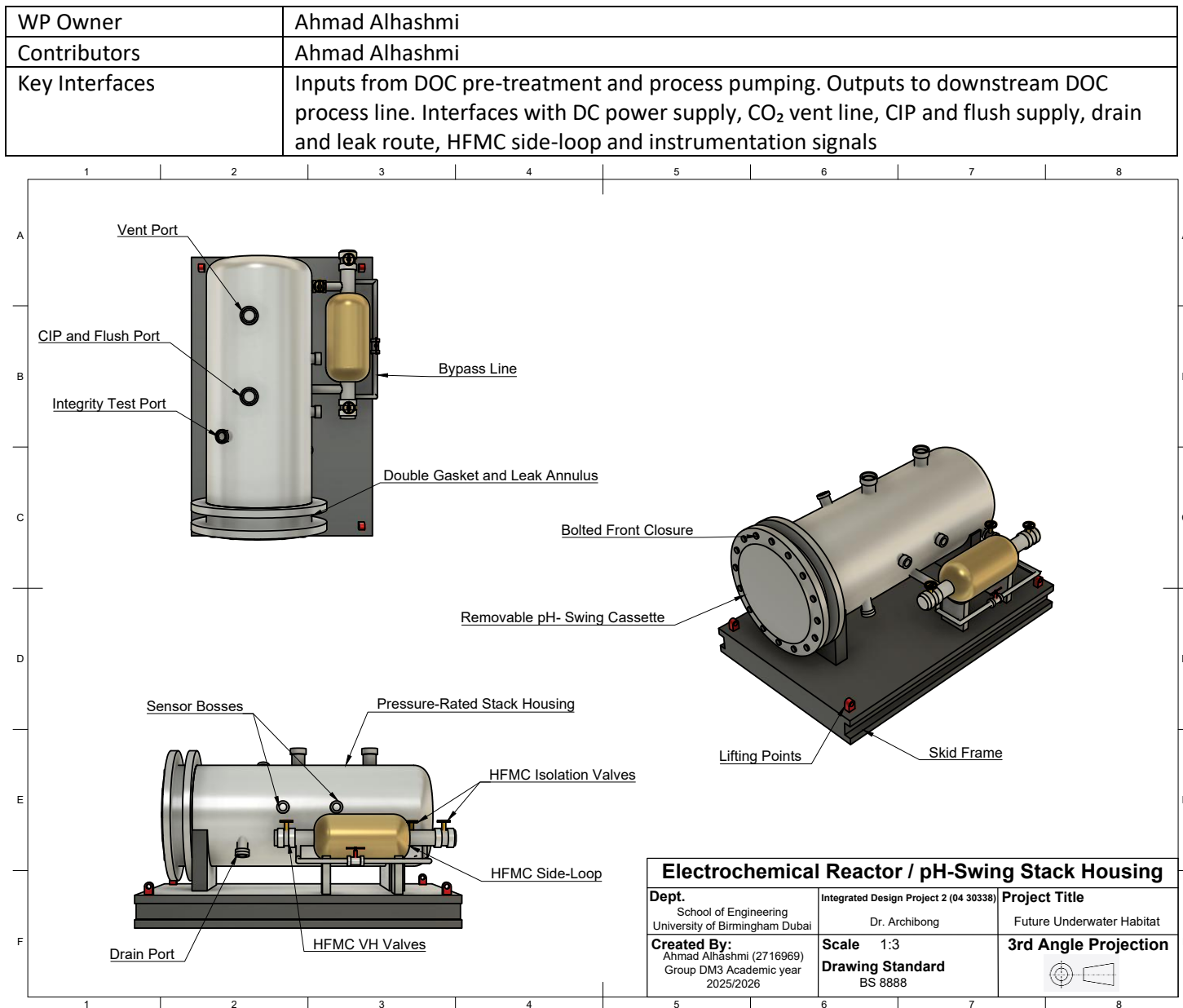


Figure 15. Comprehensive engineering drawing of the WP3 reactor stack housing. The drawing shows orthographic and isometric views of the pressure-rated housing, bolted front closure, removable cassette location, HFMC side-loop, service ports, and skid frame.

Table 93. Main components and sub-components derived from the engineering drawing illustrated in Figure X.

Component	Key subcomponent
Pressure housing assembly	Pressure-rated housing shell, bolted front closure, double gasket and leak annulus, vent port, CIP and flush port, integrity test port, sensor bosses, drain port and saddle supports
pH-swing cassette assembly	Removable pH-swing cassette located inside the pressure shell and accessed through the bolted front closure
HFMC side-loop assembly	Optional HFMC side-loop cylinder, HFMC isolation valves, HFMC bypass line, branch connections and side-loop support frame
Service and instrumentation ports	Vent port, CIP and flush port, integrity test port, sensor bosses and low-point drain connection
Structural skid frame	Main skid frame, lifting points, housing supports and HFMC support frame

Table 94. Key dimensional data used for the WP3 reactor housing CAD.

Component	Envelope L × W × H mm	Notes
Main reactor skid footprint	300 × 195 × 30	Boundary for housing, optional HFMC branch and tool access
Pressure-rated stack housing	270 × 135 × 100	Working CAD envelope for the sealed stack housing
HFMC side-loop module	90 × 35 × 45	Optional contactor branch mounted parallel to the main shell
Front access allowance	600 mm axial pull-out	Allows front closure removal and cassette withdrawal
General service clearance	≥ 400 mm service sides	Provides clearance for bolts, ports and lifting points

Table 95. Integration and clash check results for WP3 electrochemical reactor stack housing.

Check	Outcome	Action
Ports vs HFMC routing	Vent, CIP, integrity-test and drain ports remain accessible without interference from the HFMC side-loop	No change required
Front closure vs cassette access	Bolted front closure remains clear for cassette access and front-end maintenance	Keep front access zone free
HFMC side-loop vs housing ports	HFMC module sits beside the main shell without blocking service ports or front closure bolts	Maintain current side-loop spacing and bypass route
Drain port vs skid supports	Low-point drain remains clear of saddle supports and skid frame members	Keep captured drain path unobstructed
Lifting points vs side-loop	Skid lifting points remain usable without interference from the HFMC support frame	Maintain corner lifting-point positions

Table 96. Integration logic and visualization summary for WP3 reactor housing and optional HFMC contactor.

PFD and P&ID-lite of the reactor housing system	Exploded assembly of the reactor housing
Selected Concept Schematic for WP3 DOC Electrochemical Reactor / pH-Swing Stack Housing and Optional HFMC Contactor Legend: — main process flow - - - HFMC bypass line — CIP and flush — drain and leak collection — DC power supply □ Isolation valve □ Flow transmitter (FT) □ Pressure transmitter (PT) □ Differential pressure (DP) □ Leak detection (LD) Notes: - Sealed housing with removable cassette - Controlled compression and double gasket sealing - HFMC side-loop is isolated and bypassable - CIP, drain and integrity-test ports included - Leak annulus supports containment Excludes neutralisation equipment, static mixing spool, dosing tee, pH and conductivity check after neutralisation, multipoint discharge diffuser, diffuser outlet ports, seabed diffuser frame and any content related to WP4 or return to ocean. Subsystem connectivity. The schematic maps the main path from DOC pre-treatment through the sealed housing and optional HFMC side-loop to the downstream DOC process line. It shows bypass, CIP, vent, drain, DC power and pressure instrumentation.	Exploded component view. The pressure housing shell, bolted front closure, gasket rings, removable cassette and optional HFMC contactor are presented as serviceable components for assembly and maintenance.

Table 97. Maintainability and safety considerations for the WP3 reactor housing CAD package.

Maintainability	Safety
Bolted front closure supports front-end access for cassette removal and inspection	Double gasket land and leak annulus reduce uncontrolled acidic or alkaline leakage
HFMC isolation and bypass valves allow contactor cleaning or replacement without dismantling the main housing	Integrity-test port supports pressure checks before operation and after maintenance
Vent, CIP and drain ports support fouling removal in seawater BPMED operation (Sharifian et al., 2023)	Controlled compression improves gasket seating and lowers local over-compression risk (Moreno Soriano et al., 2021)
Service ports, lifting points and skid supports simplify retrieval and shop maintenance	Titanium or super-duplex wet parts with dielectric breaks reduce galvanic corrosion risk in seawater (Hao et al., 2024)

4. WP4 – DOC – Neutralisation/Mixing + Discharge Diffuser

WP Owner	Sarah Arafa
Contributors	Sarah Arafa
Key Interfaces	Inputs: Acid + base from WP3. Outputs: Neutralised seawater to ocean via multiport diffuser. Interfaces: WP2 (pump head), WP1 (upstream), WP9 (SCADA).

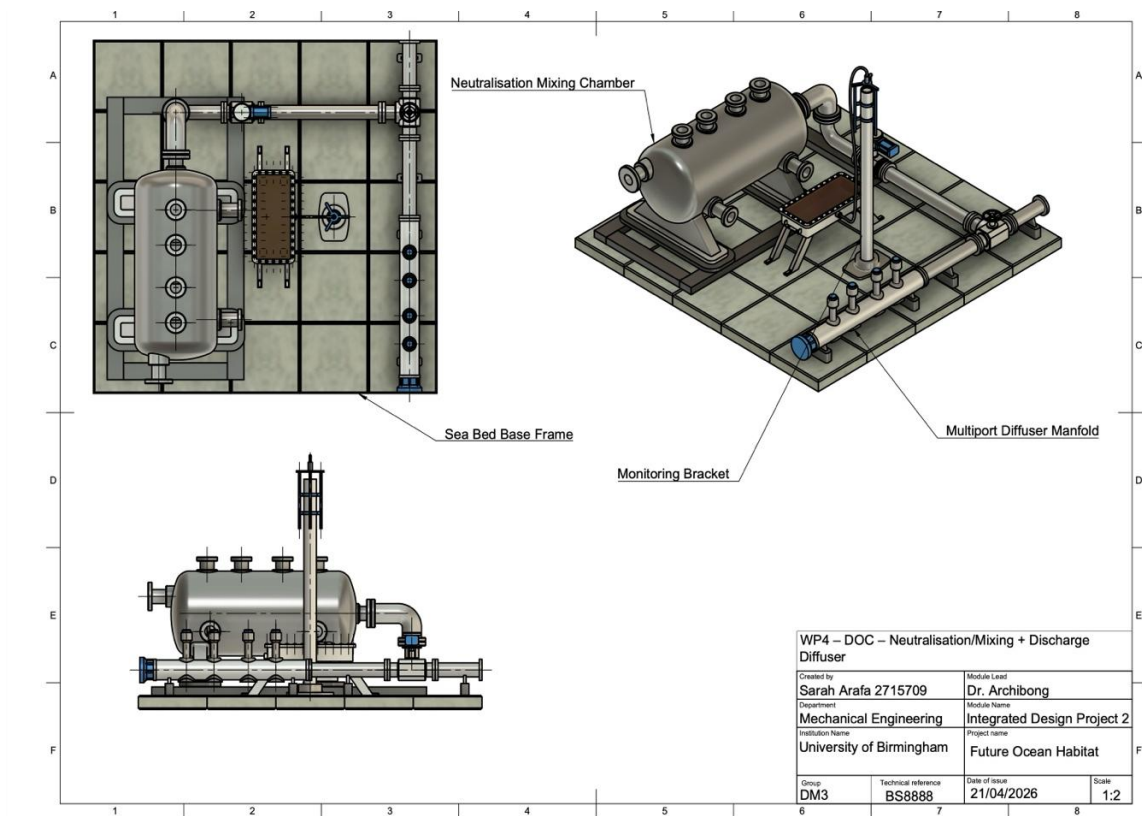


Figure 16. Comprehensive Engineering Drawing of the Neutralisation System in 3rd Angle Projection to BS 8888 standards.

Table 98. WP4 CAD Component Breakdown.

Component	Key Sub-Components
Component 1: Neutralisation Mixing Chamber (V-401)	Vessel shell (horizontal cylinder + dished ends); 8 nozzles with flanges and reinforcing pads; 4 internal baffles (alternating top/bottom, serpentine flow); 2 saddle supports with baseplates and PTFE wear pads
Component 2: Discharge Pipeline (Line 401-C)	DN300 pipe with vertical drop and horizontal run; manual gate valve, motorised ESD valve, check valve, additional isolation valve (simplified geometry); flanged connections throughout; pipe supports at regular spacing
Component 3: Multiport Diffuser Manifold	DN350 header pipe with inlet and end closure; 12 angled outlet ports with reinforcing pads at each branch; removable nozzle inserts with sealing and retaining mechanism
Component 4: Seabed Base Frame	Structural frame using square hollow sections; cross-members along length; pipe support cradles; levelling feet; pad eyes; ballast pockets; anti-scour mattresses beneath
Component 5: Monitoring Bracket	Vertical mast; sensor mounting rail; cable routing provision; subsea junction box mounted to frame

Table 99. WP4 Key Dimensional Assumptions for CAD.

Component	Envelope (L×W×H mm)	Notes
V-401 Mixing Chamber	5800 × 1800 × 1800	Incl. dished ends + saddle height
Line 401-C	50,000 × 400 × 400	Supports at 3 m spacing
Diffuser Manifold	18,000 × 400 × 400	12 ports at 1500 mm centres
Seabed Base Frame	20,000 × 2000 × 800	Cross-members, levelling feet, pad eyes
Monitoring Bracket	200 × 200 × 1500	Mast, sensor rail, junction box
Anti-scour mattresses (×2)	6000 × 3000 × 150 each	Beneath base frame
Maintenance clearance	≥ 600 mm all sides V-401	Tool access + sensor replacement
ROV corridor	≥ 1500 mm above diffuser	Nozzle replacement access

Subsystem Integration Logic

V-401 sits on the DOC process skid inside the habitat dome. The discharge pipeline exits via hull penetration (sea-chest valve at boundary) and runs 50 m to the diffuser, positioned perpendicular to prevailing current for maximum dispersion (Roberts et al., 1989). WP1 intake is on the opposite side of the habitat (≥ 100 m separation). WP5 UDC outfall is ≥ 30 m from diffuser. Assembly sequence: base frame (origin) → diffuser on cradles → pipeline to diffuser inlet → V-401 upstream → monitoring bracket at frame end → umbilical cable along rail.

Table 100. WP4 Integration and Clash Check Results.

Check	Outcome	Action
V-401 inlets vs WP3 outlets	DN150 flanges aligned; coordinates confirmed via ICD.	No change required.
Pipeline vs UDC piping	Opposite side of habitat, ≥ 2 m separation.	No clash.
Diffuser frame vs anchors/UDC outfall	> 30 m separation confirmed.	No clash.
V-401 in process bay	≥ 600 mm clearance confirmed; flanged joints ≥ 100 mm for bolt tools.	No change required.
Cables vs ROV access	Umbilical routed along frame rail, clear of ROV corridor.	No clash.
Pipe supports vs seabed interface	U-cradle spacing matches DN300/350 pipe OD; cross-member pitch = 1500 mm.	Verified against frame geometry.

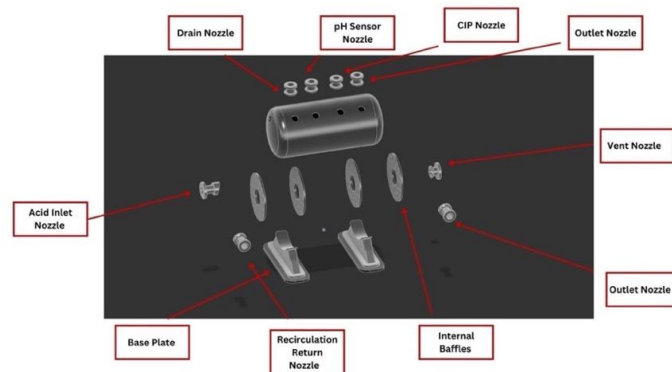


Figure 17. Exploded View of Neutralisation Mixing Chamber.

5. WP5 – UDC – Thermal Management & Heat Rejection

WP Owner	Matteo Karam
Contributors	Matteo Karam
Key Interfaces	Inputs: Hot coolant from UDC systems; Seawater for cooling. Outputs: Cooled coolant back to UDC systems; Heated seawater discharge

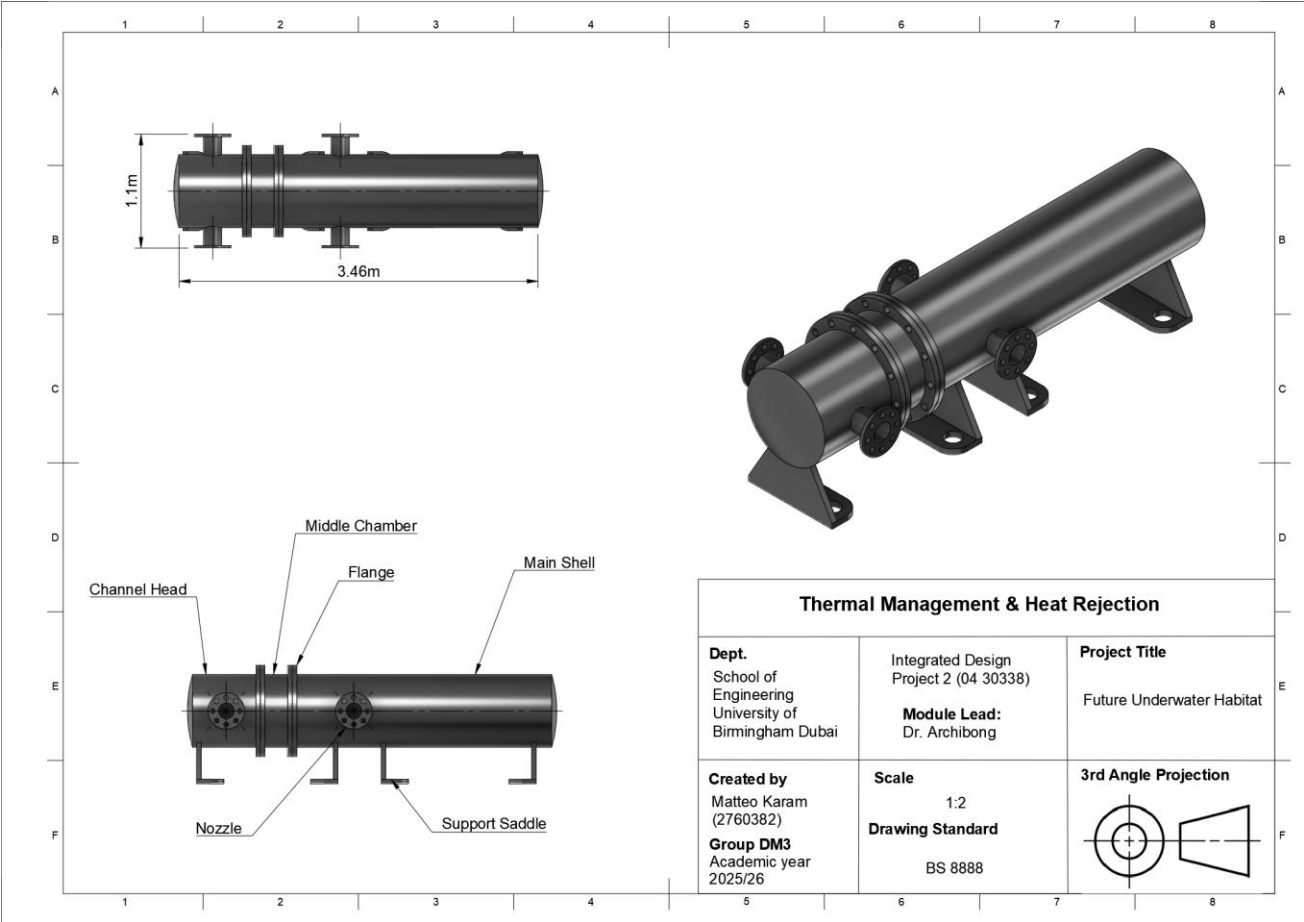


Figure 18. Comprehensive Engineering Drawing of the Heat Exchanger, presented in 3rd Angle Projection to BS 8888 standards. The drawing integrates Isometric, Plan, and Front views.

Table 101. Main Components and Sub-components derived from the Engineering Drawing Illustrated in Figure 18.

Component	Key Subcomponent
Channel Head Assembly	Front cover plate; Channel head body; Tube-side inlet nozzle; Tube-side outlet nozzle
Main Shell Assembly	Main cylindrical shell; Internal tube bundle housing; Rear shell section
Flange Connection Assembly	Bolted flange rings; Gaskets; Fasteners; Middle chamber connection
Nozzle Assembly	Shell-side inlet nozzle; Shell-side outlet nozzle; Flanged pipe connections
Structural Support Assembly	Support saddles; Base mounting holes; Load-bearing brackets

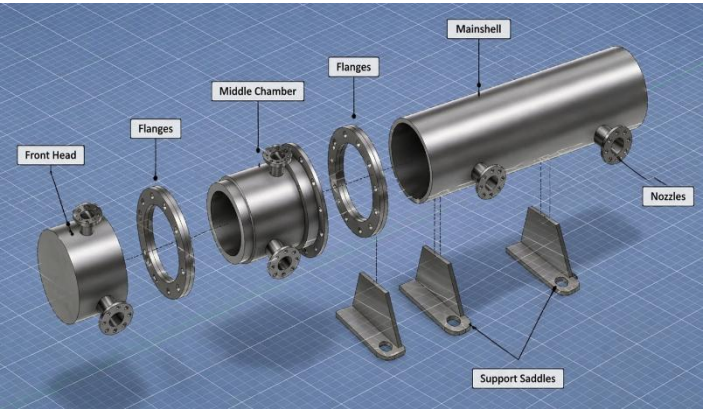
Table 102. Key Dimensional Assumptions for Thermal Management & Heat Rejection CAD.

Component	Envelope (L × W × H mm)	Notes
Full Heat exchanger	3460 x 700 x 1100	Based on final CAD dimensions
Main shell	2550 x 700 x 700	Primary heat transfer body
Channel Head	650 x 500 x 500	Houses tube-side flow
Flange Section	260 x 620 x 620	Allows maintenance/disassembly
Support Saddles	300 x 250 x 400	Structural support and mounting

Table 103. Integration and Clash Check Results for Thermal Management & Heat Rejection.

Check	Outcome	Action
Channel Head vs Nozzle Clearance	Front nozzles positioned without interference	No change required
Shell Nozzle vs Support Saddles	Side nozzle clearance verified from saddle supports	No change required
Flange Accessibility	Bolt spacing allows maintenance access	Verified in CAD
Saddle Load Distribution	Four saddles evenly distribute structural load	No change required
Installation/Lifting Access	Sufficient clearance around exchanger body for transport and installation	Confirmed during CAD review

Table 104. Exploded Assembly of the UDC Heat Exchanger

Exploded Assembly of the UDC Heat Exchanger	
 The image shows an exploded view of the UDC Heat Exchanger components. From left to right, the components are: Front Head, a Flange, Middle Chamber, another Flange, Mainshell, Nozzles, and Support Saddles. The components are arranged to show their relative positions and how they fit together. The Mainshell is the largest cylindrical component, and the Front Head is a smaller cylindrical component that fits into one end. The Middle Chamber is a smaller cylindrical component that fits into the other end of the Mainshell. The Flanges are used to connect the Front Head, Middle Chamber, and Mainshell. The Nozzles are attached to the Mainshell for inlet and outlet flow. The Support Saddles are used to mount the Mainshell to a base structure.	Exploded Assembly Description: The exploded assembly view illustrates how the major heat exchanger components are arranged and connected during assembly. The front head is positioned at the inlet side and connects to the middle chamber through bolted flanges, allowing easy disassembly for inspection and maintenance. The main shell forms the primary heat transfer body and houses the internal tube bundle region. Multiple nozzles provide inlet and outlet flow connections for the working fluids, while the support saddles transfer the structural load of the exchanger to the base structure. The exploded configuration demonstrates component hierarchy, assembly sequence, and maintenance accessibility within the thermal management system.

6. WP6 – UDC – Rack/Chassis, Power Interface, Corrosion Protection, Deployment/Retrieval Base frame

WP Owner	Talal Abu Ghazaleh
Contributors	Talal Abu Ghazaleh
Key Interfaces	Inputs: Electrical power, data signals. Outputs: Rack/chassis support, connector panel, corrosion protection, base frame. Interfaces: Pressure hull, skid/base frame, all WPs.

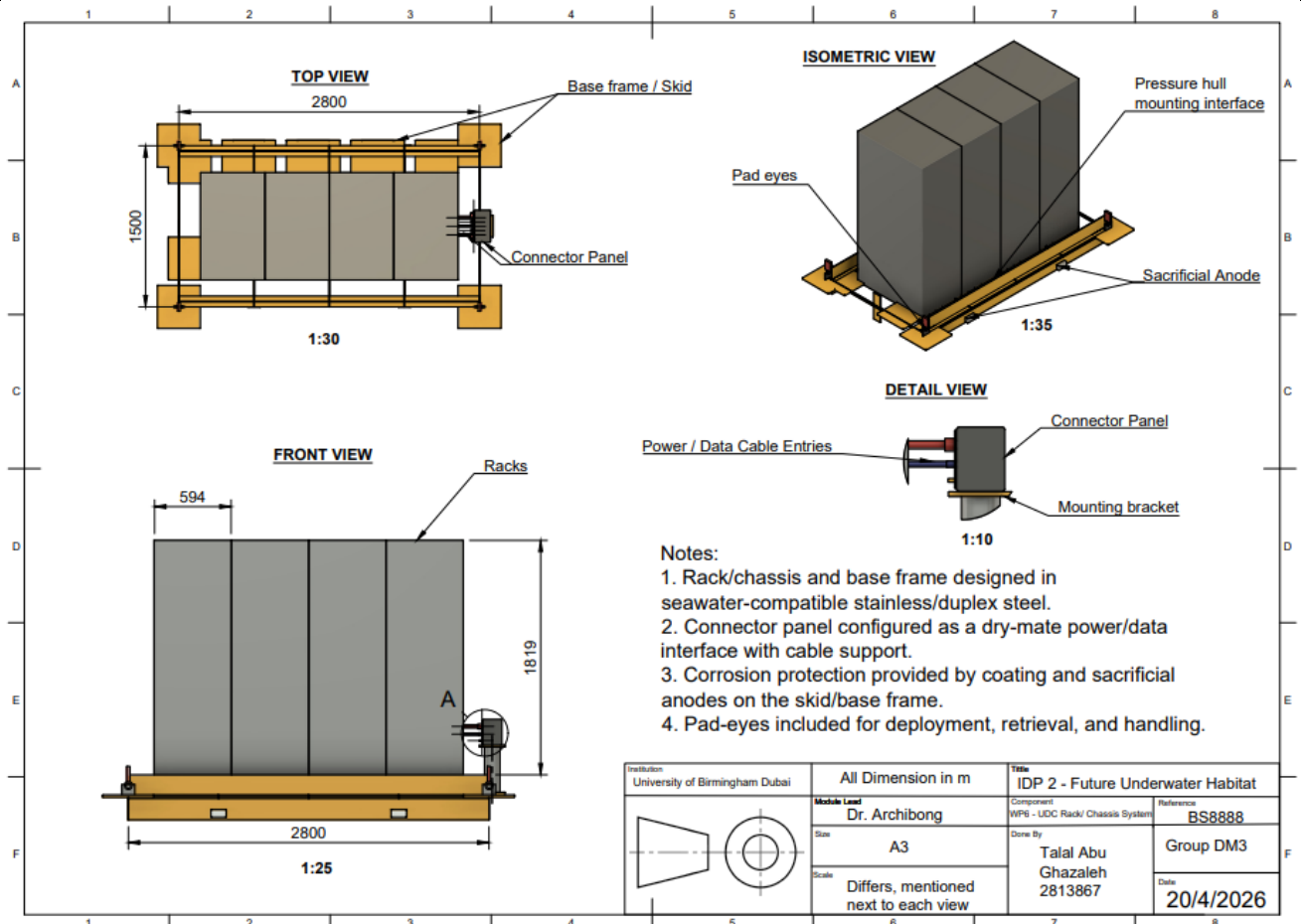


Figure 19. Comprehensive Engineering Drawing of the UDC Rack/ Chassis System in 3rd Angle Projection to BS 8888 standards.

Table 105. Main Components and Sub-components for WP6 – UDC Rack/ Chassis System.

Component	Key Subcomponents
Rack / Chassis Assembly	4x rack bays/modules (~594 mm bay width each); 1x four-bay rack arrangement; overall rack assembly height 1819 mm; top and bottom rack support members; rack mounting interface to skid
Base Frame / Skid	1x skid/base frame (2800 × 1500 mm footprint); 2x longitudinal skid rails; transverse support members; corner/end plates; rack support structure; 2x sacrificial anodes mounted on side rails
Connector Panel Assembly	1x connector panel housing; 3x power/data cable entry points; 1x mounting bracket; side-mounted interface connection to rack/chassis
Pressure Hull Mounting Interface	1x pressure hull mounting interface; skid-integrated support/mounting region; structural connection between rack/chassis assembly and enclosure support arrangement
Deployment / Retrieval Features	4x pad eyes / lifting lugs; skid-integrated lifting points for handling, deployment, and retrieval
Corrosion Protection System	2x sacrificial anodes; 2x anode mounting locations/brackets on skid side rails; corrosion protection provision for subsea deployment

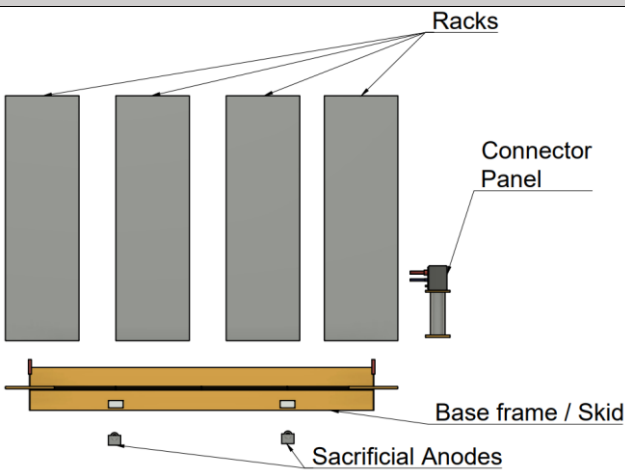
Table 106. Key Dimensional Assumptions for WP6 UDC Rack/ Chassis CAD

Component	Envelope / Value	Notes
Skid Footprint	2800 × 1500 mm	Main base frame footprint used for support and subsystem integration.
Rack / Chassis Assembly	4 rack bays; overall height = 1819 mm; bay width ≈ 594 mm	Four-rack scaled arrangement shown in the CAD.
Connector Panel	1 external panel; 3 cable entry points	Dry-mate power/data interface mounted externally for access and connection.
Cable Routing	Minimum bend radius = 150 mm	Cable routing assumption taken from the WP6 design calculations.
Deployment / Retrieval Features	4 pad-eyes	Lifting/handling features integrated into the base frame.
Corrosion Protection	2 sacrificial anodes	Cathodic protection provision mounted on the skid/base frame.

Table 107. Integration and Clash Check Results for WP6 Rack/ Chassis.

Check	Outcome	Action Taken
Rack Assembly vs Skid Footprint	Four rack bays were accommodated within the 2800 × 1500 mm skid footprint, with the rack/chassis remaining aligned to the base frame for stable support and integration.	Final rack arrangement retained on the skid with no major footprint change required.
Connector Panel vs Rack Access	The connector panel was positioned externally on the side of the rack assembly, allowing access for inspection, disconnection, and retrieval-based maintenance.	External side-mounted dry-mate connector panel retained in the final CAD layout.
Cable Routing vs Bend Radius	Power/data cable routing at the connector interface was checked against the calculated minimum bend radius of 150 mm and treated as an integration requirement.	Cable entry arrangement retained with bend-radius compliance used as the design basis.
Pad-Eyes vs Base Frame Handling	Pad-eyes were positioned on the skid/base frame as dedicated lifting points, supporting the deployment/retrieval requirement without obstructing the main rack envelope.	Four pad-eyes retained as the final lifting arrangement; pad-eye design load checked separately in calculations.
Sacrificial Anodes vs Skid Integration	Sacrificial anodes were placed on the skid side rails as cathodic-protection features without conflicting with the rack volume or the main handling points.	Side-rail anode placement retained as the corrosion-protection provision in the CAD.

Table 108. Exploded Assembly of the UDC Rack / Chassis Module.

Exploded Assembly of the UDC Rack / Chassis Module	
	Exploded Assembly Description: This exploded view shows the WP6 UDC rack/chassis subsystem, highlighting the modular four-bay rack arrangement mounted on the base frame/skid. The layout illustrates how the external connector panel is separated from the main rack structure to support inspection, disconnection, and retrieval-based maintenance, while the sacrificial anodes are mounted independently on the skid for cathodic protection. The exploded arrangement also identifies the skid as the primary structural support element, integrating the rack bays, lifting features, and mounting interface required for subsea deployment and handling. The modular separation of the rack bays demonstrates the intended maintainability strategy, allowing removal or replacement of individual units without redesigning the full support assembly.

7. WP7 – UDC – Pressure Hull / Enclosure (Pressure vessel under external pressure)

WP Owner	Talal Abu Ghazaleh
Contributors	Talal Abu Ghazaleh Matteo Karam
Key Interfaces	Inputs: Heat from UDC electronics/battery systems; Seawater for cooling. Outputs: Rejected heat to seawater; Cooled fluid returned to UDC systems

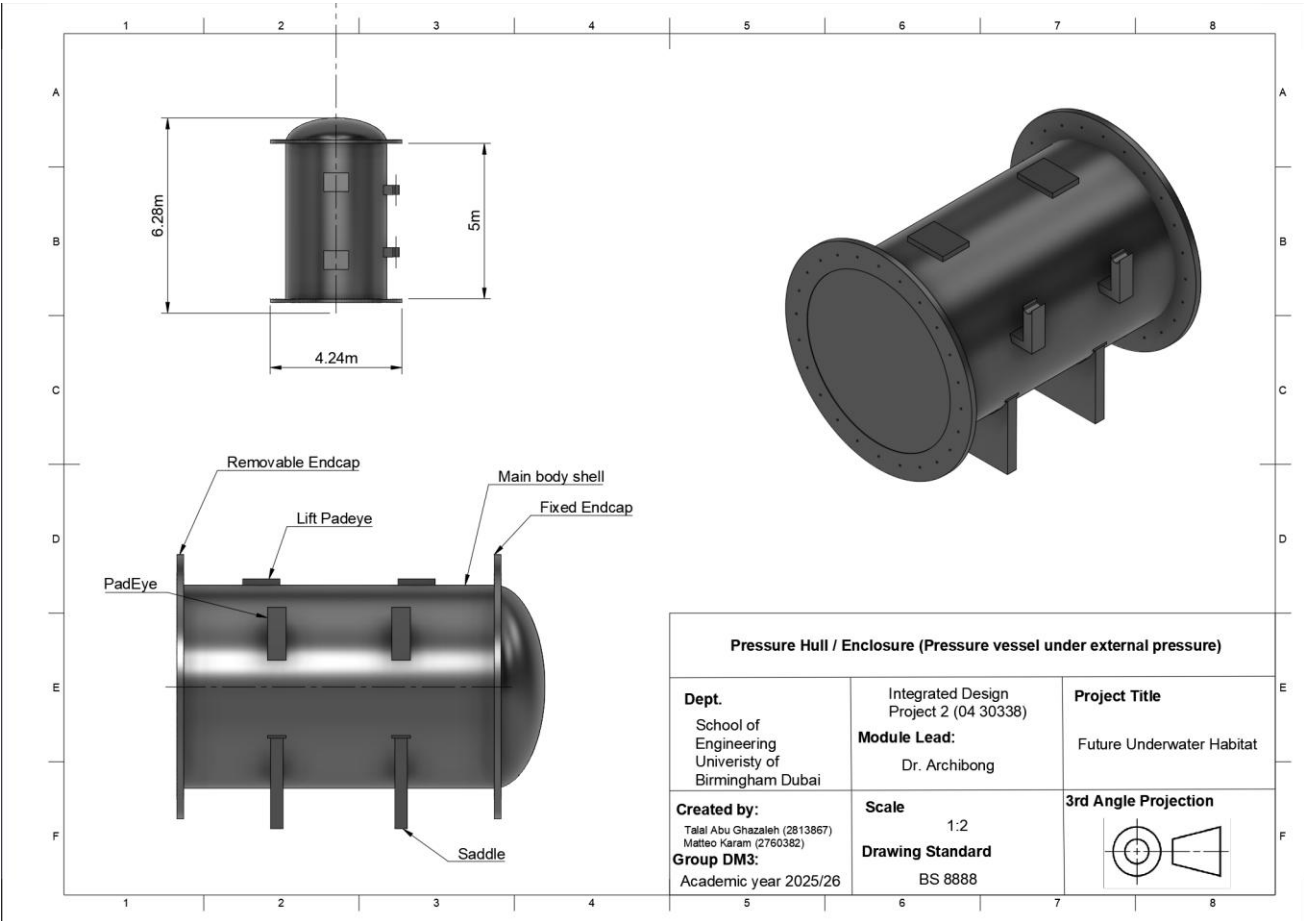


Figure 20. Comprehensive Engineering Drawing of the Pressure hull, presented in 3rd Angle Projection to BS 8888 standards. The drawing integrates Isometric, Plan, and Front views.

Table 109. Main Components and Sub-components derived from the Engineering Drawing are illustrated in Figure 20.

Component	Key Subcomponent
Main Pressure Hull Body	Cylindrical pressure shell; corrosion- resistant outer body; internal equipment housing space
Removable Endcap Assembly	Removable endcap; bolted flange connection; sealing gasket/ O-ring
Fixed Endcap Assembly	Fixed hemispherical endcap; welded connection to shell
Support Structure	Saddle supports; mounting brackets; seabed load transfer points

Lifting & Deployment System	Lift padeye; side padeyes; crane attachment points
Access / Maintenance Features	Inspection access points; removable cover for maintenance

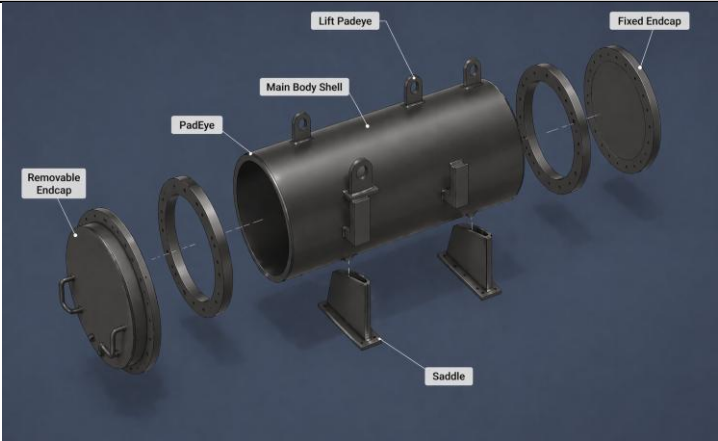
Table 110. Key Dimensional Assumptions for Pressure Hull / Enclosure (Pressure vessel under external pressure) CAD.

Component	Envelope (L × W × H mm)	Notes
Pressure Hull Footprint	6280 x 4240 x 5000	Overall CAD envelope
Main Cylindrical Shell	5000 x 4240 x 4240	Primary equipment housing section
Removable Endcap	1000 x 4240 x 4240	Allows internal maintenance access
Support Saddles	1200 x 800 x 1500	Structural seabed support
Lift Padeyes	400 x 200 x 300	Used for deployment/retrieval lifting

Table 111. Integration and Clash Check Results for Pressure Hull / Enclosure (Pressure vessel under external pressure).

Check	Outcome	Action
Hull vs Internal Equipment Space	Verified sufficient internal clearance for electronics/storage systems	No change required
Padeyes vs Lifting Clearance	Lift points positioned symmetrically for stable lifting	No change required
Endcap vs Maintenance Access	Removable endcap allows internal component servicing	Verified accessibility
Support Saddles vs Seabed Stability	Saddles evenly distribute structural loads	No change required
Hull vs Adjacent Systems	Pressure hull positioned to avoid interference with thermal/power systems	Confirmed during CAD integration

Table 112. Exploded Assembly of the UDC Heat Exchanger Module.

Exploded Assembly of the UDC Heat Exchanger	
	Exploded Assembly Description: Figure X shows the exploded view of the UDC Heat Exchanger and how its main parts fit together. The main body shell is the central section where heat exchange takes place. A fixed endcap closes one side permanently, while a removable endcap allows access for maintenance and cleaning. The parts are connected using flanges for secure assembly, and the unit is supported by two saddles that keep it stable during operation. The exploded view highlights the simple modular design for easy assembly and maintenance.

8. WP8 – Energy System – Load Audit + Sustainable Supply + Storage selection/sizing

WP Owner	Sarah Arafa – Ahmad Alhashmi
Contributors	Sarah Arafa – Ahmad Alhashmi
Key Interfaces	Inputs: Power demand from DOC (~300 kW), UDC (~1225 kW), habitat (~35 kW). Outputs: 800 V DC bus, battery energy storage (BESS), distribution. Interfaces: All WPs

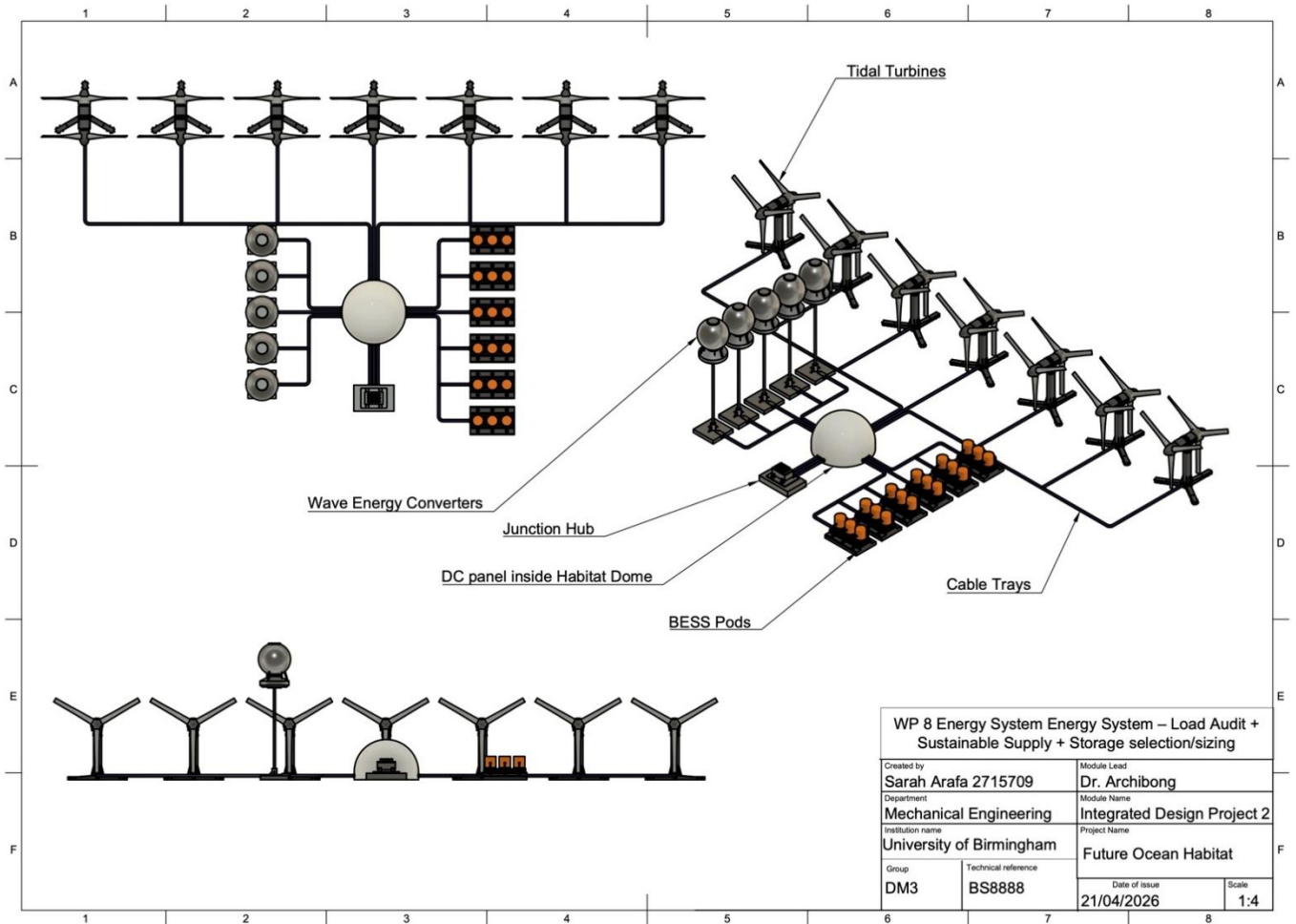


Figure 21. Comprehensive Engineering Drawing of the Energy System in 3rd Angle Projection to BS 8888 standards.

Table 113. WP8 CAD Component Envelopes and Locations.

Component	Envelope	Location
BESS Pod (×6)	Ø2000 × L 6000 mm (cylindrical)	Outside dome — gravity frame, 200–300 m from habitat
BESS gravity frame (×2)	10,000 × 4000 × 1200 mm	Outside dome — ocean floor
Junction hub (×1)	1500 × 1000 × 800 mm	Outside dome — gravity base, ~50 m from habitat
DC/DC converter pod (×4)	1200 × 800 × 800 mm each	Outside dome — mounted on hub frame
Tidal turbine (×7)	D=10 m rotor, nacelle 4 m	Outside dome — gravity base, 200–500 m from habitat
Submerged WEC (×5)	D≈8 m buoy, tether 20–50 m	Outside dome — anchor on ocean floor
DC bus / breaker panel	800 × 600 × 400 mm	Inside dome — habitat energy bay
Cable trays / terminations	Various	Inside dome — hull penetrators to DC bus panel

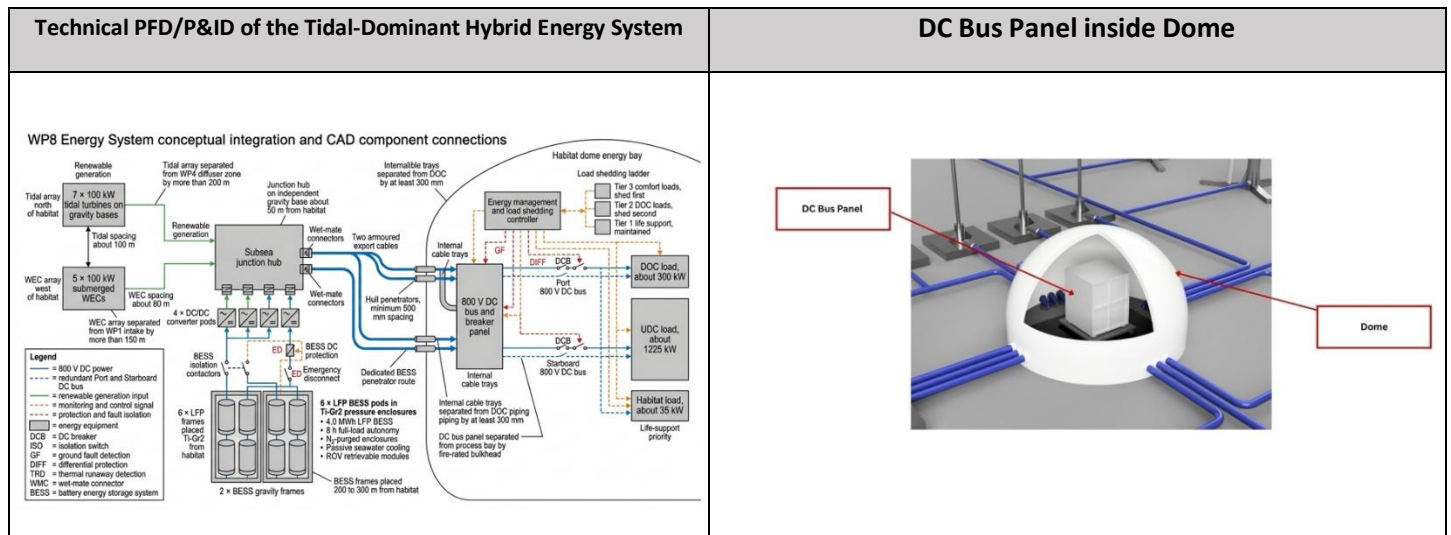
Subsystem Integration Logic

External layout: habitat dome at center; tidal array to north (7 turbines, staggered row, 100 m spacing); WEC array to west (5 devices, 80 m spacing); BESS frames to east (2 frames × 3 pods); junction hub 50 m to south. All generators are cable-connected to the junction hub via 500 mm² Cu XLPE; the hub houses 4×150 kW DC/DC converters (N+1) stepping to 800 V DC. Two armored cables from the hub enter the habitat via J-tube hull penetrators. Internal: cable trays route from penetrators to the sectionalized DC bus panel in the energy bay. BESS connected via a separate 2-cable run through dedicated penetrators. Shore cable for UDC (1225 kW) enters via an independent J-tube.

Table 114. WP8 Integration and Clash Check Results.

Check	Outcome	Action
Tidal array vs WP4 diffuser	Turbines north; diffuser south-east; > 200 m separation.	No clash.
WEC array vs WP1 intake	WECs west; intake east; > 150 m separation.	No clash.
BESS frames vs habitat dome	200–300 m east; clear of all subsea infrastructure.	No clash.
Junction hub vs anchor systems	Hub on independent gravity base; 50 m south; no anchor interference.	No clash.
Hull penetrators (J-tubes)	4 penetrators required (2 hub, 2 BESS); min 500 mm spacing; structural reinforcement confirmed with WP7.	Coordinated with WP7 ICD.
Internal cable trays vs DOC piping	Cable trays ceiling-mounted; DOC piping floor/wall; ≥ 300 mm vertical separation.	No clash.
DC bus panel vs process bay	Energy bay adjacent to but separated from DOC bay; fire-rated bulkhead.	Verified with habitat layout.

Table 115. System Integration: Technical P&ID and DC Bus Panel CAD Layout



9. WP9 – Integration – ICD + Integrated CAD + V&V + Risk

WP Owner	Muhammad Hammad
Contributors	Muhammad Hammad Saba Falah
Key Interfaces	Inputs: Power from Tidal modules; Raw seawater from Pre-treatment; Data from Habitat sensors. Outputs: Regulated power to Habitat; Processed water to DOC; Unified structural stability. Interfaces: Physical mounting for all WPs; Centralized Manifold for fluid distribution; Master Control Bus.

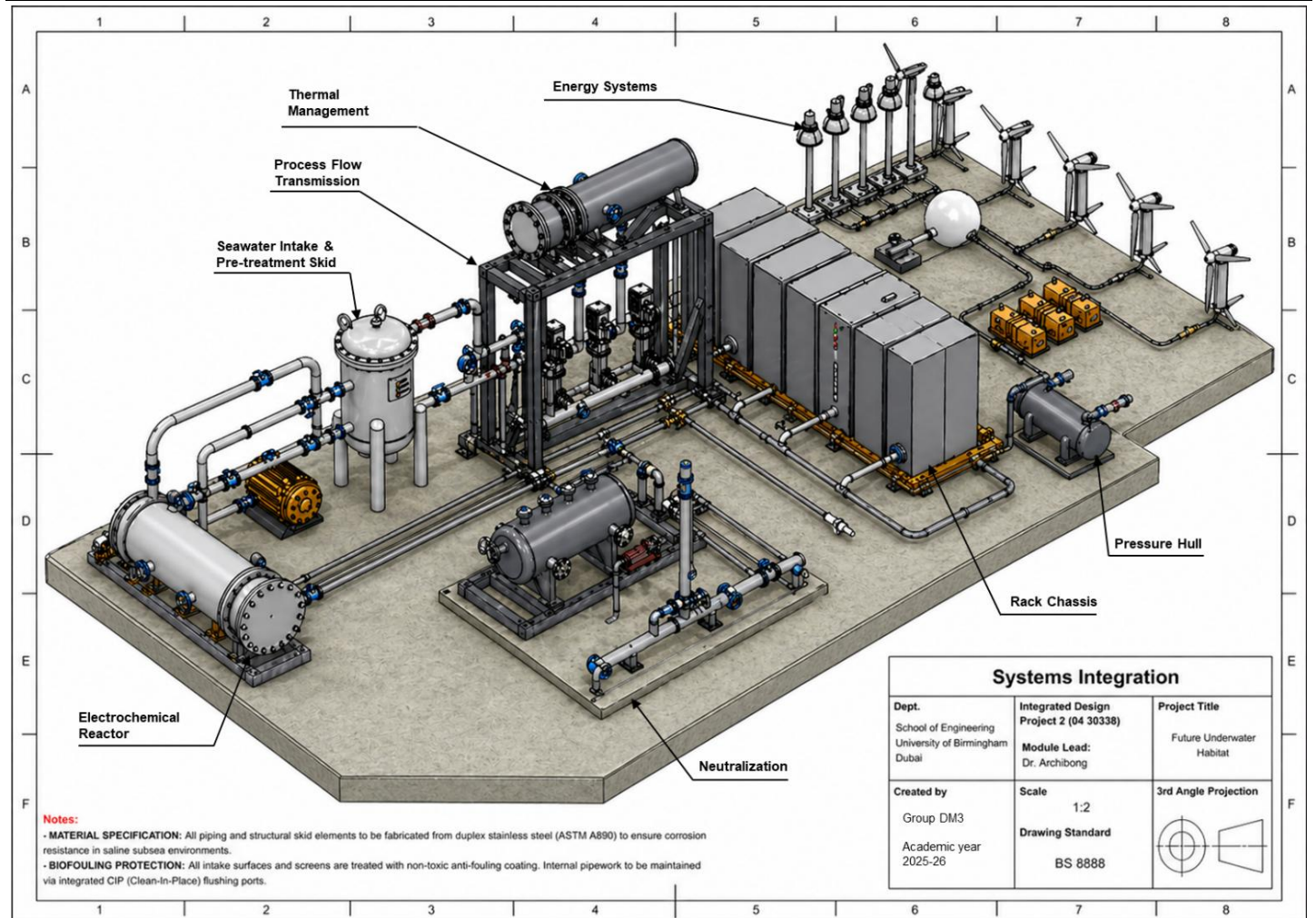


Figure 22. General Arrangement of Integrated Subsurface Infrastructure. Isometric representation of the IDP2 habitat systems integration. The layout utilizes a 300 m² rectangular footprint, engineered to balance ICD (envelopes, mounts, access zones), DOC, and UDC thermal/power interface.

Table 116. Main Integration Components and Sub-components derived from the Engineering Drawing, illustrated in Figure 22.

Component	Key Subcomponent
Central Distribution Manifold	Main header pipes (DN100); Flow control skids; Isolation valves for each module; Pressure regulation valves
Power Management Skid	Step-down transformers; Battery storage racks (grey enclosures); Inverters; Subsea electrical umbilicals
Primary Structural Base	Rectangular reinforced gravity base; Module mounting rails; Lifting lugs for deployment; Sacrificial anodes
Communication Backbone	Fiber optic cable routing; Central PLC (Programmable Logic Controller) housing; Environmental sensor array

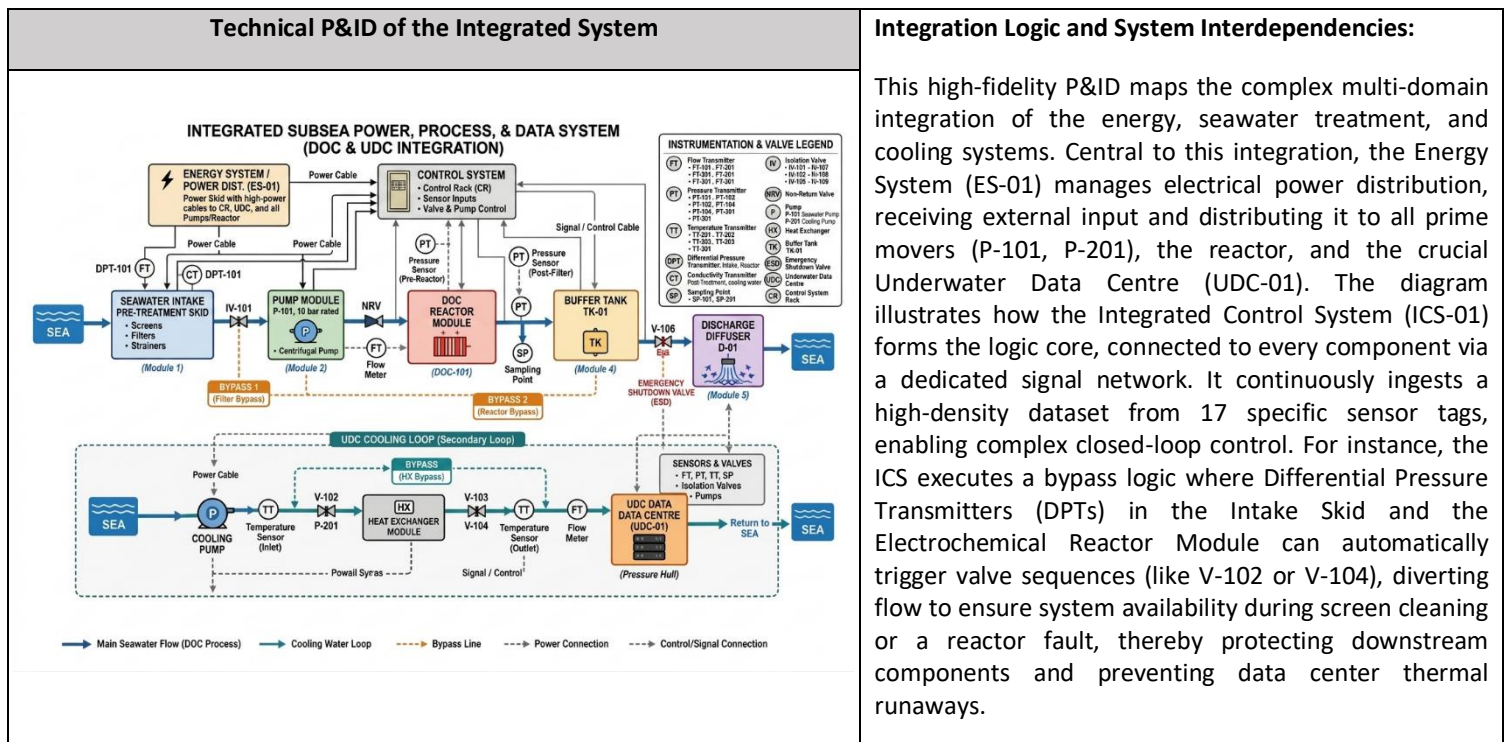
Table 117. Key Dimensional Assumptions for System Integration CAD.

Component	Envelope (L × W × H mm)	Notes
Integrated Base Footprint	20000 × 15000 × 1500	Total Area: 300 m². Reinforced gravity base sized for 60 m² of internal module space
Main Process Piping	150 mm OD (DN150)	Scaled up from DN50 to handle the aggregate flow of all integrated sub-modules
Primary Pressure Vessels	1200 × 1200 × 3500	Scaled for industrial-scale DOC/Pre-treatment throughput on a 300 m² site
Sub-Module Skid Units	5000 × 3000 × 4000	Standardized modular "building blocks" that mount onto the primary base
ROV Access Corridor	Width ≥ 2500 mm	Expanded corridor to allow Work-Class ROVs to turn and maneuver between modules

Table 118. Integration and Clash Check Results for System Integration.

Check	Outcome	Action
Thermal Interference	Power inverters' heat output potentially affects Life Support cooling	Re-positioned Power Management 2m down-current from LSS modules
Piping Alignment	Pre-treatment outlet (DN50) mismatched with DOC Process inlet (DN80)	Integrated a conical reducer and flexible coupling to manage vibration
Structural Loading	Heavy Battery modules created asymmetric loading on the base	Distributed battery enclosures into two clusters of three to balance CoG
ROV Access (Visual)	Structural braces blocked the camera view of the pressure gauges	Rotated gauge faces 90° toward the main access corridor

Table 119. Integration Logic and Visualization Summary.



Section 4 – Overall Integration of Systems/Subsystems

1. DOC seawater intake → pre-treatment → DOC process modules → discharge/reject handling

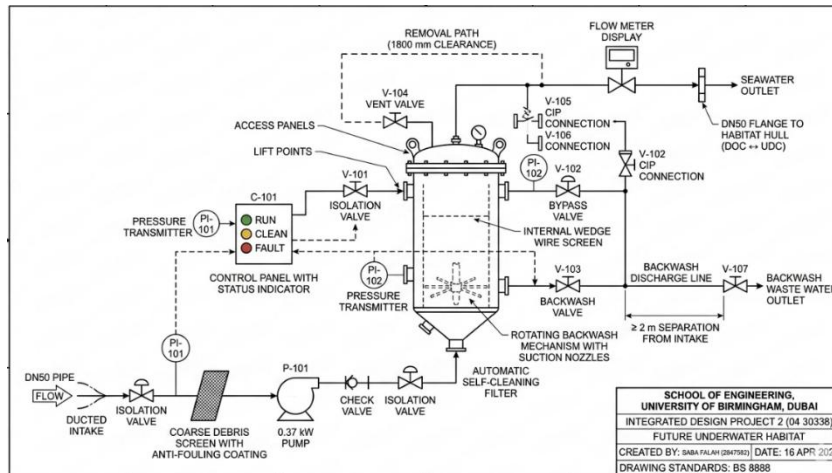


Figure 23. The Seawater Intake and Pre-treatment Skid manages the fluidic path from intake through automatic filtration and DN50 manifolding, utilizing a rotating backwash mechanism for waste disposal and integrated bypass/CIP lines to ensure uninterrupted operation during maintenance.

2. UDC thermal management loop → seawater heat rejection interface

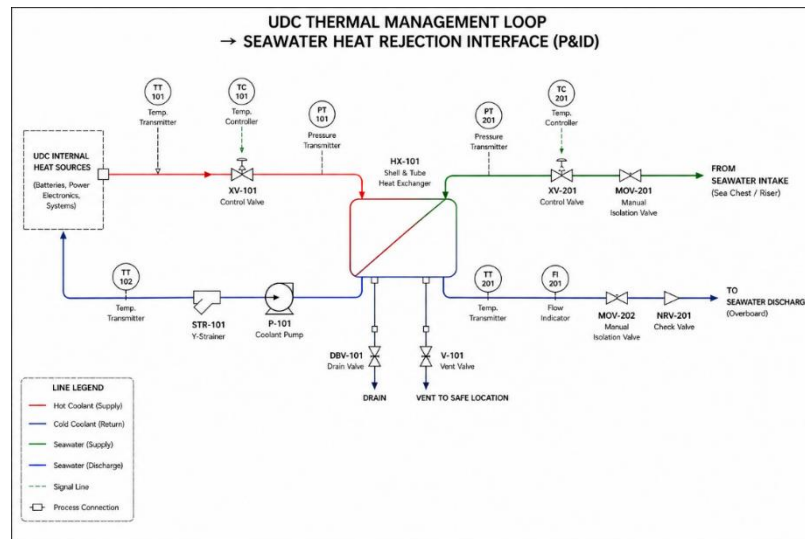


Figure 24. P&ID of the UDC thermal management loop illustrating how the coolant loop carries heat generated by internal electronics and battery systems to the shell-and-tube heat exchanger, where it is transferred to seawater and safely discharged for external heat rejection.

3. Fluid Transmission System (FTS): main headers, valves, bypass lines

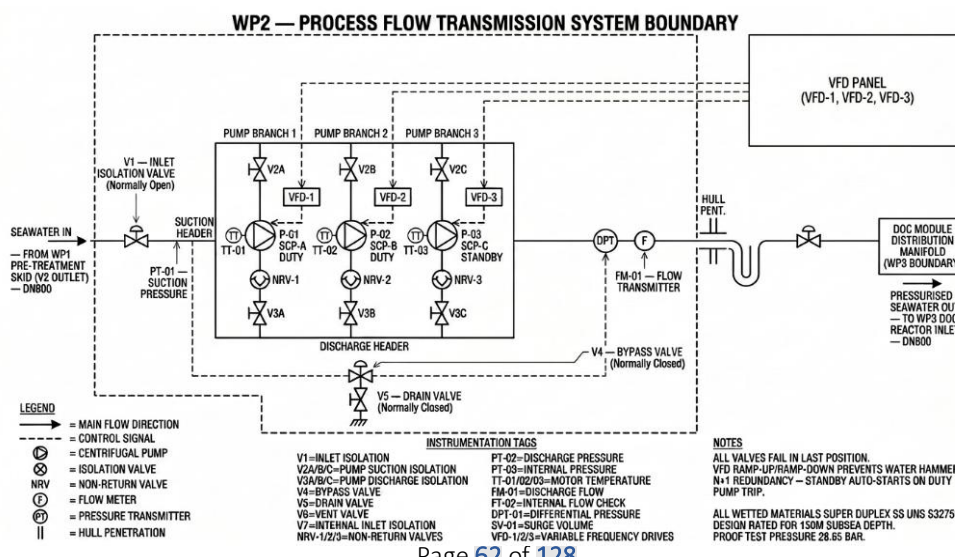


Figure 25. P&ID illustrating integrated fluid transmission, linking intake, pumping, reactor, and discharge with control and safety systems. Pre-treated seawater is drawn from intake, boosted through N+1 pumps, delivered to the electrochemical reactor for CO₂ capture.

4. Energy/power distribution: generation/storage → critical loads

Table 120. Energy distribution.

Critical Load	Source → Bus Path	Rated	Protection / Control
WP2 DOC circulation pumps	Tidal (primary) / Wave (backup) / BESS (8 h autonomy) → Junction Hub → DC/DC (N+1) → 800 V DC Bus (Port)	70 kW	CB-801; VFD speed control; overload relay; auto-switchover on source dip
WP5 UDC cooling pumps + fans	Shore cable (primary) / BESS (backup) → J-tube → 800 V DC Bus (Stbd) via CB-803	15 kW	CB-803; thermal interlock; N+1 pump redundancy; Tier 2 load shedding
Controls & instrumentation: flow sensors, pressure sensors, temperature sensors, ΔP across screens/filters, alarms	Any source (Tier 1 priority) → 800 V DC Bus → 24 V DC PSU (N+1 redundant) → UPS-backed	< 1 kW total	UPS-backed; SIL-1 rated (IEC 61511); 4–20 mA loops; hardwired ESD fail-safe de-energise-to-trip; high-ΔP alarm triggers backwash; RTD drift alarm > 2°C

5. Integration narrative

Table 121. Integration narrative.

Work Package	Function	Key Components	Inputs → Outputs	Key Features / Notes
WP1 – Intake & Pre-treatment	Seawater intake and cleaning	Screens, filters, backwash, CIP	Ocean water → Filtered seawater	Protects downstream systems; enables maintenance without shutdown
WP2 – Flow Transmission	Hydraulic transport	Pumps, valves, manifolds, sensors	Filtered water → Pressurised flow to WP3	Redundancy, isolation, flow/pressure control
WP3 – DOC Reactor	Electrochemical CO ₂ capture	pH-swing stack, sealed housing	Pressurised seawater → Acid/base streams	Leak control, modular cassette, optional HFMC loop
WP4 – Neutralisation & Discharge	Safe discharge to ocean	Mixing chamber, diffuser, pH control	Process streams → Neutralised discharge	Ensures environmental compliance, controlled release
WP5 – Thermal Management	UDC cooling	Heat exchangers, cooling loop	Heat from UDC → Seawater	Maintains safe operating temperature
WP6 – UDC Structure	Data centre support	Racks, chassis, cabling	Power/data → IT operations	Structural support, corrosion protection
WP7 – UDC Pressure Hull	Protection & sealing	Pressure vessel, seals	External pressure → Protected internal space	Prevents seawater ingress
WP8 – Energy System	Power generation & distribution	Renewables, batteries, DC grid	Energy → All subsystems	800 V DC, load prioritisation, load shedding
WP9 – Integration & Verification	System coordination	ICD, CAD, risk tools	All subsystems → Integrated system	Ensures compatibility, safety, verification

6. Design rationales for integration

Table 122. Design rationales for integration.

Principle	Application
Modularity	DOC and UDC are arranged as discrete subsystems on common interface frames, so each unit can be inspected or replaced without disturbing the rest of the habitat.
Maintainability	Service envelopes, drain points, flushing/CIP lines, and ROV-friendly access are reserved around every owned unit to allow in-situ inspection and cleaning.
Redundancy	UDC cooling and DOC pumping follow an N+1 philosophy with parallel lines, cross-tied manifolds, and protected control paths for safe operation under single-point failure.
Safe Isolation	Power, data, thermal, and fluid services are routed through clear interface points with isolation valves and bypass loops so any subsystem can be locked out without a full shutdown.
Corrosion & Biofouling	Marine-grade materials, sacrificial anodes, protective coatings, and sealed connectors throughout, with biofouling-prone surfaces kept accessible for cleaning.

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APPENDICES

APPENDIX A: Meeting Log

Table A. Group DM3 Meeting Logs.

Date	Agreed Actions	Attendees
21.01.2026	- A Group Coordinator was nominated to liaise with the academic mentor. - All candidate concept designs from Assignment 1 were reviewed and presented to the group. - A single, optimized candidate concept was selected by combining the best features from individual designs. - Work package ownership was allocated to group members based on the Work Breakdown Structure. - Group communication channels were established, and a regular meeting schedule was set.	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Saba Falah ✓ Sarah Arafa ✓ Talal Abu Ghazaleh
04.02.2026	- The Product Design Specification was drafted with measurable targets for all key subsystems. - Research topics were assigned to group members to investigate critical technology challenges. - The final hybrid candidate concept was confirmed and its design rationale was documented. - Key physical and functional interfaces between the DOC, UDC, and Energy systems were identified. - Clarifying questions regarding scope and deliverables were compiled for the academic mentor meeting.	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Saba Falah ✓ Sarah Arafa ✓ Talal Abu Ghazaleh
18.02.2026	- Preliminary mass and energy balances were completed for the DOC process and UDC thermal loop. - Suitable pump types were researched, and potential OEM suppliers were identified. - The initial head-loss budget was calculated for the DOC seawater intake and pre-treatment skid. - A candidate material for seawater piping was selected and justified based on corrosion resistance. - Preliminary sizing was performed for screens, strainers, and key manifolds.	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Saba Falah ✓ Sarah Arafa ✓ Talal Abu Ghazaleh
04.03.2026	- Pump duty sizing was performed using system curve analysis, including NPSH checks. - A thermal resistance network was developed to size the UDC heat rejection system. - Initial CAD models were created for individual components to establish spatial envelopes. - Material selections were reviewed to ensure galvanic compatibility across all subsystems.	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Saba Falah ✓ Sarah Arafa ✓ Talal Abu Ghazaleh

	Sealing strategies for the UDC pressure hull and DOC reactor housing were outlined.	
29.04.2026	- All calculations and CAD models were reviewed to ensure design consistency. - An annotated Piping and Instrumentation Diagram was developed to show system integration. - Individual CAD models were assembled into a preliminary integrated facility model. - The verification matrix and risk register were populated with key risks and test methods. - The final group report was structured according to the template, and sections were assigned.	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Saba Falah ✓ Sarah Arafa ✓ Talal Abu Ghazaleh
04.05.2026	- The group presentation slides were finalized, and audio narration was recorded for each slide. - The group report was proofread to ensure consistency across all sections and adherence to the report template. - All final CAD models and integrated schematics were compiled into the report and presentation. - A final design consistency check was performed to align the P&ID, calculations, and CAD assembly. - The Generative AI statement was completed, and all submission files were prepared for upload. - The group presentation video and final report were submitted on Canvas before the deadline.	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Saba Falah ✓ Sarah Arafa ✓ Talal Abu Ghazaleh

APPENDIX B: ACADEMIC MENTOR MEETING FORMS

Table B. Meeting form with academic mentor and PGTA for week 7.

IDP2 2025/26: Weekly academic mentor meeting record	
Stage-Gate (you should have a meeting form for each week/stage-gate) Numbers refer to weeks.	
☐ 1 INTRO to Plan ☐ 2 PLAN to Research ☐ 3 RESEARCH to Requirements ☐ 4 REQUIREMENTS to Design ☐ 5 DESIGN to Assignment 1	☒ 7 INTRO to Plan ☐ 8 PLAN to RESEARCH ☐ 9 RESEARCH to Requirements ☐ 10 REQUIREMENTS to Design ☐ 11 DESIGN to Assignment 2
Stage Progress: - A final candidate concept was selected by combining features from individual Assignment 1 designs. - Work packages were allocated to all group members according to the Work Breakdown Structure. - A Group Coordinator was nominated and regular meeting schedules were established.	
Group Health: - Initial hesitancy was observed among members unfamiliar with working in a new group composition. - Scheduling a consistent meeting time proved difficult due to conflicting personal timetables. - Understanding the scope boundaries of the detailed design required repeated clarification.	
Mentor/PGTA Feedback & Agreed Actions: - Mentor advised that the selected concept must be realistically achievable within subsea constraints. - It was agreed that a clear design rationale for the hybrid concept would be documented before Week 8. - The group was instructed to compile a list of clarifying questions on calculations and CAD expectations for the next meeting.	
Initialled present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Saba Falah, Sarah Arafa, Talal Abu Ghazaleh Absent: N/A Group number: DM3	
Initialled academic mentor: Dr. Archibong Date: 21.01.2026	Initialled PGTA: N/A Date: N/A

Table C. Meeting form with academic mentor and PGTA for week 8.

IDP2 2025/26: Weekly academic mentor meeting record	
Stage-Gate (you should have a meeting form for each week/stage-gate) Numbers refer to weeks.	
☐ 1 INTRO to Plan ☐ 2 PLAN to Research ☐ 3 RESEARCH to Requirements ☐ 4 REQUIREMENTS to Design ☐ 5 DESIGN to Assignment 1	☐ 7 INTRO to Plan ☒ 8 PLAN to RESEARCH ☐ 9 RESEARCH to Requirements ☐ 10 REQUIREMENTS to Design ☐ 11 DESIGN to Assignment 2
Stage Progress: - A comprehensive Product Design Specification was drafted with measurable targets for all subsystems. - Key interfaces between DOC, UDC, and Energy systems were clearly identified and documented. - Research topics were assigned to investigate critical challenges such as biofouling and subsea corrosion.	
Group Health: - Differing interpretations of PDS metrics led to minor disagreements on performance targets. - Some members struggled to balance research depth with the need to maintain progress on calculations. - Communication delays were experienced when waiting for external research outputs from individual members.	
Mentor/PGTA Feedback & Agreed Actions: - Mentor emphasized that the PDS should include verification methods, not just performance targets. - It was agreed to hold a mid-week check-in to ensure research outputs were shared promptly. - The group was advised to begin preliminary calculations concurrently with research to avoid bottlenecks.	
Initialled present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Saba Falah, Sarah Arafa, Talal Abu Ghazaleh Absent: N/A Group number: DM3	
Initialled academic mentor: Dr. Archibong Date: 04.02.2026	Initialled PGTA: N/A Date: N/A

Table D. Meeting form with academic mentor and PGTA for week 9.

IDP2 2025/26: Weekly academic mentor meeting record	
Stage-Gate (you should have a meeting form for each week/stage-gate) Numbers refer to weeks.	
☐ 1 INTRO to Plan ☐ 2 PLAN to Research ☐ 3 RESEARCH to Requirements ☐ 4 REQUIREMENTS to Design ☐ 5 DESIGN to Assignment 1	☐ 7 INTRO to Plan ☐ 8 PLAN to RESEARCH ☒ 9 RESEARCH to Requirements ☐ 10 REQUIREMENTS to Design ☐ 11 DESIGN to Assignment 2
Stage Progress: - Preliminary mass and energy balances were completed for the DOC and UDC systems. - The head-loss budget was calculated for the seawater intake and pre-treatment skid. - Candidate pump types were researched, and OEM suppliers were identified for key components.	
Group Health: - Uneven distribution of calculation workloads was observed, with some members lagging. - Technical difficulties were encountered when applying Darcy-Weisbach and NPSH equations correctly. - Maintaining consistent assumptions across independently performed calculations required repeated alignment.	
Mentor/PGTA Feedback & Agreed Actions: - Mentor noted that assumptions must be explicitly stated and justified in the final report. - It was agreed that all calculation sheets would be shared and cross-checked by another group member. - The group was instructed to schedule a dedicated session to standardize all input parameters.	
Initialled present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Saba Falah, Sarah Arafa, Talal Abu Ghazaleh Absent: N/A Group number: DM3	
Initialled academic mentor: Dr. Archibong Date: 18.02.2026	Initialled PGTA: N/A Date: N/A

Table E. Meeting form with academic mentor and PGTA for week 10.

IDP2 2025/26: Weekly academic mentor meeting record	
Stage-Gate (you should have a meeting form for each week/stage-gate) Numbers refer to weeks.	
☐ 1 INTRO to Plan ☐ 2 PLAN to Research ☐ 3 RESEARCH to Requirements ☐ 4 REQUIREMENTS to Design ☐ 5 DESIGN to Assignment 1	☐ 7 INTRO to Plan ☐ 8 PLAN to RESEARCH ☐ 9 RESEARCH to Requirements ☒ 10 REQUIREMENTS to Design ☐ 11 DESIGN to Assignment 2
Stage Progress: - Pump duty sizing was completed using system curve analysis, including NPSH margin verification. - A thermal resistance network was developed to size the UDC heat rejection system. - Initial CAD models were created for individual components to establish spatial envelopes.	
Group Health: - CAD skill levels varied significantly, slowing the development of integrated models. - Software compatibility issues were encountered when sharing CAD files between different versions. - Some members prioritized calculations over CAD, risking delays in the overall integration timeline.	
Mentor/PGTA Feedback & Agreed Actions: - Mentor advised that CAD models should show key features such as mounting points and access clearances, not just external shapes. - It was agreed that a shared CAD repository would be established with version control protocols. - The group was instructed to allocate additional time for members with less CAD experience to receive peer support.	
Initialled present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Saba Falah, Sarah Arafa, Talal Abu Ghazaleh Absent: N/A Group number: DM3	
Initialled academic mentor: Dr. Archibong Date: 04.03.2026	Initialled PGTA: N/A Date: N/A

Table F. Meeting form with academic mentor and PGTA for week 11.

IDP2 2025/26: Weekly academic mentor meeting record	
Stage-Gate (you should have a meeting form for each week/stage-gate) Numbers refer to weeks.	
☐ 1 INTRO to Plan ☐ 2 PLAN to Research ☐ 3 RESEARCH to Requirements ☐ 4 REQUIREMENTS to Design ☐ 5 DESIGN to Assignment 1	☐ 7 INTRO to Plan ☐ 8 PLAN to RESEARCH ☐ 9 RESEARCH to Requirements ☐ 10 REQUIREMENTS to Design ☒ 11 DESIGN to Assignment 2
Stage Progress: - All calculations and CAD models were reviewed and aligned to ensure design consistency. - An annotated P&ID was developed to show full integration of the DOC, UDC, and Energy systems. - Individual CAD models were assembled into a preliminary integrated facility model.	
Group Health: - Spatial clashes were discovered during integration, requiring last-minute design adjustments. - Finalizing the verification matrix and risk register required extended group discussions. - Some members expressed concerns about meeting the submission deadline due to remaining report writing tasks.	
Mentor/PGTA Feedback & Agreed Actions: - Mentor recommended that the P&ID should include all isolation valves, bypass lines, and instrumentation points for clarity. - It was agreed that a clear division of report writing sections would be finalized immediately. - The group was instructed to submit a draft integration model to the mentor for early feedback before Week 12.	
Initialled present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Saba Falah, Sarah Arafa, Talal Abu Ghazaleh Absent: N/A Group number: DM3	
Initialled academic mentor: Dr. Archibong Date: 29.04.2026	Initialled PGTA: N/A Date: N/A

APPENDIX C: RESEARCH JOURNAL CLUB REPORTS

Group: DM3

Reviewer: Muhammad Hammad

Journal Paper / Book Chapter / Article citation: Gülich, J. F. (2019). *Centrifugal Pumps (3rd ed.)*. Springer. Available at: <https://link.springer.com/book/10.1007/978-3-030-14788-4>

Summary:

The paper is an engineering analysis of the types of pumps used in industrial systems. Centrifugal pumps are the most common type of pump used in the transportation of seawater due to their constant flow rates, mechanical design, and efficiency.

Axial pumps are used for high flow rates and low heads, while positive-displacement pumps, such as progressive-cavity pumps, are used for exact flow rates.

Centrifugal pumps are the most common type of pump used in the transportation of seawater due to their huge volume and mechanical design that can handle seawater. In mission-critical systems, redundancy is achieved through the N+1 method.

Questions (aims):

1. Which pump type is most appropriate for seawater transmission in DOC systems?
2. What advantages do centrifugal pumps provide compared to axial or positive displacement pumps?
3. What redundancy configuration is typically used in critical pumping systems?
4. How can flow rate be controlled in centrifugal pump systems?

Answers (knowledge):

1. Centrifugal pumps are preferred for seawater transmission because they handle large flow rates efficiently.



Figure 26. Centrifugal pump

2. Axial pumps are better for very low head applications, while progressive cavity pumps are mainly used for viscous fluids.
3. Critical systems use N+1 redundancy, meaning one additional standby pump is installed in case of failure.
4. Flow control in centrifugal pump systems can be achieved using several approaches:
 - **Control valves** on the discharge line regulate flow by increasing system resistance.
 - **Variable Speed Drives (VSDs)** adjust the rotational speed of the pump motor, which directly changes the pump head and flow rate according to the pump affinity laws.
 - **Bypass lines** allow some flow to recirculate back to the suction line, preventing the pump from operating at very low flow conditions that could cause overheating or instability.

Impact:

This research supports selecting centrifugal pumps with N+1 redundancy for the DOC system to ensure continuous seawater flow even during pump failure or maintenance.

Actions:

1. Estimate the required system flow rate (Q) for CO₂ removal.
2. Select centrifugal pump models capable of meeting the required head and flow
3. Design N+1 pump arrangement in system layout.

Journal Paper / Book Chapter / Article citation: Gouin, E., Blatteau, J.-E., Dugrenot, E., Guerrero, F., & Gardette, B. (2023). *Scientific shallow saturation dive expedition using diving rebreathers and a specific dry habitat: medical management of the “Capsule” programme*. International Maritime Health, 74(1), 36–44. Available at: journals.viamedica.pl/international_maritime_health/article/view/IMH.2023.0004

Summary: This paper describes a compact dry underwater habitat used for shallow saturation diving at 20 m depth for 3 days, followed by a 245 min decompression procedure. While the paper is mainly medical, it is still useful for thermal-management sections because it discusses the habitat as an isolated, submerged environment with strict environmental and safety constraints that require careful control of internal conditions.
Questions (aims): 1. How can a compact underwater habitat safely support human occupancy for multi-day missions? 2. What environmental constraints must be managed inside a submerged dry habitat? 3. What operational conditions influence occupant safety and tolerance in isolation?
Answers (knowledge): 1. The study shows that short-duration occupation of a compact underwater habitat is feasible when the environment is rigorously controlled. 2. Isolation and environmental constraints require dedicated procedures and contingency planning. 3. For the project, this supports the need for reliable internal environmental control, including temperature and humidity management, even if the paper is not a pure heat-transfer study.
Impact: 1. Provides a direct underwater-habitat precedent. 2. Supports the need for environmental control as part of life support. 3. Strengthens the case for thermal control as a safety-critical subsystem.
Actions: 1. Use this paper to justify why underwater habitats need controlled internal environmental conditions. 2. Reference it when discussing operational constraints and emergency planning. 3. Link it to your habitat’s HVAC and thermal-management design decisions.

Journal Paper / Book Chapter / Article citation: Tetra Tech (2016) Ocean Water Desalination Intake Biofouling: Material Selection and Biofouling/Corrosion Study for West Basin Municipal Water District. Technical Report, West Basin Municipal Water District, California. Available at: <https://www3.epa.gov/region1/npdes/schillerstation/pdfs/AR-413.pdf>

Summary:

This technical report documents a long-term field study on the biofouling and corrosion of candidate materials for ocean-water desalination intake screens and piping at the West Basin demonstration facility. The study combines literature review, design and operation of a seawater pipe test skid, and immersion testing of different alloys and screen materials in Pacific Ocean water. The corrosion and fouling behaviour of copper-nickel alloys and duplex stainless steel is quantified to guide material selection for seawater intake systems, including screens, coupons, and flat plates. The report concludes with material recommendations and notes the need for operational strategies, such as chemical dosing, air-bursting, and regular inspection intervals, to control long-term biofouling and corrosion in intake structures. A pipe test facility was built and installed at the SEA Lab Facility in Redondo Beach, California (Figure 27).



Figure 27. Intake Pipe Testing Facility.

Questions (aims):

1. How can findings on intake screen and pipeline biofouling and corrosion from this study inform the selection of materials and protective measures for a DOC seawater intake and pre-treatment skid, including bell-mouth or scoop inlets and trash racks?
2. Which screen and piping materials from the report (e.g., specific copper-nickel grades or duplex stainless steels) are best suited to withstand long-term immersion while minimizing biofouling on seawater intake and coarse screening equipment?
3. How should intake design and pre-treatment hardware (screens, strainers, and associated piping) be configured so that biofouling and corrosion rates remain manageable using recommended maintenance approaches such as periodic inspection, cleaning, and antifouling strategies described in the report?
4. What practical recommendations from the study (e.g., choice of screen alloy, use of non-metallic intake piping, and reliance on air bursting or chemicals) can be integrated into the hydraulic and maintainability design of a seawater intake and pre-treatment skid to ensure reliable long-term operation?

Answers (knowledge):

1. The report shows that intake components exposed to seawater experience significant biofouling and corrosion, and it emphasizes that intake materials must be chosen based on measured fouling and

corrosion rates rather than cost alone, which directly supports careful material specification for bell-mouths, intake pipes, and screens on a DOC skid.

2. Long-term immersion tests indicate that 70-30 copper-nickel provides the lowest corrosion rate among tested alloys for intake screens while also exhibiting favourable resistance to marine biofouling, leading the authors to recommend 70-30 Cu-Ni as the preferred screen material for ocean intakes.
3. The study recommends non-metallic intake piping in submerged applications to avoid severe corrosion and to benefit from smoother internal surfaces with lower friction, suggesting that composite or plastic piping can be advantageous upstream of pre-treatment equipment.
4. The report concludes that durable seawater intake design must pair appropriate alloys or coated steels with operational measures such as chemical addition, air-burst cleaning, and scheduled inspections (e.g., no longer than five-year intervals for coated duplex steel) to keep biofouling and corrosion at acceptable levels, which can be translated into CIP, flushing, and maintenance provisions for a pre-treatment skid.

Impact:

- By providing comparative biofouling and corrosion data for common intake materials, the study enables more robust specification of screens and submerged components, improving reliability and service life of seawater intake systems.
- The recommendation to use non-metallic piping and specific copper-nickel or coated duplex alloys for screens offers a clear path to reducing corrosion-related failures and maintenance interventions in intake and pre-treatment skids.
- Highlighting the necessity of combined material and operational strategies (chemicals, air bursting, inspections) helps designers incorporate realistic maintenance regimes into intake and pre-treatment layouts, rather than relying on materials alone to control biofouling.

Actions:

1. Use the material performance data and recommendations from this report to select screen and intake pipe materials (e.g., 70-30 Cu-Ni screens and non-metallic intake piping) for your DOC seawater intake arrangement.
2. Incorporate inspection intervals and cleaning strategies suggested in the study into your maintenance philosophy, ensuring that design features such as access, drains, and flushing or air-burst lines are included on the pre-treatment skid.
3. Reflect the report's guidance on corrosion and biofouling in your materials and corrosion-protection specification, especially for intake screens, coupons, and any metallic wetted parts.
4. When performing hydraulic design and layout of the intake and pre-treatment units, reference this study to justify selected screen materials and to argue for non-metallic piping where compatible with pressure and structural requirements.

Journal Paper / Book Chapter / Article citation: Xu, H., Liu, H., Li, Q., Zhang, B., Xu, X., Zhang, D., & Qian, P. (2026). Structural and sealing health monitoring of subsea watertight connectors using embedded fiber Bragg grating sensors. *Measurement*, 258, 119367. Available at: <https://doi.org/10.1016/j.measurement.2025.119367>

Summary:

This paper studies subsea watertight connectors (WTCs), which are important parts in subsea systems because they provide sealed interfaces for electrical and optical signals. The paper proposes a monitoring method using two embedded fiber Bragg grating (FBG) sensors: one for strain and one for humidity. This allows the connector to be checked in real time for both structural damage and sealing failure. The authors tested the method under high pressure and showed that it can detect faults effectively. This is useful for WP6 because my work includes the UDC power interface, so connector reliability and sealing are important design issues.

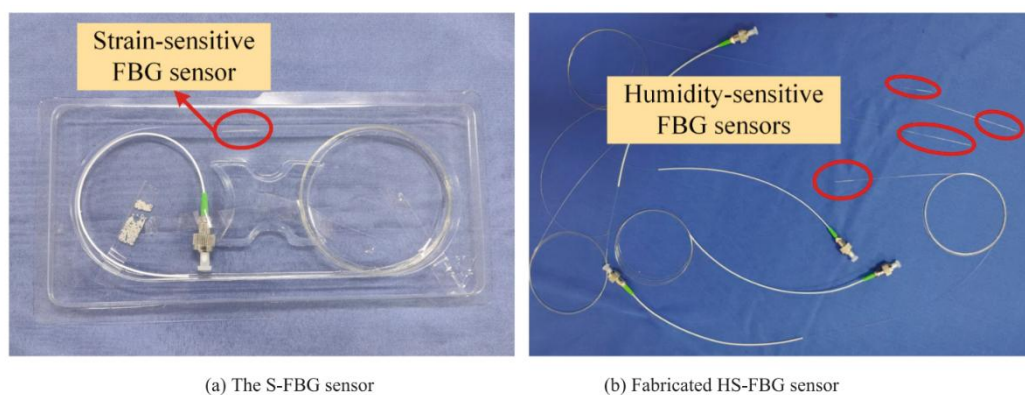


Figure 28. FBG sensors embedded in the WTC.

Questions (aims):

1. Why are subsea watertight connectors important in subsea systems?
2. What are the main failure risks for these connectors?
3. How does the proposed FBG-based method monitor the connector condition?
4. How can this paper help the design of the UDC power interface in WP6?

Answers (knowledge):

1. WTCs are important because they provide a sealed and secure path for power and signal transmission in deep-sea systems.

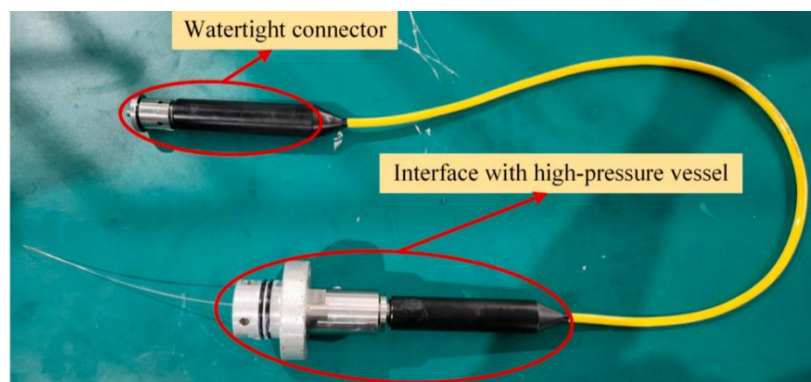


Figure 29. The completed WTC sample.

2. The main risks are structural damage to the shell and sealing degradation, which can cause malfunction, data loss, and hardware damage in subsea systems.

3. The paper uses a strain-sensitive FBG and a humidity-sensitive FBG embedded inside the connector. The strain sensor tracks structural behaviour, while the humidity sensor helps detect leakage or sealing failure.
4. This helps WP6 because the UDC power interface must stay sealed, reliable, and safe underwater. The paper shows that connector sealing and condition monitoring should be treated as important design requirements, not small details.

Impact:

This paper is useful for my WP6 because it supports the design of a reliable subsea connector/interface for the UDC. It shows that connector failure can affect the whole subsea system, so the connector area needs good sealing, protection, and an inspection strategy. It is not about the full rack chassis, but it is a strong support for the power interface part of WP6.

Actions:

1. Use a watertight subsea connector approach for the UDC power/signal interface.
2. Include sealing reliability as a design requirement in WP6.
3. Add protection and inspection access around the connector area.
4. Consider connector monitoring or at least a condition-check strategy in the final design.

Journal Paper / Book Chapter / Article citation: Sharifian, R., Boer, L., Wagterveld, R. M., & Vermaas, D. A. (2022). Oceanic carbon capture through electrochemically induced in situ carbonate mineralization using bipolar membrane. Chemical Engineering Journal, 438, 135326. Available at: <https://www.sciencedirect.com/science/article/pii/S1385894722008300?via%3Dihub>

Summary: This paper is directly useful for WP3 because it explains how a pH-swing route can be used for direct ocean carbon capture. The work uses bipolar membrane electrodialysis to create acid and base streams from seawater chemistry. For the mechanical design, the important point is that the electrochemical stack is not just an electrical item. It needs a sealed housing, defined flow channels, controlled compression, service access and safe routing for acidified and alkaline streams. The study supports using a modular pH-swing stack as the core item inside the WP3 pressure-rated housing.

Questions (aims):

1. Why is a pH-swing stack a suitable baseline for the WP3 reactor housing?
2. What mechanical features are needed around the electrochemical stack?
3. How does this source support a cassette-based service philosophy?
4. Which risks must be controlled before using this concept in seawater service?

Answers (knowledge):

1. The pH-swing route is suitable because it directly changes seawater carbonate chemistry and can release CO₂ without needing a separate chemical regeneration loop inside WP3.
2. The stack needs a sealed housing, controlled compression, leak containment, inlet and outlet nozzles, drain and vent access, and space for electrical and instrumentation interfaces.
3. A cassette approach is useful because the active stack can be removed or replaced without redesigning the whole pressure housing and skid.
4. The main risks are leakage, carbonate fouling, corrosion, flow imbalance and maintaining reliable sealing during operation and maintenance.

Impact: This source justifies the main WP3 architecture. It supports designing the work package as a pressure-rated electrochemical stack housing rather than a simple open plate pack. It also gives a clear reason for including a removable cassette, sealing strategy, leak management and service ports in the CAD model.

Actions:

1. Use BPMED pH-swing operation as the baseline process for WP3.
2. Keep the pressure housing and cassette as the main CAD focus.
3. Include vent, drain, CIP and integrity-test ports in the housing design.
4. Link the pH-swing concept to sealing, corrosion and maintainability requirements.
5. Use the paper to support the concept selection discussion in the report.
6. Include this completed worksheet in the project appendix.

Journal Paper / Book Chapter / Article citation: Oliver, C. J., Davidson, M. J., & Nokes, R. I. (2016). Removing the boundary influence on negatively buoyant jets. *Environmental Fluid Mechanics*, 16(5), 907–927. <https://doi.org/10.1007/s10652-016-9458-2>

Summary:

This paper investigated how water depth affects the mixing of dense jets like brine discharged upward from a desalination outfall. The researchers ran lab experiments where they changed the water depth relative to the jet's maximum rise height and used laser-based measurement tools to track mixing. They found that when the depth is less than about 2.5 times the peak rise height, the seabed and surface interfere with jet development, causing dilution to drop by 15-40%. To help engineers deal with this, they developed correction factors for shallow-water sites.

Questions (aims):

1. How deep must the receiving water be for a dense jet to mix without boundary interference?
2. Can standard dilution estimates be corrected when the water is too shallow?
3. What experimental techniques were used to study the jet behaviour?

Answers (knowledge):

1. The depth (H) needs to be at least 2.5 times the maximum rise height (Y_{max}). When H/Y_{max} drops below 1.5, dilution falls by 30–40% compared with deep water. Between 1.5 and 2.5, losses are 15–30%. Site selection should prioritise adequate depth.
2. Yes. A boundary correction factor k_b depends on H/Y_{max} . Adjusted dilution is $S_{corrected} = k_b \times S$. At $H/Y_{max} \approx 1.5$, k_b is roughly 0.6–0.7; above 2.5, k_b returns to 1.0, and no adjustment is needed.
3. They combined planar laser-induced fluorescence (PLIF) for concentration mapping with particle image velocimetry (PIV) for velocity measurement, running both at the same time to capture full entrainment data along the jet path.

Impact:

Water depth must be confirmed early in the design process. If the site is too shallow, the diffuser will underperform against predictions. The correction factors are handy for preliminary estimates, but a bathymetric survey of the actual site is still essential.

Actions:

1. Verify that the water depth at the proposed site exceeds $2.5 \times Y_{max}$.
2. Apply the k_b correction factor wherever the depth ratio falls short.
3. Obtain bathymetric survey data for all candidate diffuser locations.

Journal Paper / Book Chapter / Article citation: White, F.M. (2011). *Fluid Mechanics, 7th ed.* McGraw-Hill. Ch. 6: Viscous Flow in Ducts & Ch. 11: Turbomachinery. Available at: http://ftp.demec.ufpr.br/disciplinas/TM240/Marchi/Bibliografia/White_2011_7ed_Fluid-Mechanics.pdf

Summary:

White's Fluid Mechanics provides the hydraulic calculation framework for pipe flow and pump systems. Chapter 6 in the book derives the equation for the Darcy-Weisbach friction head loss, Moody chart, and Colebrook-White friction factor equation for turbulent flows, as well as the K-factor method for minor losses due to fittings. Chapter 11 deals with the performance curves of centrifugal pumps, the system curve, duty point determination, affinity laws, and variable speed drives. These chapters contain sufficient information to design the seawater pump circuit hydraulically.

Questions (aims):

1. How is friction head loss calculated in a pipe, and how is the friction factor determined?
2. How are losses through fittings and valves accounted for in the system?
3. How is the system curve built and the pump duty point identified?
4. How is NPSHA calculated and what margin is needed to avoid cavitation?
5. How do affinity laws help justify using a variable-speed drive (VFD)?

Answers (knowledge):

1. **Friction head loss.** The head loss in a straight pipe depends upon the pipe length, diameter, velocity of the fluid, and a friction factor, which depends upon the pipe's roughness and the flow regime. For a turbulent flow, the friction factor can be obtained directly from the Moody chart, given the Reynolds number and the relative pipe roughness. Smooth stainless steel pipes are used because they have low friction factors compared to other materials, which makes them suitable for the DOC seawater circuit.
2. **Minor losses.** Every fitting — elbow, valve, tee, inlet — adds a head loss proportional to the velocity head. Each fitting type has a resistance coefficient (K value); the table below gives typical values. Minor losses are easy to underestimate but can represent up to 40% of total system losses in a compact skid with many fittings:
3. **System curve & duty point.** The system curve plots the total head the pump must deliver at each flowrate — the sum of static head (elevation + back-pressure) and friction losses. Since friction scales with flow squared, the curve is parabolic. The duty point is where this curve crosses the pump's H-Q curve. A fouling allowance of around 15% should be added to friction losses to account for pipe roughness increasing over time. Table 122 below shows a simple example:

Table 122. Pump-system curve example and duty point data

Q (m ³ /h)	H _{pump} (m)	H _{sys} (m)	Note
0	48	10	Shut-off head
30	42	18	Low flow
55	24	35	≈ Duty point
70	10	52	Overload — avoid

4. **NPSH.** NPSH Available (NPSHA) is the pressure energy at the pump suction above the fluid's vapour pressure. If NPSHA drops below the pump's required NPSH (NPSHR), the fluid locally vapourises and cavitation occurs, damaging the impeller. For a subsea installation, the elevated water pressure at depth naturally increases NPSHA — a useful advantage. A minimum margin of 1.0 m above NPSHR is recommended for seawater service, as chloride corrosion worsens cavitation damage.

5. **Affinity laws & VFD.** When a pump's speed is reduced, flow drops proportionally, head drops with the square of speed, and — crucially — power drops with the cube of speed. This means halving the speed cuts power consumption to one eighth. A variable-speed drive (VFD) exploits this relationship, making it

far more energy-efficient than throttling a valve to reduce flow, which wastes energy as pressure drop across the valve rather than saving it at the motor.

Impact:

The friction and minor loss framework from this chapter will be applied to build the DOC seawater system curve, with a +15% fouling allowance on friction losses. The K-factor table covers all fittings in the circuit. The duty point will be confirmed by plotting the system curve against the OEM pump H-Q curve. NPSHA will be checked at operating depth against a ≥ 1.0 m margin. The VFD energy saving argument feeds directly into the group energy audit.

Actions:

1. Calculate friction losses for each pipe segment using the Darcy-Weisbach equation; use the Moody chart or Colebrook-White equation to find the friction factor.
2. Sum all K-factor minor losses for fittings in the DOC circuit; add to friction losses to get total system head at design flow.
3. Plot the system curve (with +15% fouling margin) against the OEM pump H-Q curve; confirm duty point is within $\pm 10\%$ of BEP.
4. Calculate NPSHA at the pump suction at operating depth; verify $NPSHA \geq NPSHR + 1.0$ m using manufacturer data.

Journal Paper / Book Chapter / Article citation: Mirza, J.M.J., Kanwal, F.K.F., Salaria, U.A., Ghafoor, S.G., Aziz, I., Atieh, A.A., Almogren, A.A., Ul Haq, A., & Kanwal, B. (2024). *Underwater temperature and pressure monitoring for deep-sea SCUBA divers using optical techniques*. *Frontiers in Physics*, 12, 1417293. Available at: frontiersin.org/journals/physics/articles/10.3389/fphy.2024.1417293/full

Summary:

This paper proposes an all-optical monitoring system for underwater temperature and pressure, designed for diver safety. The system was evaluated over 0-7 MPa and 40 to -2°C, with reported sensitivities of 4.3 pm/°C for temperature and 30.5 pm/MPa for pressure. For an underwater habitat, it is highly relevant to the verification plan side of thermal management, especially instrumentation and acceptance criteria.

Questions (aims):

1. How can underwater temperature and pressure be monitored accurately in real time?
2. What instrumentation ranges are suitable for deep underwater environments?
3. How can environmental data improve operational safety?

Answers (knowledge):

1. The paper shows that underwater temperature and pressure can be monitored remotely and locally using fiber-optic sensors.
2. The tested range covers harsh underwater conditions relevant to submerged systems.
3. This supports adding temperature and pressure instrumentation to habitat thermal-management verification plans.

Impact:

1. Provides a monitoring strategy for thermal-system verification.
2. Supports continuous environmental sensing.
3. Strengthens the instrumentation part of your heat-rejection section.

Actions:

1. Use it to justify temperature and pressure sensors in your thermal loop.
2. Reference it in your verification plan.
3. Mention it when defining acceptance criteria for monitored thermal performance.

Journal Paper / Book Chapter / Article citation: Rajagopal, S., van der Velde, G., van der Gaag, M., & Jenner, H. A. (2006) Biofouling in Cooling Water Intake Systems: Ecological Aspects and Engineering Controls. In: Dürr, S. & Thomason, J. C. (eds.), Marine and Industrial Biofouling. Springer, Berlin, Heidelberg. Available at: https://www.researchgate.net/publication/254897956_Biofouling_in_Cooling_Water_Intake_Systems_Ecological_Aspects

Summary:

This chapter reviews the ecology of biofouling communities that develop in seawater cooling-water intake systems and the associated operational problems, including blockage, reduced hydraulic capacity, and corrosion. It distinguishes microfouling (biofilms) from macrofouling (e.g., mussels, barnacles, algae) and describes how hydrodynamics, temperature, and nutrient levels in intake tunnels, bays, and pipelines influence settlement patterns. Various engineering and operational controls, such as mechanical cleaning, chlorination, thermal treatment, and fouling-release coatings, are evaluated for effectiveness and limitations in intake environments. The authors link ecological understanding of fouling organism life cycles and attachment mechanisms to practical design and maintenance strategies for reliable cooling-water intakes.

Questions (aims):

1. How do ecological characteristics of biofouling communities in seawater intake and cooling systems inform the design of intake geometry and pre-treatment (screens, tunnels, bays) to limit macrofouling accumulation?
2. Which engineering control methods discussed in the chapter (e.g., mechanical cleaning, chlorination, coatings) are most applicable to seawater intake structures, screens, and pre-treatment skids to maintain capacity and reduce blockage risk?
3. How do hydrodynamic conditions and residence times in intake tunnels and piping affect the distribution and severity of macrofouling, and what implications does this have for locating screens, strainers, and flushing or CIP provisions?
4. In what ways can knowledge of fouling organism life cycles, settlement preferences, and growth rates be integrated into inspection intervals, maintenance planning, and material selection for seawater intake and pre-treatment equipment?

Answers (knowledge):

1. The chapter shows that biofouling intensity often increases in low-velocity regions, bends, and sheltered intake structures, implying that intake geometry should avoid stagnant zones, sharp turns, and dead legs where possible, and that screens should be located where flow conditions discourage heavy settlement.
2. Mechanical cleaning and high-pressure water jetting are highlighted as essential for removing established macrofouling in intake tunnels and screens, while continuous or intermittent chlorination can suppress early settlement but must be balanced against environmental discharge constraints.
3. The authors report that prolonged exposure and moderate flow favour dense fouling communities in intake conduits, suggesting that design should minimise residence time before coarse screening, provide access for periodic mechanical cleaning, and consider flushing strategies in sections prone to accumulation.
4. By relating recruitment and growth seasons of key fouling species to operational practice, the chapter recommends scheduling inspections and major cleaning during or after peak recruitment periods and selecting materials and coatings that resist both corrosion and biological attachment in long-term seawater exposure.

Impact:

- The ecological insight into where and how macrofouling develops in intake systems enables more targeted placement of screens, strainers, and access points, improving maintainability and reliability of seawater intakes.
- Evaluating multiple control options, mechanical, chemical, thermal, and coatings, helps designers choose a combined strategy suited to specific intake conditions, reducing unplanned outages and cleaning frequency.
- Linking biology with engineering design encourages integration of monitoring, inspection planning, and material/coating selection into early intake and pre-treatment design, rather than treating biofouling control as an afterthought.

Actions:

1. Use the chapter's discussion of flow-dependent fouling patterns to justify intake layouts that minimise low-velocity pockets and to define where inspection and cleaning access should be provided in your seawater intake and pre-treatment skid.
2. Select a combination of mechanical cleaning methods and, where appropriate, controlled chlorination or other biocides for your intake and screen design, aligning with the control strategies described in the chapter.
3. Incorporate knowledge of fouling organism seasonality into your maintenance plan by scheduling inspections and CIP or flushing operations around predicted peak fouling periods highlighted by the authors.
4. Draw on the chapter's recommendations on materials and coatings performance in fouled seawater environments when specifying alloys, linings, and fouling-release coatings for intake tunnels, screens, and pre-treatment equipment.

Journal Paper / Book Chapter / Article citation: Zhang, W., Che, J., Shu, H., Bai, Y., Liu, X., Jiang, W., Wu, J., & Chen, X. (2025). Stewart-inspired vibration isolation mechanisms for precision equipment: Current status and future research trend. *Mechanism and Machine Theory*, 214, 106153. Available at: <https://doi.org/10.1016/j.mechmachtheory.2025.106153>

Summary:

This paper reviews Stewart-inspired vibration isolation systems (SVIS) for precision equipment. It explains that unwanted vibration can reduce accuracy, accelerate fatigue failure, and even damage structures. The paper is useful for my WP6 because it reviews compact multi-DOF isolation systems that can be used on different carriers, including marine applications. It also covers dynamic modeling, parameter design, and passive/active/semi-active isolation methods. For my UDC work, this is mainly useful for the vibration-isolation side of the rack/chassis support system.

Questions (aims):

1. Why is vibration isolation important for precision equipment?
2. What is a Stewart-inspired vibration isolation system, and why is it useful?
3. What design factors affect vibration-isolation performance?
4. How can this paper help the design of the UDC rack/chassis in WP6?

Answers (knowledge):

1. Vibration isolation is important because unwanted vibrations can reduce performance accuracy, increase fatigue failure, and cause structural defects in precision equipment.
2. A Stewart-inspired vibration isolation system is a compact multi-DOF isolation mechanism based on the Stewart parallel mechanism. It is useful because it offers six-DOF isolation and structural compactness.

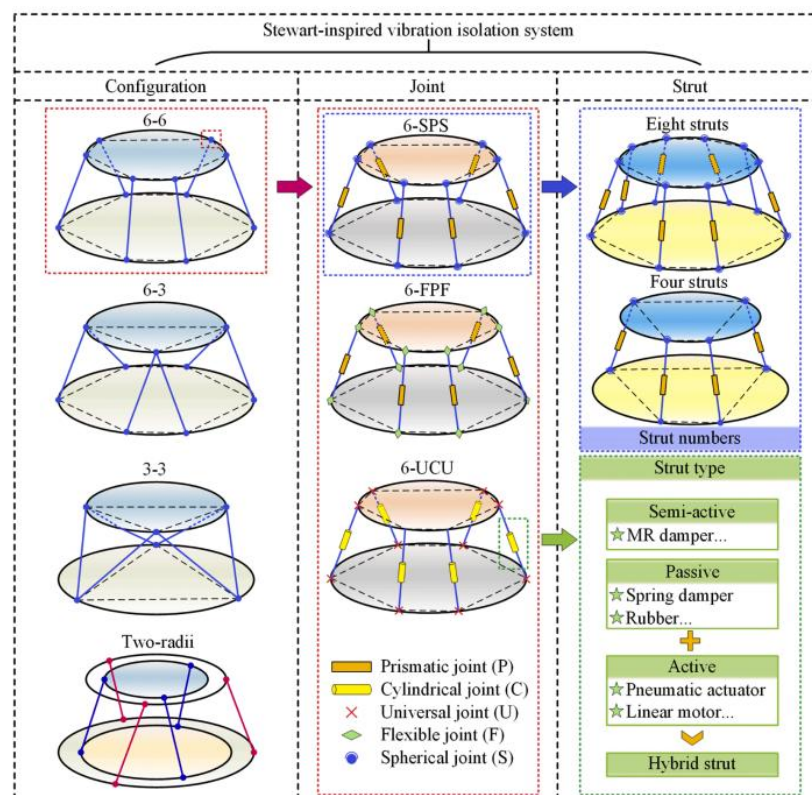


Figure 30. Structure and classification of the SVIS. Some representative configurations are demonstrated here.

3. The paper shows that performance depends on dynamic modeling, parameter design, configuration, joints, and struts. It also explains that the structure should be chosen based on the application scenario and the required isolation performance.

Table 123. Comparison of various modeling approaches.

References	Configuration	Method	Assumption	Consideration		Applicability	
				Joint Stiffness	Strut inertia	Asymmetrical configuration	Whole workspace
[97]	3-3 SPS	Kane	1.a, 1.c, 2, 3.a, 4.b, 5.a, 5.c	No	Yes	No	No
[99]	6-6 FFF	Transfer matrix	1. a, 2, 3. a, 4. a, 5. a, 5.c	Yes	Yes	Yes	No
[100]	6-6 SPS	Transfer matrix	2, 3, 4.c, 5. a, 5.c	Yes	Yes	Yes	Yes
[104]	6-6 UPS	Newton-Euler	1. a, 1.c, 2, 3. a, 4.b, 5	No	No	No	No
[105]	3-3	Newton-Euler	1. a, 2, 3. a, 4. a, 4.b, 5. a, 5.c	Yes	Yes	No	No
[109]	6-6	Lagrange	1. a, 2, 3. a, 4.b, 5	No	No	No	No
[110]	6-6 UPS	Lagrange	1. a, 2, 3. a, 4.b, 5. a, 5.c	No	Yes	No	No
[111]	6-6 FPF	Virtual work	1. a, 2, 3. a, 4. a, 5.c	Yes	Yes	Yes	No
[112]	6-6 UPU	Virtual work	1.b, 3, 5. a	No	Yes	Yes	Yes
[114]	3-3 SPS	Hamilton & virtual work	1. a, 1.c, 2, 3, 4.b, 5	No	No	No	No
[115]	3-3 FPF	Kane & virtual work	1. a, 1.b, 2, 3, 5. a, 5.c,	Yes	Yes	No	No
[117]	Multiple type	Screw theory	1. a, 3. a, 4.c, 5.c	Yes	Yes	Yes	No
[119]	Two radii	Frequency-domain modeling	1. a, 2, 3. a, 4.b, 5	No	No	No	No

4. This helps WP6 because the UDC rack/chassis will hold sensitive electronics, so the support system may need vibration/shock isolation, compact mounting, and careful structural design instead of being treated as only a simple frame.

Impact:

This paper is useful for my WP6 because it supports the vibration-isolation side of the UDC rack/chassis design. It shows that sensitive equipment can be affected by multi-directional vibration, so the support system should not be treated as only a simple frame. The paper also gives design support for considering shock/vibration isolation features, compact mounting, and careful structural layout. However, it is not specifically about subsea UDC systems, so it should be used as a supporting source rather than the only design source.

Actions:

1. Consider adding vibration-isolation features between the UDC frame and sensitive equipment.
2. Treat shock/vibration as a design requirement for the rack/chassis.
3. Use compact support geometry and proper mounting design for the UDC equipment.
4. Use this paper as support for why sensitive electronics need structural vibration protection.

Journal Paper / Book Chapter / Article citation: Aliaskari, M., Wezstein, J., Saravia, F., & Horn, H. (2024). A systematic analysis of operating parameters for CO₂ capture from seawater by Bipolar Membrane Electrodialysis. Separation and Purification Technology, 339, 126679. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S1383586624004180?via%3Dihub>

Summary: This article investigates the operating parameters that affect seawater CO₂ capture by bipolar membrane electrodialysis. It is useful for WP3 because it connects flow rate, salinity, electrical driving conditions and process performance. The study shows that the reactor housing must be designed around an operating window rather than one fixed point. For mechanical design, this means the cassette, nozzles, manifolds and service ports must support different operating modes, including normal operation, flushing and cleaning.

Questions (aims):

1. What operating variables affect the WP3 pH-swing stack?
2. How should the CAD model reflect an operating window rather than a single fixed duty?
3. What interfaces are needed between WP3 and the pumping or pre-treatment systems?
4. How does this source affect the sizing of the stack housing and ports?

Answers (knowledge):

1. Flow rate, salinity, voltage or current density, channel velocity and fouling tendency all affect the stack performance.
2. The CAD should include accessible nozzles, isolation space and service ports so the housing can operate, flush and be checked under more than one condition.
3. WP3 needs stable inlet flow from pre-treatment, a downstream process outlet, electrical power, pressure measurement and cleaning connections.
4. The housing should not be designed too tightly around a single value. The ports and side-loop should allow flow control, cleaning and future adjustment.

Impact: The source strengthens the assumptions table and helps explain why WP3 uses conservative screening values. It supports the inclusion of flow control, service access, pressure monitoring and bypass thinking. This makes the design more realistic for a seawater DOC module.

Actions:

1. State the design flow rate and operating envelope clearly in the report.
2. Use the operating window to justify the selected housing size and port layout.
3. Add pressure and flow monitoring to the PFD and P&ID-lite.
4. Keep the HFMC branch and stack cassette serviceable in the CAD model.
5. Avoid claiming one perfect operating point without further testing.
6. Include this completed worksheet in the project appendix.

Journal Paper / Book Chapter / Article citation: Abessi, O., & Roberts, P. J. W. (2017). Multiport diffusers for dense discharges. Journal of Hydraulic Engineering, 143(6), 04017012. [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0001279](https://doi.org/10.1061/(ASCE)HY.1943-7900.0001279)

Summary:

Abessi and Roberts tested multiport diffusers that release concentrated brine through angled ports. Their lab work showed that when neighbouring jets are too close together, they compete for the same ambient water and each one entrains less fresh water than it would alone. Dilution dropped 20-50% for closely spaced ports compared with an isolated jet. They also found that dense return flows on the seabed merge into a thicker layer when jets are packed tightly. The minimum spacing needed is three to four times the jet length scale ($d \cdot F$).

Questions (aims):

1. What port spacing is needed to stop neighbouring jets from interfering?
2. What happens to dilution performance when jets merge?
3. Are there design equations for predicting multiport dilution?

Answers (knowledge):

1. The ratio $s/(d \cdot F)$ must exceed 3-4 to keep jets independent. In physical terms, s/d should be roughly 15-25, depending on the Froude number. Below that, each jet recycles the other's mixed fluid instead of drawing in clean water.
2. Merged jets share the same entrainment pool, cutting dilution 20-50%. The dense return flows on the seabed also combine into a thicker and saltier layer, increasing the near-bed environmental impact.
3. $S_{\min} = c \times F \times f(s/(d \cdot F))$, where c is about 0.3-0.5, and f is a spacing correction that approaches 1.0 once $s/(d \cdot F)$ exceeds 3-4. At that point, single-jet formulas can be used directly.

Impact:

Port spacing is a controlling variable for diffuser performance. Cramming too many ports into a short pipe saves material but kills dilution. The spacing rules let us balance port count against total pipe length for cost-effective design.

Actions:

1. Calculate minimum port spacing using $s/(d \cdot F) > 3-4$.
2. Determine the total diffuser length needed for the target number of ports.
3. Evaluate the trade-off between port count, spacing, and overall cost.

Journal Paper / Book Chapter / Article citation: Towler, G. & Sinnott, R. (2013). *Chemical Engineering Design, 2nd ed.* Butterworth-Heinemann. Ch. 5: Pumping Systems & Ch. 11: Safety and Loss Prevention. Available at: <https://elmoukrie.com/wp-content/uploads/2022/06/chemical-engineering-design-principles-practice-and-economics-of-plant-and-process-design-gavin-towler-ray-sinnott-3edition.pdf>

Summary:

Towler & Sinnott's Chemical Engineering Design covers pump system design, the need for redundancy, control strategies, and valve selection. Chapter 5 deals with the arrangement of the pump system to ensure operability and maintainability. Chapter 11 deals with the safe isolation of the system, pressure relief in the system, and shutdown in the system. All this provides the basis for the safe and reliable design of the DOC seawater pump system.

Questions (aims):

1. What does N+1 redundancy mean, and how is it applied to the DOC pumping circuit?
2. What valve types are used for isolation, control, and non-return duties?
3. What is a minimum-flow bypass loop, and when is it needed?
4. What instrumentation is required to monitor and protect the pumps?
5. How should the system be arranged to allow maintenance without shutting down the DOC process?

Answers (knowledge):

1. N+1 redundancy. One duty pump runs normally; one standby pump sits ready and auto-starts if the duty pump fails. This ensures the DOC process continues without manual intervention.

2. Valve types.

Valve Type	Role	DOC Application
Gate / Ball valve	Isolation (on/off)	Pump suction and discharge isolation
Globe valve	Throttling / flow control	Bypass and flow regulation
Butterfly valve	Isolation or coarse control	Large-bore seawater headers
Swing check valve	Non-return / backflow prevention	Each pump discharge
Relief valve	Overpressure protection	Pump discharge header

3. Minimum-flow bypass. A centrifugal pump must never run at zero flow or it overheats and cavitates. A small bypass line from the pump discharge back to the suction vessel ensures a minimum flow is always maintained, particularly during start-up or when the main process valve is closed.

4. Instrumentation.

Instrument	Purpose
Pressure transmitter — suction	Detect low suction pressure / cavitation risk
Pressure transmitter — discharge	Confirm pump is delivering; detect blockage
Flow transmitter — discharge	Monitor flowrate; trigger standby on low flow
Temperature element — casing	Detect overheating from deadheading
Vibration switch — bearing	Early warning of bearing or impeller fault

5. Maintainability. Each pump should have isolation valves on suction and discharge, a check valve on the discharge, a drain valve on the casing, and a removable spool piece so it can be unbolted and lifted out without disturbing the pipework. At least 1 m clear access space around each pump is required for maintenance

Impact: The N+1 duty/standby arrangement, valve selection, minimum-flow bypass, instrumentation list, and maintainability features will all be incorporated into the WP2 P&ID and pump skid CAD model.
Actions: 1. Show duty/standby pump pair on the P&ID with auto-start logic on low pressure or low flow. 2. Select valve types for each location in the DOC circuit using the table above. 3. Add a minimum-flow bypass line with a restriction orifice or globe valve to the pump discharge.

Journal Paper / Book Chapter / Article citation: Lucas, F., & Sanjivy, K. (2024). *Sea Water Air Conditioning (SWAC) Technology: Performance and Worldwide Potential*. In *Air Conditioning Technologies and Applications*. IntechOpen. Available at: [intechopen.com/chapters/1192910](https://www.intechopen.com/chapters/1192910)

Summary:

This open-access book chapter reviews Sea Water Air Conditioning (SWAC), which uses cold seawater drawn from depth to provide building cooling. It reports that SWAC can achieve COP values from 25 to 150, depending on loop length and configuration. For underwater habitation, this is highly relevant because it directly supports the use of seawater as a heat sink for active cooling loops and habitat heat rejection.

Questions (aims):

1. How can seawater be used as a cooling source?
2. What are the thermal-performance benefits of seawater-based cooling?
3. How effective is deep seawater as a heat sink?

Answers (knowledge):

1. SWAC systems use seawater as a direct cooling source for high-efficiency thermal control.
2. Reported COP values are far higher than typical vapor-compression cooling in suitable conditions.
3. This makes seawater heat exchangers a strong concept choice for underwater habitats

Impact:

1. Gives a strong theoretical basis for seawater-based cooling.
2. Supports your choice of seawater heat exchangers.
3. Helps justify passive vs pumped-loop comparisons.

Actions:

1. Cite it when explaining why seawater is used for heat rejection.
2. Use it to support your heat-exchanger concept selection.
3. Reference it in discussions of energy-efficient habitat cooling.

Journal Paper / Book Chapter / Article citation: Ghaffour, N., Missimer, T. M., & Amy, G. (2013) Technical review and evaluation of the economics of water desalination: Current and future challenges for better water supply sustainability. Desalination, 309, 197–207. Available at: <https://doi.org/10.1016/j.desal.2012.10.015>

Summary:

This review paper discusses the design and operation of seawater desalination systems, emphasizing how intake configuration and pre-treatment strategy influence plant performance, fouling risk, and overall cost. It highlights typical pre-treatment trains for seawater reverse osmosis (SWRO), including coagulation–flocculation, sedimentation, media filtration, and membrane filtration to control suspended solids, organics, and biofouling before the RO stage. The authors analyse how intake type (open intake versus beach wells) strongly affects raw seawater quality, with open intakes generally requiring more robust pre-treatment due to higher turbidity and biological loads. They also stress the importance of monitoring indices such as the silt density index (SDI) to confirm pre-treatment effectiveness and maintain stable RO operation over the plant's lifetime.

Questions (aims):

1. How can insights from this review on intake type and raw seawater quality be used to justify the selection of an open intake with a bell-mouth or ducted arrangement versus alternative intake concepts for a DOC seawater intake skid?
2. Which pre-treatment steps described in the paper (e.g., coagulation-flocculation, sedimentation, media filtration, membrane filtration) are most relevant to designing an effective coarse screening and pre-treatment system ahead of downstream equipment?
3. How does the paper's discussion of fouling, SDI control, and RO performance inform the hydraulic design, head-loss budgeting, and allowable approach velocities across intake and pre-treatment components?
4. In what ways can the paper's recommendations on long-term plant reliability and operational costs influence choices of materials, corrosion protection, and maintainability features such as flushing, CIP, and access provisions on a seawater intake and pre-treatment skid?

Answers (knowledge):

1. The review explains that open seawater intakes typically experience higher levels of suspended solids and biological activity than beach wells, which justifies designing robust intake geometries and upstream screening to manage variable debris and biofouling loads and protect downstream pre-treatment and RO units.
2. It identifies conventional pre-treatment processes, such as coagulation-flocculation followed by sedimentation and granular media filtration, as proven methods to reduce turbidity and particulate matter, often supplemented by membrane filtration in high-performance SWRO plants, providing a framework for layering coarse screening with subsequent finer pre-treatment steps.
3. The authors emphasise that inadequate pre-treatment leads to rapid RO fouling and increased cleaning frequency, underscoring the need to control SDI and manage head losses so that pumps operate efficiently while maintaining sufficient driving pressure and avoiding excessive energy consumption.
4. By linking material and process choices to life-cycle cost, the paper supports specifying corrosion-resistant materials for intake and pre-treatment equipment and integrating CIP, flushing, and easy access for maintenance to sustain performance and reduce operational expenses over the plant's service life.

Impact:

- The paper provides a high-level framework connecting intake configuration, seawater quality, and pre-treatment design to long-term SWRO plant reliability and cost, which can be adapted to a DOC intake and skid concept.

- Its discussion of pre-treatment schemes and fouling control helps justify multi-stage screening and filtration strategies that protect downstream membranes or heat-transfer surfaces.
- The emphasis on SDI monitoring, materials selection, and operational reliability encourages designers to integrate hydraulic design, corrosion protection, and maintainability features from the outset of the intake and pre-treatment skid design.

Actions:

1. Use the review's comparison of intake options and seawater quality impacts to justify your choice of open intake geometry and associated coarse screening requirements in the design basis for the DOC skid.
2. Map your planned pre-treatment sequence (coarse screening, strainers, filters) against the typical trains described in the paper to ensure adequate removal of particulates and fouling precursors before downstream treatment.
3. Incorporate SDI targets and fouling considerations from the paper into your hydraulic design and head-loss budget, ensuring approach velocities and pressure margins are compatible with stable long-term operation.
4. Reflect the paper's life-cycle perspective in your materials and maintenance philosophy by specifying corrosion-resistant components and integrating CIP, flushing, and access features that support efficient operation over the plant's design life.

Journal Paper / Book Chapter / Article citation: Liu, S., Liu, Y., Wang, J., Pan, C., Zhao, X., Su, X., Zhang, W., Qiao, Z., Zhou, F., & Qiao, G. (2026). Underwater acoustic release: A comprehensive review and HAR11000 designed for the Mariana Trench. Ocean Engineering, 343, 123156. Available at: <https://doi.org/10.1016/j.oceaneng.2025.123156>

Summary:

This paper reviews underwater acoustic releases, which are remote-control mechanical release devices used for the deployment and retrieval of marine instruments and equipment. The paper explains the working principle, common communication methods, common release mechanisms, and recent industrial developments. It also presents the HAR11000 ultra-deep acoustic release for the Mariana Trench, including its release capacity, operating depth, and successful deployment/recovery missions. This is useful for my WP6 mainly because it supports the deployment/retrieval part of the UDC system and shows how release and recovery can be designed more reliably in underwater conditions.

Questions (aims):

1. Why are underwater acoustic releases important in marine systems?
2. What main release mechanisms are commonly used?
3. What design features improve underwater release reliability?
4. How can this paper help the deployment/retrieval part of WP6?

Answers (knowledge):

1. Underwater acoustic releases are important because they are used for the deployment and retrieval of marine instruments and equipment through remote acoustic control. They are widely used in marine monitoring, scientific projects, and underwater networks.

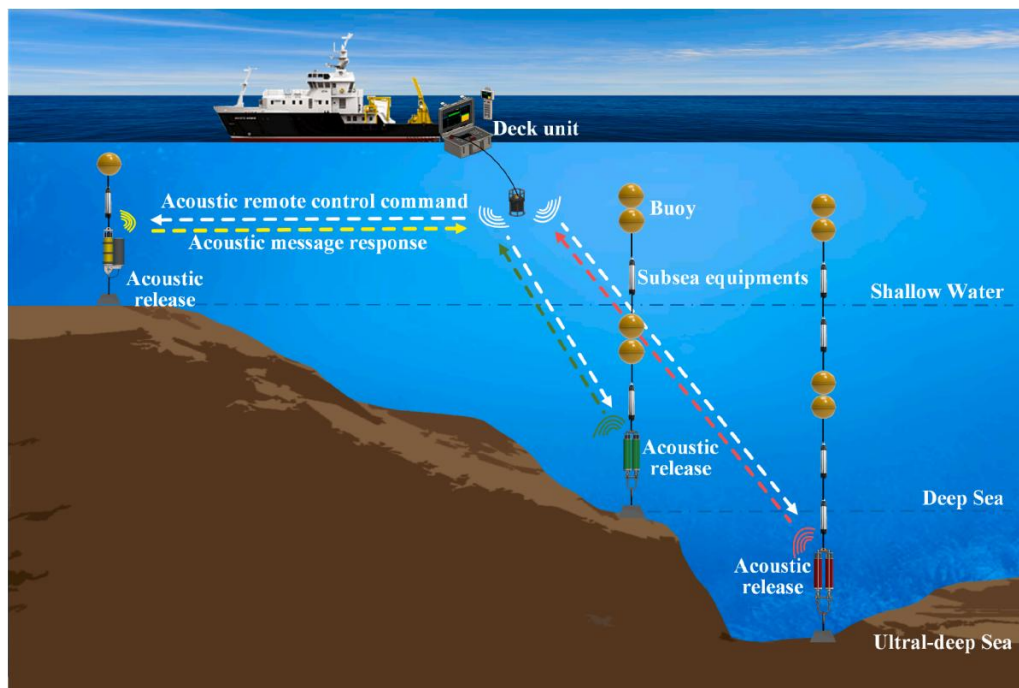


Figure 31. Acoustic release and working scenario.

2. The paper shows that common release mechanisms include motor-driven pelican hooks, motor-driven push-off systems, burn wires, and electrolytic erosion. It compares them in terms of reliability, load capacity, release speed, complexity, and application level.

Table 124. Release mechanisms for underwater acoustic release.

Release mechanism	Reliability	Load capacity	Release speed	Complexity	Application level
Motor-driven Pelican Hook	high	high	medium	medium	popular
Motor-driven Push-off	high	medium	medium	medium	popular
Burn Wire	medium	low	fast	low	less popular
Electrolytic Erosion	medium	low	slow	low	less popular

3. The paper explains that reliability can be improved by using heavy-duty release frames, mooring recovery systems, parallel or series redundant structures, and integrated monitoring/communication functions. It also shows that different products are designed for different depths and load capacities.
4. This helps WP6 because the UDC will need a safe deployment/retrieval approach, and the paper gives useful ideas about release/recovery strategy, load handling, remote release, and redundancy. It does not design the UDC base frame directly, but it supports the retrieval and release side of the system.

Impact:

This paper is useful for my WP6 because it supports the deployment and retrieval side of the UDC concept. It shows that underwater recovery systems need reliable release mechanisms, suitable load capacity, and redundancy for safe operation. It also gives practical examples of industrial products and real deep-sea use, which makes it useful for understanding how underwater recovery can actually be done. However, it is not specifically about UDC rack/chassis or detailed base-frame structural design, so it should be used as a supporting source rather than the main design source.

Actions:

1. Consider a reliable release/retrieval method as part of the UDC deployment concept.
2. Consider redundancy or fail-safe ideas if recovery is critical.
3. Use the paper as support for load handling and underwater retrieval planning.
4. Do not rely on this paper alone for the detailed frame/CAD design.

Journal Paper / Book Chapter / Article citation: Rivero, J., Lieber, A., Snodgrass, C., Neal, Z., Hildebrandt, M., Gamble, W., & Hornbostel, K. (2023). Demonstration of direct ocean carbon capture using hollow fiber membrane contactors. Chemical Engineering Journal, 470, 143868. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S1385894723025998>

Summary: This paper is important for the optional HFMC part of WP3. It demonstrates that hollow fiber membrane contactors can be used in a direct ocean capture system to remove CO₂ from seawater-related streams. For the mechanical design, the HFMC cannot be treated as a small detail. It needs isolation valves, bypass control, drain and purge access, integrity checking and a layout that does not block the main reactor cassette from being removed. This supports the side-loop module shown beside the main reactor housing in the CAD.

Questions (aims):

1. Why is an HFMC side-loop relevant to WP3?
2. What hardware is needed around the HFMC module?
3. How should the side-loop be arranged so it remains maintainable?
4. What should be avoided when adding the HFMC to the reactor skid?

Answers (knowledge):

1. It is relevant because the WP3 brief allows a membrane or contactor if used, and the HFMC provides a route for CO₂ transfer from the pH-swing stream.
2. The module needs inlet and outlet isolation, a bypass line, drain and purge route, pressure taps and cleanable connections.
3. It should sit as a compact side-loop next to the main housing, with clear access to valves and no obstruction of the front cassette route.
4. The side-loop should not block service ports, lifting points, the removable closure or the cassette pull-out path.

Impact: This source supports keeping the HFMC as an optional, bypassable and isolated module in the CAD and PFD. It helps justify the small secondary cylinder in the model, the valve locations and the drain or purge valve between the small pipe sections.

Actions:

1. Show the HFMC as a separate side-loop module in the CAD assembly.
2. Add two isolation valves and one bypass route around the HFMC module.
3. Include a drain and purge valve in the side-loop schematic.
4. Keep the HFMC module removable without dismantling the main pressure housing.
5. Use this paper to justify the optional contactor in Table 4 and Figure 2.
6. Include this completed worksheet in the project appendix.

Journal Paper / Book Chapter / Article citation: Lai, C. C. K., & Lee, J. H. W. (2018). Initial mixing of inclined dense jet in perpendicular crossflow. *Environmental Fluid Mechanics*, 18(4), 985–1014. <https://doi.org/10.1007/s10652-018-9583-z>

Summary:

Lai and Lee studied what happens when an inclined dense jet enters water that has a sideways current. Using 60-degree jets at various crossflow speeds with laser fluorescence, they found that even a weak current improves mixing dramatically. A crossflow ratio (u_a/v_{port}) of just 0.05 roughly doubled the dilution measured in still water, and a ratio of 0.1 increased it three- to four-fold. The current pushes the jet sideways, giving it more surface area to mix with clean water and keeping the return flow away from the diffuser.

Questions (aims):

1. How much does a crosscurrent improve dense jet dilution?
2. What current speed should be selected for conservative design?
3. What predictive formulas were developed?

Answers (knowledge):

1. Even weak currents help substantially. By bending the jet sideways, the current exposes more jet surface to clean ambient water and boosts entrainment. It also stops the dense return plume from falling back onto the diffuser, reducing re-entrainment.
2. The key parameter is $u_r = u_a/v_{port}$. At $u_r = 0.05$ dilution roughly doubles; at 0.1 it triples or more. The authors recommend using the 10th-percentile current speed from site data as the conservative design input.
3. $S_{min} = a \times F^b \times u_r^c$, with fitted constants from experiments. They also parameterised the lateral displacement of the plume, so both dilution and footprint can be estimated from site data.

Impact:

Site currents will help mixing, which is good news. However, actual current measurements are essential. Using the 10th-percentile speed is a smart conservative approach. The lateral shift must also be mapped relative to any sensitive habitats nearby.

Actions:

1. Collect current speed statistics for the site, including the 10th-percentile value.
2. Apply the crossflow dilution correlations for conservative design estimates.
3. Map the predicted lateral plume footprint relative to sensitive environments.

Journal Paper / Book Chapter / Article citation: Wylie, E.B. & Streeter, V.L. (1993). *Fluid Transients in Systems*. Prentice Hall. & Chaudhry, M.H. (2014). *Applied Hydraulic Transients*, 3rd ed. Springer. Available at: <https://zeppvargas.wordpress.com/wp-content/uploads/2014/08/applied-hydraulic-transients-chaudhry.pdf>

Summary:

Wylie & Streeter's *Fluid Transients in Systems* is the foundational reference on hydraulic transients. It covers pressure wave propagation from sudden flow changes (water hammer), the Joukowsky equation for peak pressure rise, and practical mitigation strategies. Chaudhry's *Applied Hydraulic Transients* adds worked design examples. Together, they support the transient risk assessment for the DOC seawater circuit on pump trip or emergency valve closure.

Questions (aims):

1. What is water hammer, and when does it occur in the DOC circuit?
2. How is the peak transient pressure rise estimated?
3. What is the critical valve closure time?
4. What mitigation measures suit a compact subsea skid?
5. What risk arises on a sudden pump trip, and how is it managed?

Answers (knowledge):

1. Water hammer. A pressure surge from a sudden velocity change, rapid valve closure, or pump trip. Fluid kinetic energy converts to a pressure wave that travels through the pipe, risking joint leaks or check valve slam.

2. Pressure rise. Joukowsky equation: $\Delta P = \rho \times a \times \Delta V$, where $a \approx 1200\text{--}1400$ m/s for steel pipe. Must be checked against the pipe pressure rating.

3. Critical closure time. If a valve closes faster than $2L/a$ (wave return time), the full pressure spike occurs. Closing slower than this threshold reduces the transient significantly.

4. Mitigation measures.

Mitigation	How it Helps	DOC Recommendation
Slow-closing actuated valves	Extends closure beyond the critical time	≥ 5 s closure on all ESD valves
Non-slam check valves	Close gently; limit backflow on trip	On each pump discharge
Pressure relief valve	Caps peak pressure at the set point	Discharge header at $1.1 \times$ operating pressure
Surge accumulator	Absorbs wave energy in a gas bladder	Consider if pipe runs exceed 10 m

5. Pump trip risk. A rapid trip can drop suction pressure below vapour pressure, causing column separation. When columns rejoin, the impact exceeds normal Joukowsky values. Prevented by good NPSHA margins and non-slam check valves.

Impact:

Joukowsky equation used to screen transient pressure against pipe rating. Slow-closing ESD valves (≥ 5 s), non-slam check valves, and a PRV on the discharge header form the mitigation package — all shown on the WP2 P&ID.

Actions:

1. Apply the Joukowsky equation at design velocity; confirm within the pipe pressure rating.
2. Specify ESD valve closure ≥ 5 s; verify this exceeds $2L/a$ for the DOC pipework.
3. Specify non-slam check valves on each pump discharge; add to P&ID.

Journal Paper / Book Chapter / Article citation: Sanjivy, K., Marc, O., Davies, N., & Lucas, F. (2023). *Energy performance assessment of Sea Water Air Conditioning (SWAC) as a solution toward net zero carbon emissions: A case study in French Polynesia*. *Energy Reports*, 9, 437–446. Available at: [sciencedirect.com/science/article/pii/S2352484722025884](https://www.sciencedirect.com/science/article/pii/S2352484722025884)

Summary:

This paper presents a real-world 2.4 MW SWAC system and analyzes 3 years of data. It reports that the system achieved COP values of 20 to 150, depending on the secondary loop, and significantly reduced emissions compared with conventional cooling. For your habitat, this is a strong practical reference for sizing and justifying seawater heat rejection.

Questions (aims):

1. How well does seawater cooling perform in practice?
2. What real-world efficiency can be achieved using seawater thermal sinks?
3. Can seawater cooling significantly reduce conventional cooling demand?

Answers (knowledge):

1. The paper demonstrates long-term real-world operation of a seawater cooling system.
2. It shows very high thermal performance compared with conventional cooling systems.
3. For underwater habitats, it supports the practicality of seawater-based thermal rejection rather than relying solely on conventional HVAC logic.

Impact:

1. Adds real operational evidence to your design section.
2. Supports energy-efficient heat rejection.
3. Strengthens your justification for seawater cooling as a feasible solution.

Actions:

1. Use this paper when discussing thermal efficiency and heat-rejection concept selection.
2. Reference its real-world performance when comparing options.
3. Cite it in your heat-balance and thermal-management rationale.

Journal Paper / Book Chapter / Article citation: Zarzo, L., & Campos, E. (2011). Modern experience in large desalination plants: The Ras Al-Khair project. *Desalination and Water Treatment*, 36(1-3), 214-227. Available at: <https://ida.memberclicks.net/assets/docs/1023.1.pdf>

Summary:

This paper details the engineering of large-scale seawater intake systems for desalination plants, focusing on the Ras Al-Khair facility with multiple intake pipelines and screening stations. It describes offshore intake risers using velocity cap arrangements with bar screens to prevent marine organism ingress while maintaining hydraulic efficiency. The design includes hydraulic modeling to verify flow distribution across parallel intake pipes under clean and fouled conditions, ensuring adequate pump suction head and minimal air entrainment.

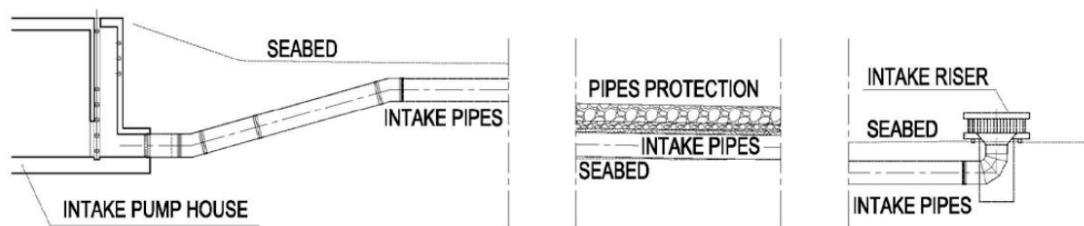


Figure 32. Intake system scheme - From left to right, pump-house, intake pipes, and raiser.

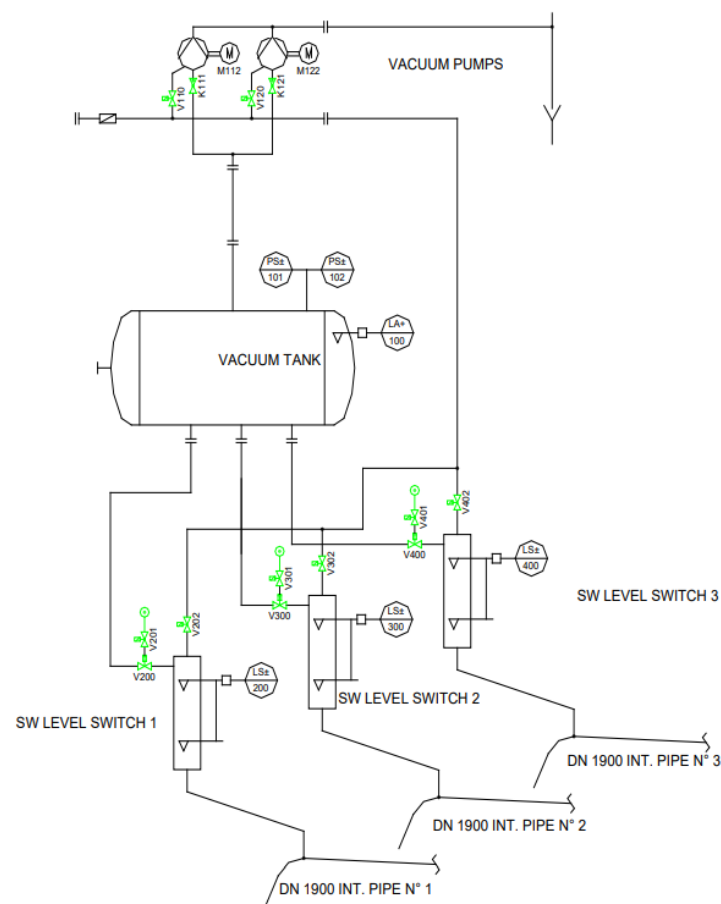


Figure 33. Air extraction schematic P&ID.

Questions (aims): 1. How should intake riser geometry and screening be configured to balance marine life protection with debris management in high-flow desalination intakes? 2. What hydraulic modeling approach ensures even flow distribution across multiple parallel intake pipes serving a common pump house? 3. How can intake systems accommodate both clean and biofouled conditions while maintaining NPSH margins at the pump station? 4. What construction and material strategies support long-term reliability of submerged intake components in aggressive seawater environments?
Answers (knowledge): 1. Velocity cap risers with bar screens at intake heads effectively mitigate large organism impingement while smooth pipe geometry reduces head losses to the pump house. 2. Physical hydraulic modeling of pump house and intake basin geometry validates water levels and flow uniformity when operating multiple intake lines in parallel. 3. Pipe sizing accounts for 20-30% additional head loss due to biofouling, ensuring NPSH available exceeds pump requirements across tide ranges and fouling scenarios. 4. Concrete pipe protection with armour stone, combined with chemical dosing and air extraction systems, maintains long-term structural integrity and operational reliability.
Impact: - Proven scalability: The Ras Al-Khair design validates multi-pipe intake configurations for capacities exceeding 50,000 m³/h, directly applicable to skid-based systems. - Fouling resilience: Incorporating worst-case biofouling losses in hydraulic design prevents capacity degradation and pump cavitation issues. - Maintenance integration: Screening stations and air extraction features enable reliable operation with planned cleaning intervals.
Actions: 1. Reference the paper's velocity cap and bar screen design for your intake riser specification, adapting pipe diameters to your design flow requirements. 2. Conduct similar hydraulic modeling of your intake-to-pump path, including fouling factors, to verify NPSH margins and flow uniformity. 3. Specify armour stone protection and chemical dosing provisions matching the paper's approach for your site's seabed conditions. 4. Integrate air extraction systems at intake pipe inversions as shown in the paper's scheme to prevent pump air locking during startup.

Journal Paper / Book Chapter / Article citation: Gao, H., Xu, H., Ma, Y., Liu, Y., & Liang, X. (2025). Simulation study on the hydrodynamic characteristics and stability of an Open-Frame underwater towed body. Ocean Engineering, 337, 121902. Available at: <https://doi.org/10.1016/j.oceaneng.2025.121902>

Summary:

This paper studies a novel open-frame underwater towed body using CFD to understand its hydrodynamic performance and motion stability. The authors investigate how pitch angle and towing speed affect drag, lift, lateral force, and overall stability. The paper shows that the open-frame layout causes complex internal flow interactions, and these can create asymmetric turbulence and unstable lateral forces. This is useful for my WP6 because my work includes the UDC support/base-frame around electronics, so understanding how an open-frame structure behaves in water helps with thinking about frame stability, layout, and safe operation underwater.

Questions (aims):

1. Why is hydrodynamic stability important for an open-frame underwater structure?
2. How does the open-frame layout affect the surrounding flow field?
3. What factors increase instability in the structure?
4. How can this paper help the base-frame stability part of WP6?

Answers (knowledge):

1. Hydrodynamic stability is important because unstable motion can reduce performance, affect payload operation, and make it hard to keep a safe straight-line trajectory underwater. The paper says this is especially important for systems carrying sensitive equipment.
2. The open-frame layout creates complex fluid-structure interactions inside and around the frame. The paper shows that the head structure and internal frame members can disturb the flow and create asymmetric turbulence, which then affects the downstream flow.
3. The paper shows that larger pitch angles and higher speeds increase instability. These conditions increase lateral hydrodynamic force and make the wake more uneven, which can stop the structure from keeping a stable linear trajectory.

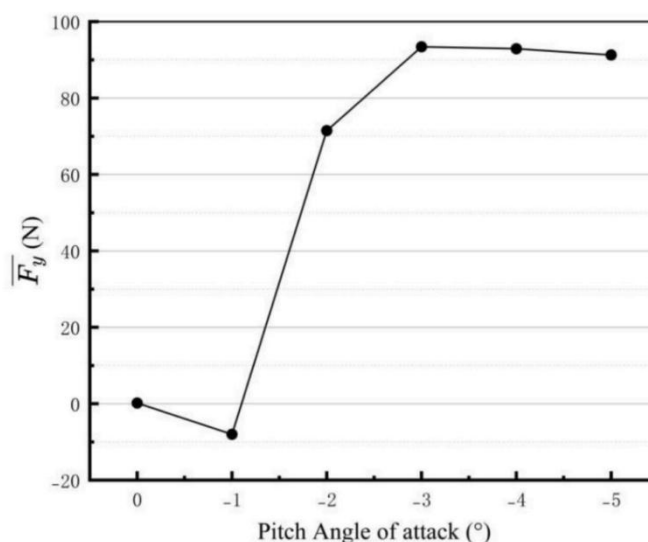


Figure 34. Variation of mean lateral force with pitch angle at speed of 3.09 m/s.

4. This helps WP6 because the UDC base/support frame should not only hold equipment, but also stay stable in water. The paper supports the idea that frame geometry, internal layout, and exposed structure can all affect stability, so the support frame should be designed carefully.

Impact:

This paper is useful for my WP6 because it supports the base-frame stability side of the UDC support design. It shows that an open-frame structure in water can suffer from unstable lateral forces if the geometry and internal layout are not designed well. It also gives support for thinking carefully about frame arrangement, exposed members, and hydrodynamic behaviour rather than treating the support frame as only a simple static structure. However, it is not specifically about a UDC frame, so it should be used as a supporting source rather than the only design source.

Actions:

1. Consider hydrodynamic stability as a design requirement for the UDC support/base frame.
2. Avoid frame layouts that may create strong asymmetric flow and unstable wake effects.
3. Use a smoother and more stable structural arrangement where possible.
4. Use this paper as support for why frame geometry and layout matter in underwater operation.

Journal Paper / Book Chapter / Article citation: Sharifian, R., van der Wal, H. C., Wagterveld, R. M., & Vermaas, D. A. (2023). Fouling management in oceanic carbon capture via in-situ electrochemical bipolar membrane electrodialysis. Chemical Engineering Journal, 458, 141407. Available at: <https://www.sciencedirect.com/science/article/pii/S1385894723001389?via%3Dihub>

Summary: This paper focuses on fouling management in seawater BPMED operation, which is central to the WP3 mechanical design. The study shows that carbonate scaling and fouling can change voltage, pressure drop and recovery behaviour. For WP3, the main lesson is that cleaning cannot be added after the CAD is finished. The reactor housing needs CIP and flush ports, drain points, pressure monitoring and a cassette replacement route. The design must also avoid trapping gas or stagnant seawater in the stack.

Questions (aims):

1. Why does WP3 need CIP and flush connections?
2. What measurements can show that fouling is increasing?
3. Should the stack be cleaned in place or replaced as a cassette?
4. How does fouling affect the housing and service design?

Answers (knowledge):

1. CIP and flush connections are needed because seawater BPMED can form scale and deposits that reduce performance and increase pressure drop.
2. Differential pressure, voltage rise, flow reduction and unusual drain or leak readings can indicate fouling or blockage.
3. Both are needed. Cleaning supports normal maintenance, while cassette removal supports deeper repair or replacement.
4. Fouling requires accessible service ports, low-point drains, clear pressure taps and enough front access to remove the cassette.

Impact: The source directly supports the WP3 maintainability features. It explains why the CAD must show vent, CIP, drain, integrity-test and sensor features, not only the main pressure shell. It also supports the decision to use a removable cassette instead of a fully sealed non-serviceable unit.

Actions:

1. Keep CIP and flush ports visible on the reactor housing.
2. Include bottom drain and leak collection in the PFD.
3. Add differential pressure monitoring across the stack.
4. Keep a front cassette route free of pipework and supports.
5. Add fouling and cleaning as a maintainability point in the report.
6. Include this completed worksheet in the project appendix.

Journal Paper / Book Chapter / Article citation: Missimer, T. M., & Maliva, R. G. (2018). Environmental issues in seawater reverse osmosis desalination: Intakes and outfalls. *Desalination*, 434, 198–215. <https://doi.org/10.1016/j.desal.2017.07.012>

Summary:

Missimer and Maliva reviewed global regulations and environmental concerns around brine discharge. They compared frameworks in Australia, the United States, and Europe, looking at salinity limits, chemical constituents of concern, and monitoring requirements. The central finding is that there is no single international standard. Allowable salinity increases range from +1 ppt to +5% above background, depending on the jurisdiction. They also stress that brine contains more than just salt, antiscalants, residual chlorine, and trace metals, all of which need assessment.

Questions (aims):

1. What salinity thresholds do different regions apply at the mixing zone boundary?
2. Is there a unified international standard for brine discharge?
3. What other brine constituents require environmental assessment?

Answers (knowledge):

1. Australia typically uses +1 ppt above ambient at 50–100 m from the outfall. Mediterranean countries often apply +5% above background salinity. Near protected habitats such as seagrass, the limit can be as tight as +0.5 ppt. The exact number depends on local rules and ecology.
2. No universal standard exists. Each country has its own framework, the US EPA, the Australian EPA, and the EU Water Framework Directive, all with different requirements. The applicable regulations must be identified at the start of any project.
3. Brine can contain antiscalants, coagulants, residual chlorine, heavy metals leached from plant equipment, and water with reduced dissolved oxygen. Each substance needs to be checked against local water quality criteria separately.

Impact:

Identifying the regulatory framework early is a prerequisite for setting dilution targets. This paper provides a useful range of thresholds while we confirm the specific rules for our site. It also reminds us to characterise the full chemical make-up of the discharge.

Actions:

1. Determine the salinity threshold applicable at the project location.
2. Define the permitted mixing zone dimensions under local regulations.
3. Compile all brine constituents and check each against water quality limits.

Journal Paper / Book Chapter / Article citation: Becht, C. (2009). *Process Piping: The Complete Guide to ASME B31.3, 3rd ed.* ASME Press. Available at: <https://doi.org/10.1115/1.802861>

Summary:

It covers pipe sizing, manifold layout, support spacing, thermal expansion, and vibration isolation. It explains how piping loads transfer to the structure and how to design systems that stay leak-free and maintainable, directly applicable to the DOC pump skid.

Questions (aims):

1. How should a manifold be sized and laid out for parallel pumps?
2. What pipe support types are used, and how is spacing determined?
3. How is thermal expansion in the piping accommodated?
4. How is pump vibration isolated from the structure and pipework?
5. What flexible connections are used at pump nozzles?

Answers (knowledge):

1. Manifold layout. Keep the header velocity below 1.5 m/s for even pressure at each branch. Space branches evenly and fit an isolation valve on each so pumps can be taken offline independently.

2. Pipe supports.

Support Type	Function	DOC Skid Use
Resting/sliding	Carries weight; allows axial movement	Straight horizontal runs
Anchor	Fixes the pipe fully in all directions	Either side of the expansion loops
Guide	Allows axial movement; prevents sway	Long runs between anchors
Clamp + elastomer liner	Weight and vibration damping	Near pump nozzles

Spacing guide: DN50 pipe, support every 2.5-3 m; DN100, every 3.5-4.5 m.

3. Thermal expansion. Restrained expansion builds stress that can crack welds. Accommodate with pipe loops (U- or L-bends) or axial bellows where space is tight.

4. Vibration isolation. Anti-vibration mounts (spring-elastomer pads) under the pump baseplate stop vibration from reaching the skid structure. Elastomer-lined clamps on nearby pipework dampen any residual vibration.

5. Flexible nozzle connections. Bellows at suction and discharge nozzles absorb pipe loads and vibration, protect impeller alignment, and allow pump removal without disturbing adjacent pipework.

Impact:

Manifold header velocity kept below 1.5 m/s. Support types from the table are applied to the piping layout. Expansion loops or bellows are included in the skid design. Anti-vibration mounts and flexible bellows are shown on the CAD model and P&ID.

Actions:

1. Size manifold header for <1.5 m/s; fit isolation valves on each branch.
2. Select support types from the table; space within diameter-based limits.
3. Calculate thermal expansion ($\alpha \cdot L \cdot \Delta T$); add pipe loops or bellows to the skid layout.

Journal Paper / Book Chapter / Article citation: Periola, A.A., Alonge, A.A., & Ogudo, K.A. (2022). *Heat Wave Resilient Systems Architecture for Underwater Data Centers*. Scientific Reports, 12, 17161. Available at: [nature.com/articles/s41598-022-21293-2](https://www.nature.com/articles/s41598-022-21293-2)

Summary:

This open-access paper studies submerged systems that rely on surrounding ocean water for cooling and shows that marine heat waves can reduce the availability of cold water for heat rejection. The authors propose reservoir-based mitigation, improving operational duration by about 5.5–12.3% or 5.2–11.5% depending on the scenario. This is highly useful for underwater habitats because it directly addresses contingency planning for when external seawater is warmer than expected.

Questions (aims):

1. What happens when seawater cooling conditions become less favorable?
2. How can submerged systems remain thermally stable during ocean warming events?
3. What contingency strategies improve cooling resilience?

Answers (knowledge):

1. Submerged systems that depend on seawater cooling are vulnerable to ocean warming and marine heat waves.
2. Thermal resilience can be improved by including stored cold-water or reservoir-assisted cooling strategies.
3. For underwater habitats, this supports adding redundancy and contingency to the thermal-management loop

Impact:

1. Supports fault-contingency planning in your thermal section.
2. Helps justify redundancy and buffering.
3. Adds depth to your verification and resilience discussion.

Actions:

1. Cite this when discussing thermal-management contingency.
2. Use it to justify backup cooling or thermal buffering.
3. Mention it in your failure-handling section.

Journal Paper / Book Chapter / Article citation: Missimer, T. M., & Maliva, R. G. (2018). Environmental issues in seawater reverse osmosis desalination: Intakes and outfalls. *Desalination*, 434, 198–215. Available at: https://www.increa.eu/gestor/recursos/archivos/sea_water_intake_structures.pdf

Summary:

This paper evaluates intake technologies for seawater reverse osmosis plants, comparing open ocean intakes, subsurface intakes, and beach wells based on water quality, environmental impact, and cost. Open intakes use velocity caps or screened risers to limit marine life entrainment, while coarse bar screens at intake heads block large debris. Subsurface options like infiltration galleries naturally filter particulates and reduce biofouling loads before reaching onshore pre-treatment skids. The study emphasizes hydraulic design criteria, including approach velocities below 0.15 m/s, screen openings sized for local debris, and head loss budgets that maintain pump NPSH margins.



Figure 35. Most habitual types of direct intake structures on the seabed.

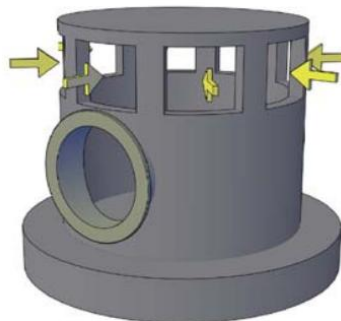


Figure 36. Recommended entry flow.

Questions (aims):

1. Which intake configuration best balances capital cost, water quality improvement, and environmental compliance for a coastal seawater pre-treatment skid?
2. How should coarse screening and strainer sizing account for local marine debris profiles and biofouling seasonality?
3. What hydraulic criteria ensure adequate NPSH margins across intake screens, piping, and pump suction under tidal variation and fouling conditions?
4. Which material combinations and maintenance features optimize long-term performance of submerged intake screens and pre-treatment structures?

Answers (knowledge):

1. Velocity cap intakes with 50-100 mm bar spacing effectively exclude fish and macro-debris while maintaining acceptable through-screen velocities of 0.3-0.5 m/s.

2. Dual parallel screen channels with isolation valves enable continuous operation during cleaning, with backwash systems sized for 10-15% of design flow.
3. Head loss calculations include clean screen losses (0.1-0.3 m), fouled conditions (up to 1.5 m), and pipe friction to verify NPSHa exceeds NPSHr by 1-2 m across operating cases.
4. 90-10 or 70-30 Cu-Ni screens combined with HDPE or GRP intake piping, plus epoxy-coated steel structures, minimize corrosion while silicone fouling-release coatings reduce cleaning frequency.

Impact:

- Water quality gains: Proper intake screening reduces suspended solids by 70-90%, easing downstream pre-treatment burden and extending equipment life.
- Reliability focus: Designs incorporating fouling factors prevent capacity loss and cavitation, ensuring stable plant operation year-round.
- Maintenance optimization: Clean-in-place systems and material choices cut diver interventions by 50-70% compared to unprotected carbon steel screens.

Actions:

1. Select velocity cap intake geometry based on the paper's debris exclusion data, adapting bar spacing to your site's marine growth profile.
2. Size coarse screens using the recommended velocity limits and include dual-channel redundancy with isolation for safe maintenance.
3. Build head loss budget per paper guidelines (clean + 3x fouling factor) and verify NPSH margins for your pump selection.
4. Specify Cu-Ni screens with HDPE piping and fouling-release coatings as validated by the study's material performance analysis.

Journal Paper / Book Chapter / Article citation: Ma, Y., Vayakis, G., Carmona, J.-M., Pascual, Q., Walton, R., & Walsh, M. (2019). Design and development of the mechanical support structure for ITER in-vessel magnetic sensors. *Fusion Engineering and Design*, 146, 887–893. Available at: <https://doi.org/10.1016/j.fusengdes.2019.01.106>

Summary:

This paper presents the design and development of a modular mechanical support structure for ITER in-vessel magnetic sensors. The platform was designed to securely support the sensor, provide an electrical connection with service cabling, improve heat conduction, protect the sensor, and allow replacement by remote handling. The final design includes a permanent base and a removable section, alignment pins, a connector box, cable clamping, and anti-vibration/self-locking features. This is useful for my WP6 because it shows how a support structure around sensitive equipment can be designed with attention to mounting, connector integration, cable support, replacement, and practical assembly. The paper also mentions continued testing for mechanical vibration and cable clamping, which makes it even more relevant to WP6.

Questions (aims):

1. Why is a dedicated mechanical support structure needed for sensitive equipment?
2. What design features were used to improve mounting, alignment, and replacement?
3. How does the paper deal with cable support, connector integration, and strain relief?
4. How can this paper help the design of the UDC support/rack system in WP6?

Answers (knowledge):

1. A dedicated support structure is needed to securely hold the equipment during operation, provide electrical connection, protect it from the environment, and allow replacement when required. The paper shows that support structures are not only for holding components, but also for integration, protection, and maintainability.
2. The design uses a modular platform with permanent and removable sections, alignment pins, captive bolts, a connector box, and remote-handling compatibility. The paper also uses self-locking anti-vibration threaded bosses to improve fastening reliability.
3. The paper directly includes a connector box, cable clamp jaw, cable tie, and strain-relief grooves. It explains that the cable body is clamped at the connector-box entrance and also states that later testing includes cable clamping and mechanical shake/vibration tests.
4. This helps WP6 because the UDC support/rack system will also need more than a simple frame. It should support sensitive equipment, allow proper connector/cable routing, include secure clamping and strain relief, and be designed with assembly, maintenance, and replacement in mind.

Impact:

This paper is useful for my WP6 because it supports the design of a practical mechanical support system around sensitive equipment. It shows that support structures should include not only mounting features, but also alignment, connector integration, cable clamping, strain relief, anti-vibration fastening, and replacement access. Even though the application is not a subsea UDC, the design thinking is still very relevant because WP6 is also about building a reliable support system around electronics rather than only drawing a simple frame. This makes the paper a strong supporting source for the rack/chassis and cable-support side of WP6.

Actions:

1. Include proper cable clamping and strain-relief features in the UDC support design.
2. Consider modular mounting so equipment can be installed and replaced more easily.
3. Use alignment and fastening features that improve assembly accuracy and stability.

Journal Paper / Book Chapter / Article citation: Moreno Soriano, R., Rojas, N., Nieto, E., Gonzalez-Huerta, R. G., & Sandoval-Pineda, J. M. (2021). Influence of gasket materials on clamping pressure distribution in a PEM water electrolyzer. International Journal of Hydrogen Energy, 46(51), 25944-25953. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0360319921009393?via%3Dihub>

Summary: Although this paper is based on a PEM water electrolyzer, it is useful for WP3 because it studies gasket materials, clamping pressure and bolt torque effects in an electrochemical stack. The WP3 housing also depends on controlled compression to prevent leakage and maintain stable contact. The study supports the use of a bolted front closure, a controlled tightening method, compression stops and a gasket land rather than a loosely clamped or exposed stack.

Questions (aims):

1. Why is controlled compression important for WP3?
2. How does gasket behaviour influence the mechanical design?
3. Why should the closure be bolted rather than simply pressed shut?
4. What CAD features show that the sealing strategy has been considered?

Answers (knowledge):

1. Controlled compression helps prevent leakage, uneven sealing and local over-compression of the cassette or gasket regions.
2. Gasket stiffness, material and contact pressure affect how evenly the load is transferred around the seal land.
3. A bolted closure allows a defined torque sequence, repeatable preload and easier removal for cassette service.
4. The CAD should show a bolted front closure, gasket or seal land, front access route and leak annulus or captured drain philosophy.

Impact: This source supports the mechanical calculations for bolt preload and gasket pressure. It also gives evidence for the front bolted closure shown in the CAD and helps explain why the pressure housing is designed around controlled sealing rather than an open exposed stack.

Actions:

1. Use controlled bolt torque in the design calculations.
2. Show the bolted front closure clearly in the CAD drawing.
3. Include a double gasket or leak annulus concept in the design description.
4. Avoid exposed gasket edges in the selected concept.
5. Link gasket compression to safety and maintainability in Table 6.
6. Include this completed worksheet in the project appendix.

Journal Paper / Book Chapter / Article citation: Srinarayana, N., Porteous, A., & Guo, Q. (2019). Design and performance of diffuser systems for desalination plant outfalls: A review. *Water Science and Technology: Water Supply*, 19(5), 1317–1328. <https://doi.org/10.2166/ws.2019.006>

Summary:

This review compared diffuser types at real desalination plants around the world. Multiport manifolds fitted with 45-60° inclined ports and operating at Froude numbers above 15 delivered the best results, achieving near-field dilution ratios of 20:1 to 50:1. The paper also addressed flow distribution: if the manifold-to-port area ratio is too low, upstream ports get more flow and dilution suffers. An area ratio of at least 3:1 to 5:1 is recommended. Poor distribution alone can cut effective dilution by 10-30%.

Questions (aims):

1. Which diffuser type performs best based on real operational data?
2. How do you ensure even flow distribution across all ports?
3. What Froude number range should be targeted?

Answers (knowledge):

1. Multiport manifolds with 45-60° inclined ports consistently achieved the highest dilution ratios (20:1 to 50:1). Single-port outlets and rosette arrangements performed noticeably worse under comparable conditions.
2. The manifold cross-section area divided by the total port area should be at least 3:1 to 5:1. A low ratio starves downstream ports while overfeeding upstream ones. Tapering the manifold or varying port sizes can also help balance flow.
3. $F = 15-30$ is the practical sweet spot. Below 10, the jet never develops properly, and dilution is poor. Above 30, port diameters become so small that blockage becomes a risk.

Impact:

This confirms the multiport manifold as the right choice. The area ratio guideline helps size the manifold pipe, and the Froude number window of 15-30 gives clear port sizing targets. Whether tapering is needed depends on the final port count.

Actions:

1. Size the manifold pipe so the area ratio stays at or above 3:1 to 5:1.
2. Select port diameter and count to keep Froude number between 15 and 30.
3. Assess whether manifold tapering is required for the chosen configuration.

Journal Paper / Book Chapter / Article citation: Anwar, S. and Khan, F. (2021) 'Safety in Extreme Environments,' *Corrosion Behavior of Aluminium Alloys 2024 and 7075 under Simulated Marine Environment* [Preprint]. Available at: <https://doi.org/10.1007/s42797-021-00045-x>.

Summary:

A review of corrosion characteristics of common metallic materials in seawater environments is provided in the paper by Anwar, where the main types of corrosion are discussed, as well as material composition effects on corrosion resistance, in particular, chromium, molybdenum, and nitrogen content effects. The Pitting Resistance Equivalent Number is introduced as a material selection criterion. Requirements for corrosion-resistant materials in sour and marine environments are provided in NACE MR0175/ISO 15156, and ASTM G48 is the standard for the test of resistance to crevice corrosion. These sources are used for the material selection of all wetted parts of the DOC seawater pumping circuit.

Questions (aims):

1. What are the main corrosion mechanisms affecting metallic materials in seawater?
2. Which materials are suitable for pump internals and piping in seawater service?
3. How must material selection account for the DOC pH-swing process (acid/base excursions)?
4. How is galvanic corrosion between dissimilar metals prevented?

Answers (knowledge):

1. Corrosion mechanisms. The four main mechanisms in seawater are: pitting (localised chloride attack on passive film), crevice corrosion (under gaskets or flanges where oxygen is depleted), galvanic corrosion (between dissimilar metals in electrical contact), and MIC (microbiologically influenced corrosion from biofilm). All four are relevant to the DOC circuit.

2. Material selection for seawater.

Material	PREN	Seawater Resistance	Recommended Use
Stainless 316L	~24	Poor, pits in seawater	Not for wetted parts
Duplex 2205	~35	Good below 25°C	Piping, manifolds, valve bodies
Super Duplex 2507	~42	Excellent	Pump impeller, shaft, casing
Titanium Grade 2	>100	Immune	Critical wetted parts, acid zones
PTFE-lined steel	N/A	Excellent (liner protects)	pH-swing process pipework

3. pH-swing compatibility. The DOC electrochemical process generates acid and alkaline streams. Super duplex 2507 resists both seawater chlorides and mild acid/alkali. EPDM elastomers are recommended for seals and gaskets — NBR (nitrile) degrades in acidic conditions and must be avoided.

4. Galvanic corrosion prevention. When two dissimilar metals are in electrical contact in seawater, the less noble metal corrodes preferentially. Prevented by using dielectric isolation kits (insulating sleeves and washers at flanged joints) between different alloy sections, and by applying sacrificial zinc anodes to the skid structure

Impact:

Super duplex 2507 selected for pump internals (impeller, shaft, casing); duplex 2205 for piping and manifolds; EPDM for all seals and gaskets. PTFE-lined pipe used in pH-swing zones. Dielectric isolation kits specified at all dissimilar metal joints. Zinc anodes are included on the skid structure. Material selection table to be included in the WP2 design calculations section.

Actions:

1. Specify Super Duplex 2507 for all pump wetted internals; Duplex 2205 for piping, manifolds, and valve bodies.
2. Use EPDM seals and gaskets throughout; confirm compatibility with DOC acid/alkali streams.
3. Specify PTFE-lined pipework for any sections directly handling pH-swing process fluid.

Journal Paper / Book Chapter / Article citation: Hu, Y., Tang, Y., Ji, X., & Chen, Y. (2025). *A Submerged Building Strategy for Low-Carbon Data Centers in Coal Mining Subsidence Areas: System Design and Energy–Carbon Performance Assessment*. *Buildings*, 15(17), 3148. Available at: [mdpi.com/2075-5309/15/17/3148](https://www.mdpi.com/2075-5309/15/17/3148)

Summary:

This paper proposes a submerged building that rejects heat through wall-mounted plate heat exchangers using surrounding water as a heat sink. Its thermal model was validated with deviations generally below 3%, and the submerged solution reduced cooling energy use by 42.5-64.3% compared with conventional systems. Although this is not an underwater habitat paper, it is one of the best fully open-access submerged-building thermal analogs for your design.

Questions (aims):

1. How can a submerged structure reject heat effectively to the surrounding water?
2. Can plate heat exchangers integrated into the structure reduce cooling energy demand?
3. How accurately can submerged heat-exchange models be validated?

Answers (knowledge):

1. The study shows that surrounding water can be used as a low-grade heat sink through integrated plate heat exchangers.
2. The approach significantly reduces cooling energy demand.
3. Model validation stayed within a low deviation range, which supports screening-level habitat heat-exchanger sizing.

Impact:

1. Strongly supports plate heat exchanger selection.
2. Gives a good precedent for passive/semi-passive heat rejection.
3. Helps justify submerged heat-exchanger integration into the habitat envelope.

Actions:

1. Use this paper in your HX-type discussion.
2. Reference it when comparing passive vs pumped-loop concepts.
3. Cite it when explaining why plate heat exchangers are a strong option.

Journal Paper / Book Chapter / Article citation: Craig, I. (2018). Seawater intake risers for floating desalination plants. Doctoral Report, University of Sunderland. Available at: <https://sure.sunderland.ac.uk/id/eprint/10129/2/Craig,%20Ian%20Doctoral%20Report%20-%20September%202018.pdf>

Summary:

This doctoral report examines riser pipe intake systems for floating desalination platforms, focusing on hydraulic performance, debris exclusion, and biofouling mitigation. It details bell-mouthed intake geometries with trash racks and velocity caps that limit approach velocities to 0.1-0.15 m/s while screening out macro-debris larger than 50 mm. The study presents CFD modeling of multi-port intake arrays, verifying uniform flow distribution and NPSH margins under tidal variations and partial fouling conditions. Material selection emphasizes corrosion-resistant composites and coated steels, with clean-in-place systems integrated into riser designs for maintenance without diver intervention.

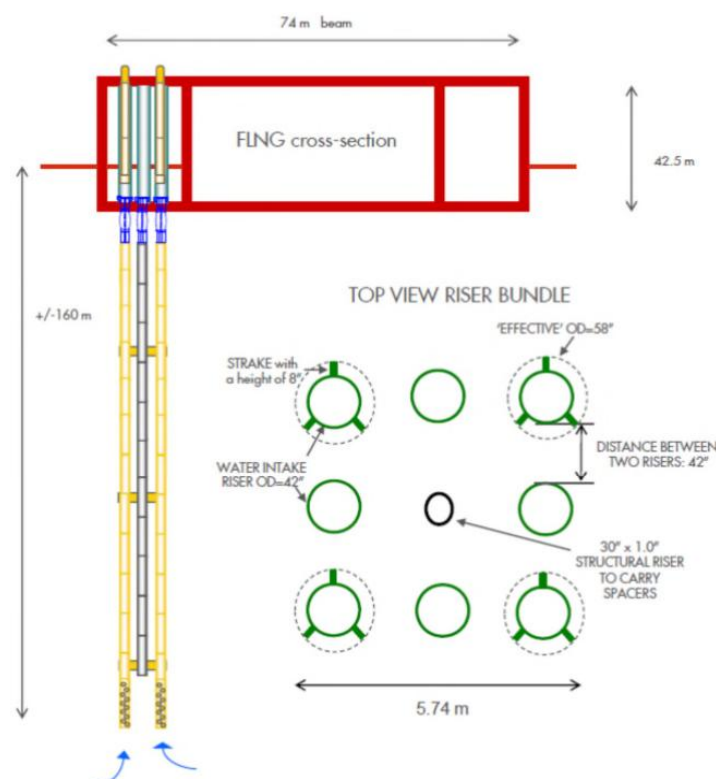


Figure 37. Shell Prelude FLNG Seawater Intake Riser (SWIR) Bundle.

Questions (aims):

1. How should multi-port intake risers be arranged and screened to achieve uniform flow distribution while excluding marine debris?
2. What CFD modeling approach validates hydraulic performance across clean, partially fouled, and peak debris loading scenarios?
3. How can intake designs incorporate fouling factors while maintaining adequate NPSH at pump suction during tidal extremes?
4. Which riser materials and surface treatments optimize corrosion resistance and reduce biofouling accumulation rates?

Answers (knowledge):

1. Symmetrical bell-mouth intake ports with 50 mm bar spacing and velocity caps achieve debris exclusion while limiting through-screen velocities to 0.3-0.5 m/s.
2. CFD analysis confirms that riser spacing greater than 3D and intake elevation above seabed prevents recirculation zones and ensure even flow split.
3. Design head loss budget includes clean condition (0.2 m), 50% fouled (0.8 m), and 100% fouled (1.8 m) scenarios to maintain $NPSH_a > NPSH_r + 1.5$ m margin.
4. HDPE riser pipes with electropolished stainless screens and silicone fouling-release coatings reduce maintenance intervals from monthly to quarterly.

Impact:

- Flow uniformity: Multi-port CFD validation prevents pump cavitation and capacity imbalance across parallel risers.
- Debris resilience: Velocity cap geometry reduces large debris loading by 85% compared to open pipe intakes.
- Maintenance reduction: Integrated CIP systems eliminate 70% of underwater inspections for floating platform intakes.

Actions:

1. Adopt the report's bell-mouth geometry and 50 mm screening for your skid intake design, scaling port diameters to match required flow.
2. Apply CFD methodology to model your intake-to-pump flow path, incorporating a 3x fouling factor for conservative NPSH verification.
3. Specify HDPE piping with electropolished screens per the material performance data to minimize corrosion and biofouling.
4. Integrate high-pressure CIP headers at intake ports following the report's maintenance philosophy for diver-free cleaning cycles.

Journal Paper / Book Chapter / Article citation: Zan, Y., Li, Z., Li, F., Qin, L., Huang, F., & Qi, B. (2025). Dynamic characteristics of subsea structures being lifted out of water in irregular waves. Ocean Engineering, 328, 121044. Available at: <https://doi.org/10.1016/j.oceaneng.2025.121044>

Summary:

This paper studies the dynamic behaviour of subsea structures during lifting out of water in irregular waves. The authors use both experiments and numerical simulations to analyse the water-exit process, with main focus on cable tension and structural motion. The results show that irregular waves strongly affect the lifting process, especially when the structure passes through the splash zone, where cable tension becomes more critical. The paper also shows that more severe sea states increase tension and motion instability. This is useful for my WP6 because it supports the lift-case / recovery / base-frame safety side of the UDC support design.

Questions (aims):

1. Why is it important to study subsea structures being lifted out of water?
2. What main factors affect cable tension during lifting?
3. What operating conditions are considered safer for lifting?
4. How can this paper help the lift-case part of WP6?

Answers (knowledge):

1. It is important because lifting subsea structures out of water is a common offshore operation, and safety problems can happen due to changing wet surface area, fluid-structure interaction, free-surface effects, and dynamic cable loading. The paper says these effects can threaten structural safety during recovery and lifting operations.
2. The paper shows that irregular waves, sea-state severity, and the structure passing through the splash zone all strongly affect cable tension. As sea conditions become worse, the maximum cable tension increases and its fluctuations become larger.

Table 125. Case setup for numerical simulations.

Case	H_s (m)	T_p (s)
1	0.5	3
2	1	4
3	1.5	5
4	2	6
5	2.5	7

3. The paper states that safer conditions happen when significant wave height $H_s \leq 1.5$ m, because cable tension stays more stable. When $H_s > 2$ m, tension increases a lot and large fluctuations appear, which can reduce operational safety.

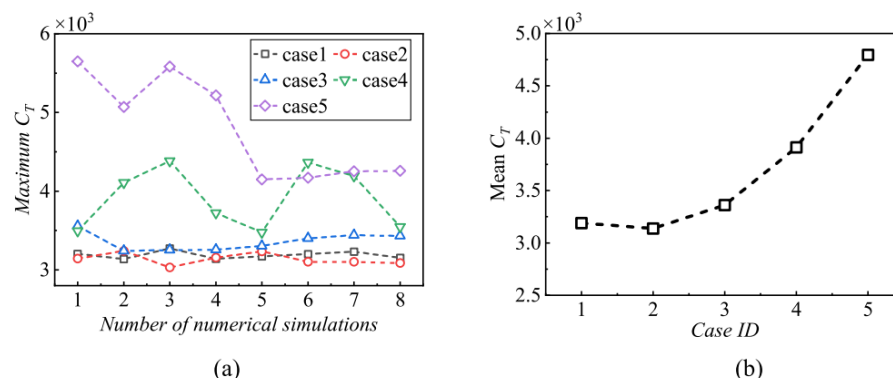


Figure 38. Dimensionless maximum tension coefficient (C_T) in five cases: (a) Maximum tension coefficient in eight simulations; (b) Mean value of the maximum tension coefficient.

4. This helps WP6 because the UDC support/base frame may need lifting and recovery, so the design should consider lift cases, cable tension, wave effects, and especially the danger of the splash zone during water exit. It supports the idea that recovery should be planned around safe environmental limits, not treated as a simple static lifting task.

Impact:

This paper is useful for my WP6 because it supports the lift-case and recovery part of the UDC support design. It shows that lifting a subsea structure out of water is strongly affected by irregular waves and that cable tension can increase significantly in bad sea conditions, especially in the splash zone. It also gives practical guidance on safer operating conditions, which is useful when thinking about recovery planning and safe handling of the UDC support/base frame. However, it is not specifically about a UDC rack system, so it should be used as a supporting source rather than the only design source.

Actions:

1. Treat lift cases as an important part of the WP6 support/base-frame design.
2. Consider splash-zone effects and dynamic cable tension during recovery.
3. Use safer environmental limits when planning lifting/retrieval operations.
4. Use this paper as support for why recovery out of water must be treated as a dynamic offshore operation, not only a simple lifting task.

Journal Paper / Book Chapter / Article citation: Shokri, A., & Sanavi Fard, M. (2022). Corrosion in seawater desalination industry: A critical analysis of impacts and mitigation strategies. Chemosphere, 307, 135640. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0045653522021336?via%3Dihub>

Summary: This review is useful for WP3 because the reactor housing, side-loop and service ports are exposed to seawater and chemically shifted streams. The paper explains that seawater systems face general corrosion, crevice corrosion, biofouling and scaling risks. For the WP3 design, the key implication is that material selection must be linked to geometry and maintenance. Flanges, gasket lands, ports and mixed-metal joints become risk areas if they trap seawater or create galvanic couples.

Questions (aims):

1. Which WP3 features create corrosion or crevice risk?
2. What material strategy is most suitable for the reactor housing?
3. Why are drain and flush routes important for corrosion control?
4. How should galvanic effects be managed in the CAD and report?

Answers (knowledge):

1. Bolted flanges, gasket grooves, sensor bosses, drain ports and dissimilar-metal supports are the main local risk areas.
2. Titanium Grade 2 or super-duplex wet parts are suitable screening choices, supported by coatings, liners or dielectric isolation where needed.
3. Drain and flush routes reduce stagnant seawater, which lowers crevice corrosion, scaling and biofouling risk during shutdown.
4. Use dielectric washers, isolation pads and material compatibility checks wherever titanium, stainless steel and structural frame materials meet.

Impact: This source strengthens the seawater suitability section of WP3. It supports the use of corrosion-resistant materials, coatings, dielectric breaks, drains and flushing features. It also explains why the CAD should include service access and not rely on a maintenance-free sealed body.

Actions:

1. Prepare a simple material and galvanic compatibility matrix for WP3.
2. Keep drain, flush and inspection access in the CAD model.
3. State titanium or super-duplex as the wet-body material basis.
4. Add dielectric isolation at dissimilar joints and skid contacts.
5. Include corrosion and fouling as risks in the WP3 safety table.
6. Include this completed worksheet in the project appendix.

Journal Paper / Book Chapter / Article citation: Palomar, P., Lara, J. L., Losada, I. J., Rodrigo, M., & Álvarez, A. (2012). Near-field brine discharge modelling part 1: Analysis of commercial tools. *Desalination*, 290, 14–27. <https://doi.org/10.1016/j.desal.2012.01.005>

Summary:

Palomar and colleagues tested how accurately CORMIX, VISUAL PLUMES (UM3), and VISJET predict dense jet behaviour by comparing them against lab data. Trajectory predictions (rise height, return distance) were within 15-20% of measurements, which is acceptable. Dilution predictions, however, showed errors of up to 30-50%. CORMIX tended to overestimate dilution, which is risky for design, while UM3 leaned toward more conservative values. The key recommendation is to run at least two models and adopt the lower dilution number.

Questions (aims):

1. How reliable are the standard modelling tools for dilution prediction?
2. Which model gives the safest (most conservative) results?
3. What is the best modelling strategy for design?

Answers (knowledge):

1. Trajectory parameters are predicted within 15-20%, which is reasonable. Dilution estimates, however, can deviate by 30–50% depending on the discharge angle, Froude number, and ambient current. Model outputs should be treated as approximate, not exact.
2. UM3 within VISUAL PLUMES was the most conservative, especially for 60° dense jets in still water. CORMIX often overpredicted dilution, meaning a diffuser sized with CORMIX alone might not meet targets in reality.
3. Run at least two independent models and base the design on the lower dilution figure. Carry out sensitivity analysis by varying angle, current speed, and brine density to see how much the prediction shifts. If models disagree significantly, physical testing may be warranted.

Impact:

Understanding model uncertainty is important since the project will rely on numerical simulations. The dual-model strategy with conservative selection provides a built-in safety margin. Access to both UM3 and CORMIX should be confirmed early on.

Actions:

1. Run near-field dilution simulations in both UM3 and CORMIX.
2. Adopt the more conservative dilution result for diffuser sizing.
3. Perform sensitivity analysis on the key input parameters.

APPENDIX D:

Rich Picture

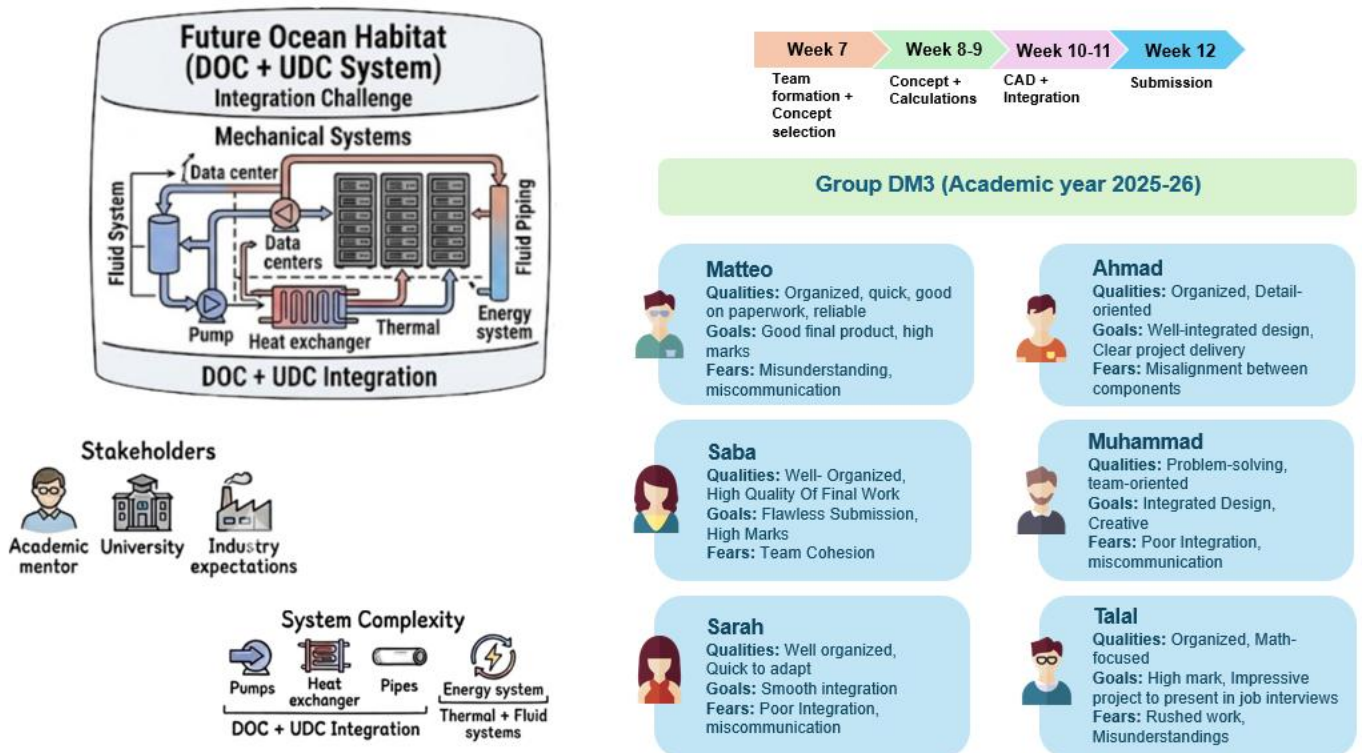


Figure 39. Rich picture of “Future Ocean Habitat” for Group DM3.

Figure 39 illustrates the team’s rich picture for the underwater habitat project, centred on delivering an innovative, well-integrated mechanical design aimed at achieving high academic performance and strong professional presentation value. The diagram captures both internal and external stakeholders, including the student team, academic mentor, and university, alongside broader stakeholders such as engineers, industry expectations, and regulatory or environmental considerations.

The rich picture emphasises key system-level challenges associated with integrating Direct Ocean Capture (DOC) and the Underwater Data Centre (UDC) within a single habitat. Major technical considerations include fluid transmission systems (pumps, pipelines, and valves), thermal management and heat rejection, energy supply and storage, corrosion and biofouling in subsea environments, and overall system reliability and maintainability.

In addition, the diagram highlights critical project risks such as miscommunication, poor subsystem integration, CAD clashes, unrealistic assumptions, and time constraints. Team dynamics are also represented through each member’s qualities, goals, and fears, reinforcing the importance of collaboration, communication, and shared responsibility in achieving a successful outcome.

Finally, the rich picture incorporates a structured project timeline aligned with Assignment 2, outlining phases from team formation and concept selection (Week 7), through conceptual design and calculations (Weeks 8-9), to CAD development, system integration, and final submission (Weeks 10-12).

WP3 – Electrochemical Reactor Calculations

Table 49. Design requirements, assumptions, and constants for WP3 - DOC electrochemical reactor and pH-swing stack housing.

System Requirements and Design Goals	Requirements	Goals	
	- Maintain continuous pH-swing operation in a sealed stack - Ensure controlled gasket compression and leak containment - Allow stack cassette removal and isolated maintenance - Provide vent, drain, CIP, flush and integrity-test access - If HFMC is used, include bypass, isolation and anti-biofouling provisions	Parameter	Target
		Stack flow rate	0.004 m ³ /s (14.4 m ³ /h)
		Housing design pressure	300 kPa gauge screening
		Gasket compression	≥ 2 MPa average seal pressure
		HFMC side-loop velocity	1 to 2 m/s
		HFMC side-loop pressure loss	≤ 50 kPa concept target
		Service philosophy	Removable cassette and bypassable HFMC
Assumptions and Constants	Parameter	Symbol	Value
	Seawater density	ρ	1025 kg/m ³
	Gravity	g	9.81 m/s ²
	DOC process flow	Q	0.004 m ³ /s
	HFMC branch flow	Q_c	0.002 m ³ /s
	Housing radius	r	0.25 m
	Housing wall thickness	t	0.012 m
	Housing straight length	L_h	1.10 m
	Design pressure	P	300 kPa gauge
	Temperature change	ΔT	15 K
	Titanium thermal expansion	α	8.6×10 ⁻⁶ 1/K
	Titanium Young modulus	E	110 GPa
	Bolt diameter	d	12 mm
	Bolt torque	T	65 N m
	Torque factor	K	0.20
	Seal land area	A_{seal}	0.016 m ²
	HFMC line diameter	D_h	0.041 m
	HFMC equivalent length	L_{hfm}	5 m
	Closure bolt count	n	8
	Leak annulus width	w	0.006 m
	Leak annulus height	h	0.004 m
	Leak annulus mean diameter	D_{ann}	0.45 m
	Assumed wet cassette mass	m_c	45 kg
	Number of guides bearing pads	n_{pad}	4
	Guide pad width	b_{pad}	0.020 m

	Guide pad contact length	l_{pad}	0.100 m
	HFMC friction factor	f	0.03
	HFMC minor loss coefficient	ΣK	8
	HFMC module allowance	ΔP_{module}	20 kPa
	Hydraulic efficiency	η	0.70
	Daily operating period	t_{day}	24 h/day

Note. Screening values reflect the WP3 brief and are used to define an initial mechanical concept. The pH-swing and HFMC assumptions follow published BPMED and HFMC DOC studies (Sharifian et al., 2022; Rivero et al., 2023; Aliaskari et al., 2024).

APPENDIX E: GENERATIVE AI

In accordance with the university's guidance on the ethical use of generative AI, this statement confirms that generative AI was used as a support tool during the completion of Integrated Design Project 2 (04 30338). All final ideas, analyses, and conclusions regarding the work packages and the Future Underwater Habitat are the result of the team's critical thinking and engineering judgment. The AI was used strictly as a thought partner and editorial assistant, as outlined below.

Declaration

Group DM3 confirms the use of generative AI in the following ways:

- Scoping out a topic or developing initial ideas: Yes
- Summarizing and understanding research: No
- To generate code used in this assignment: No
- Editing and proofreading work: Yes
- Other uses (Data structuring for CAD tables): Yes

Summary of Prompts and Application

1. Scoping Out Initial Ideas (Brainstorming)

- **Prompt Used:** "Generate ideas for an underwater habitat structure. Suggest features like hull material, structural shape, foundation type, deployment method, pressure hull design, access & egress, emergency systems, and corrosion protection."
- **How the Output Was Used:** The AI provided a list of plausible engineering options. Team reviewed these for technical feasibility, selecting Duplex Stainless Steel (ASTM A890) for corrosion resistance and BS 8888 as the drawing standard for the 3rd Angle Projections. These were further refined using engineering knowledge of subsea filtration and saline environments.

2. Editing, Proofreading, and Technical Structuring (Draft Refinement)

- **Prompt Used:** "Review the following paragraph for clarity and conciseness. Suggest improvements while maintaining a professional engineering tone. Draft a summarized table for P&ID and exploded view integration logic."
- **How the Output Was Used:** The AI acted as a technical editor to refine the clarity of the Integration and Clash Check Results. It also assisted in structuring the Integration Logic and Visualization Summary table, ensuring that the relationship between the 0.37 kW pump, Automatic Self-Cleaning Filter, and the 1800 mm maintenance envelope was clearly communicated for the project's Work Package. All final technical descriptions remain the team's original work.